模拟电子技术基础
Analog Device

2.2 Amplifier circuits based on transistors 晶体管放大电路

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电子与电气工程系 E^3

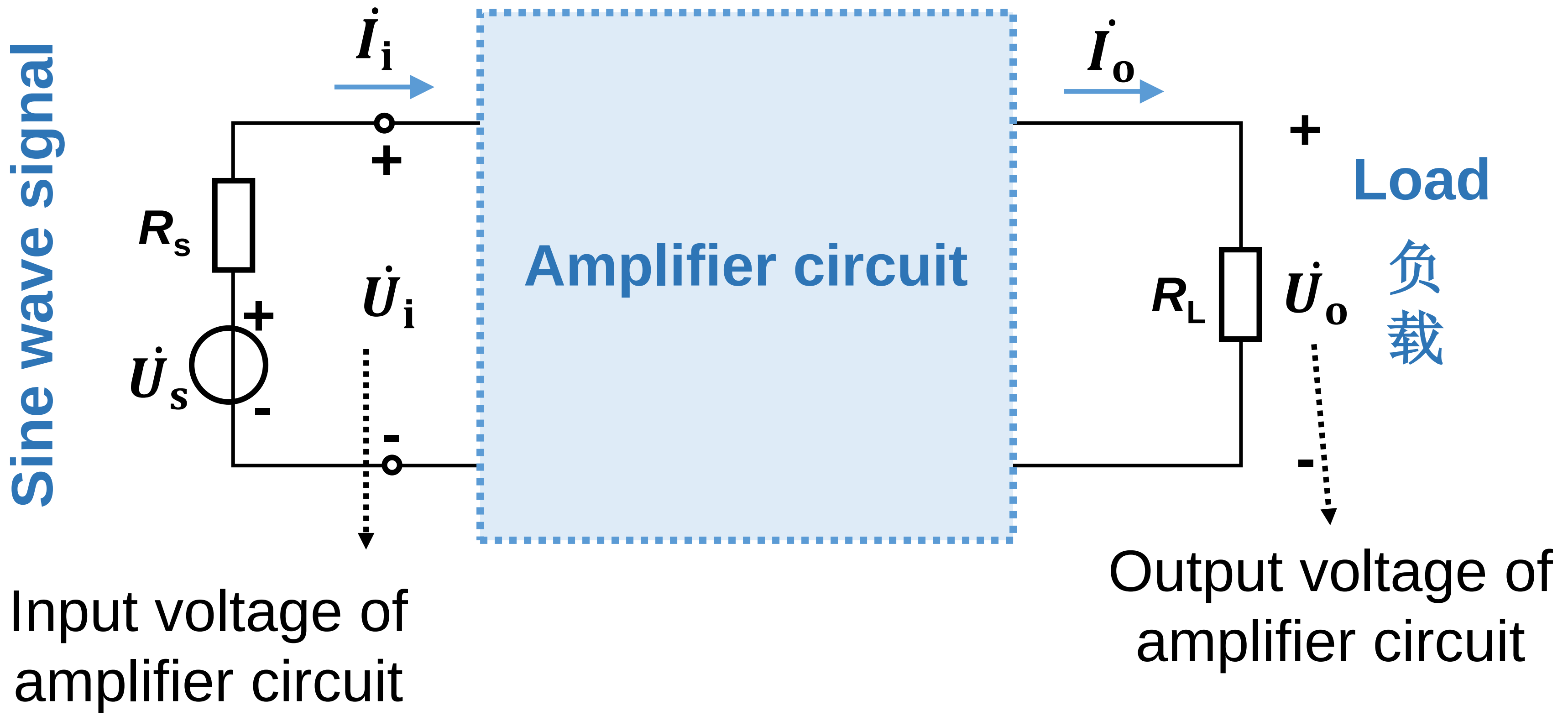
2.2.1 What is amplification

Red Thunder

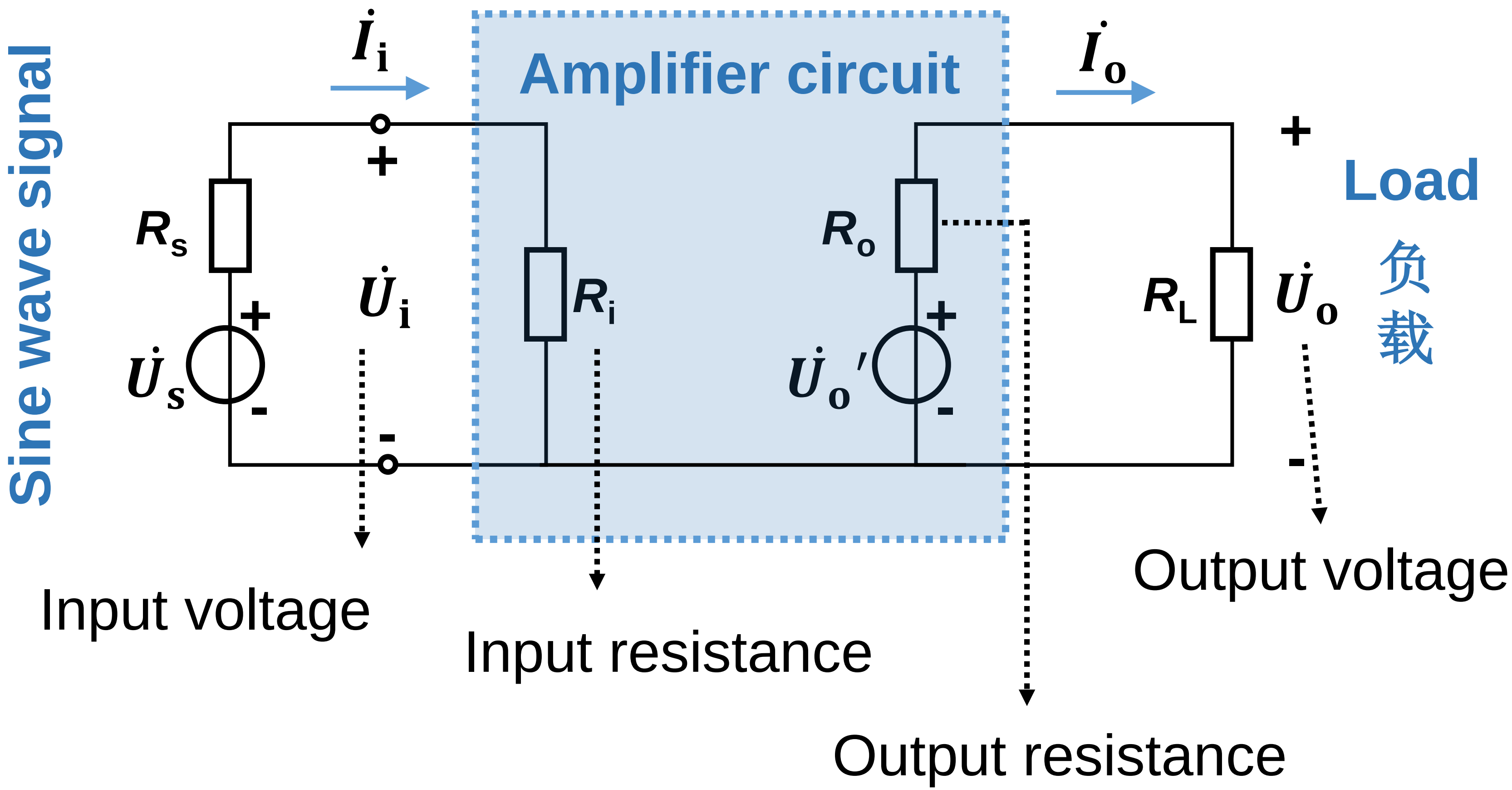
The requirement of amplification is: **Non-distortion**不失真

2.2.2 Key parameters of amplifier circuit

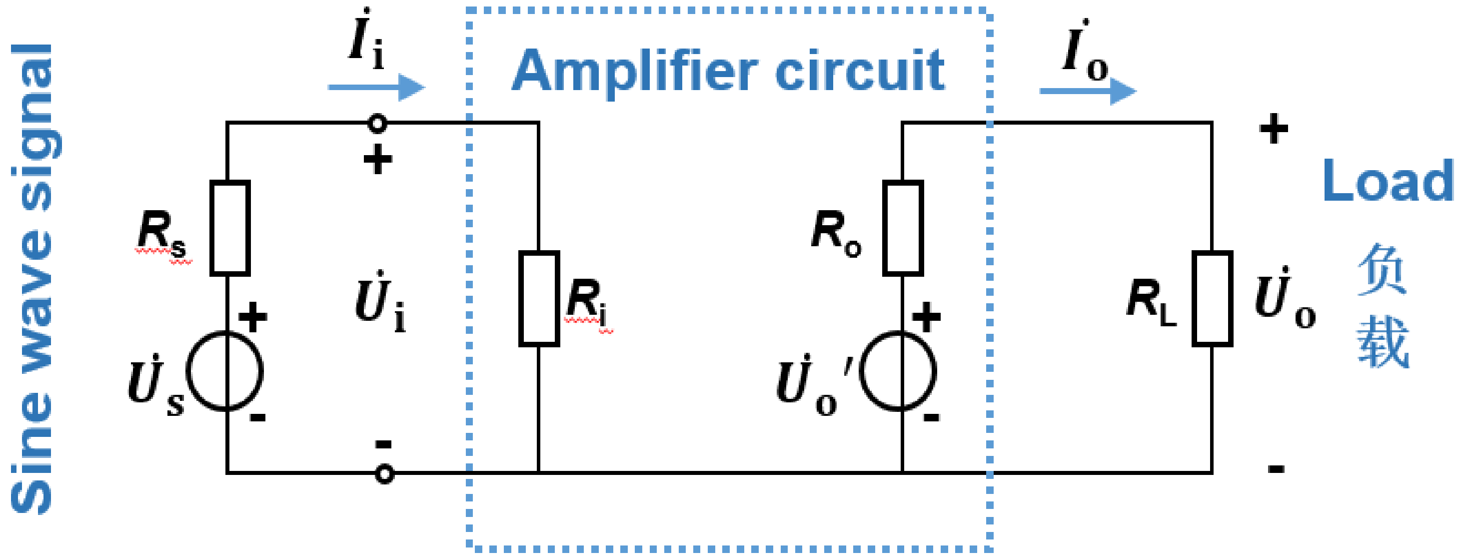
In most cases, signals are AC.



Equivalent circuit



Gain of amplifier circuit



Voltage gain

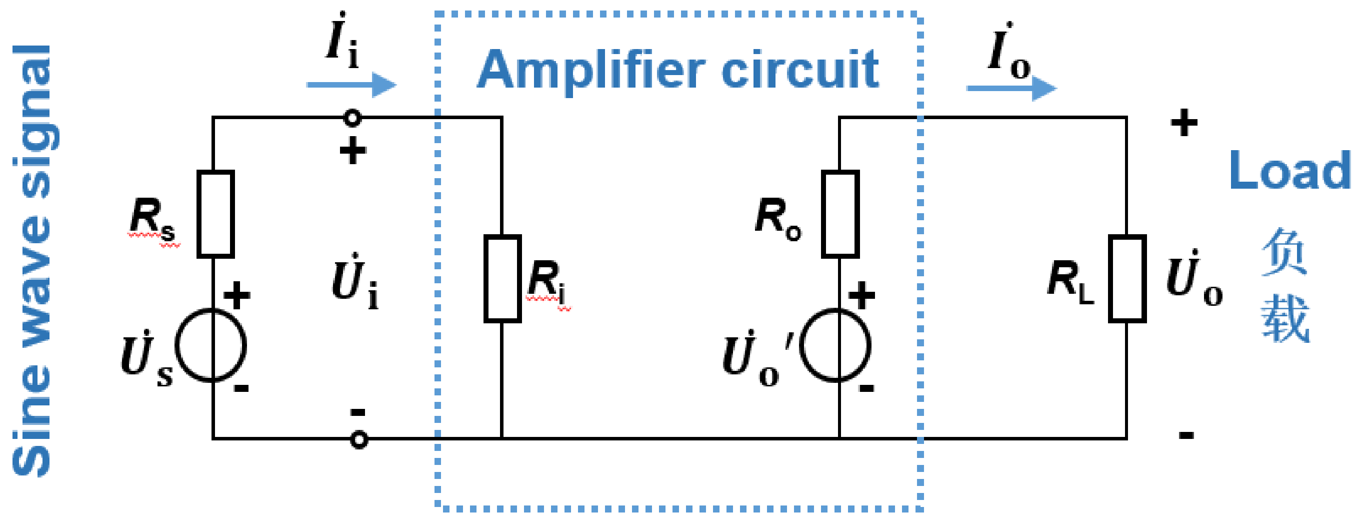
电压放大倍数

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$$

Voltage gain

电压放大倍数

$$\dot{A}_{us} = \frac{\dot{U}_o}{\dot{U}_s}$$



Current gain

电流放大倍数

$$\dot{A}_i = \frac{\dot{I}_o}{\dot{I}_i}$$

Transimpedance gain

互阻放大倍数

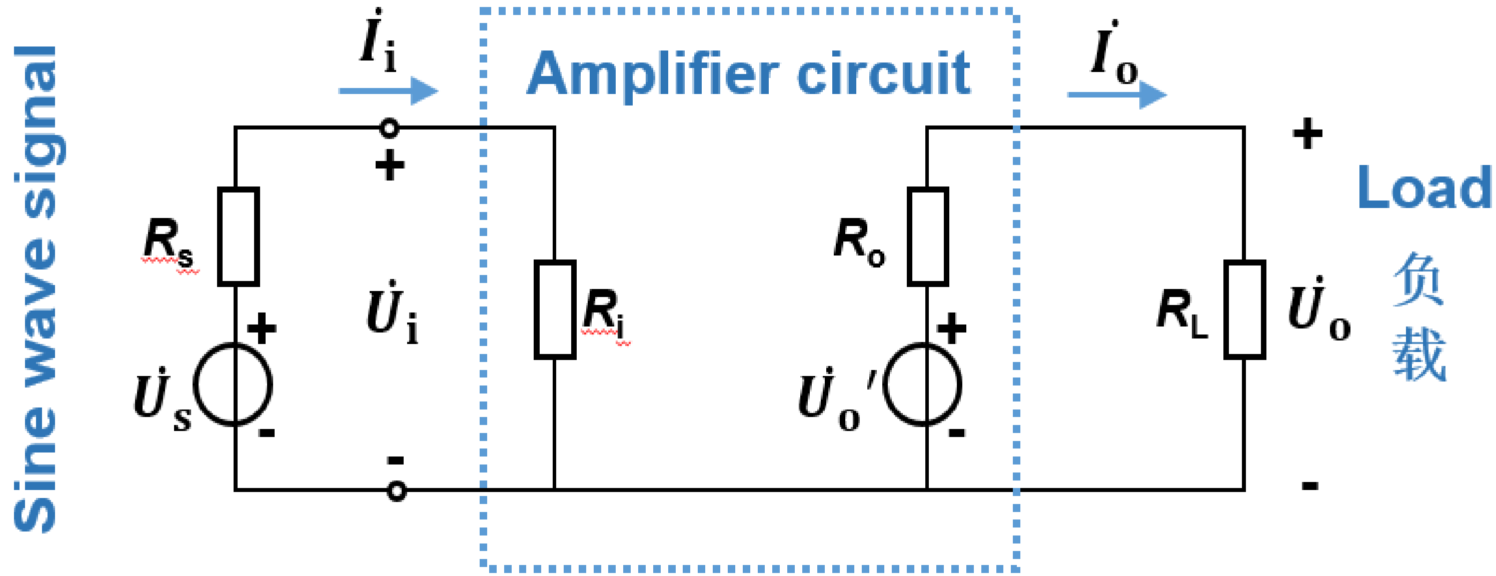
$$\dot{A}_{ui} = \dot{A}_r = \frac{\dot{U}_o}{\dot{I}_i}$$

Transconductance gain

互导放大倍数

$$\dot{A}_{iu} = \dot{A}_g = \frac{\dot{I}_o}{\dot{U}_i}$$

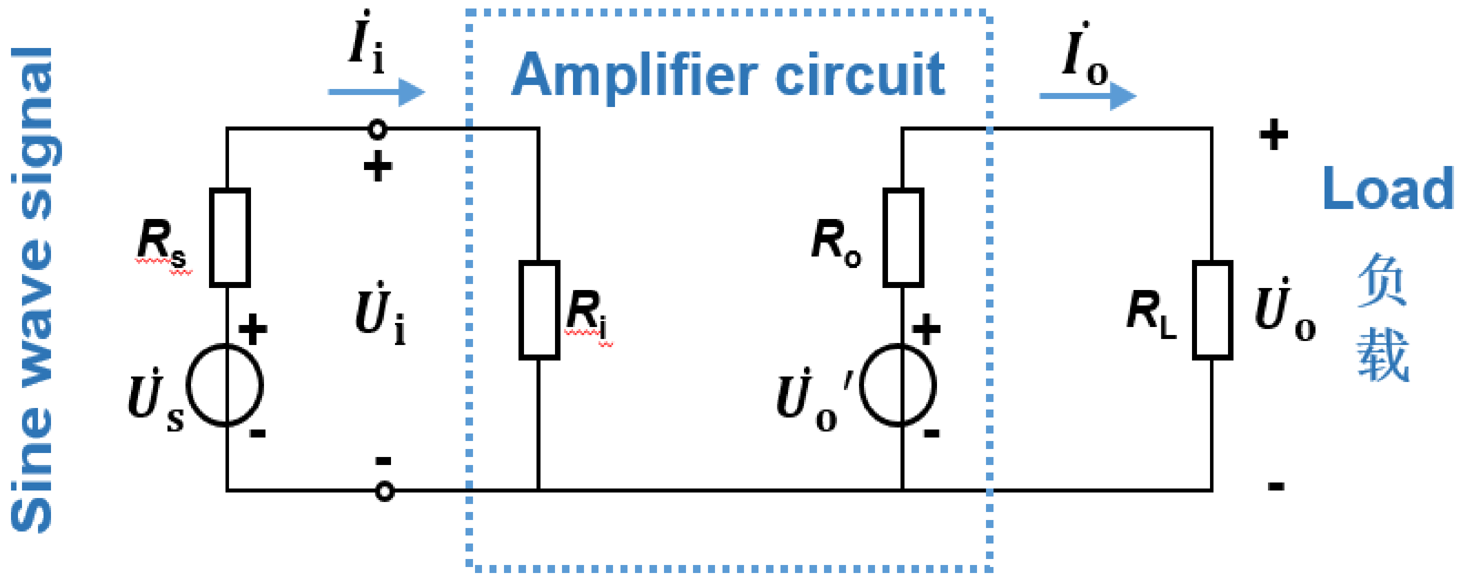
Input resistance of amplifier circuit R_i



$$R_i = \frac{\dot{U}_i}{\dot{I}_i} \qquad \dot{U}_i = \frac{R_i}{R_i + R_s}\dot{U}_s$$

R_i↑， U_i↑， better performance

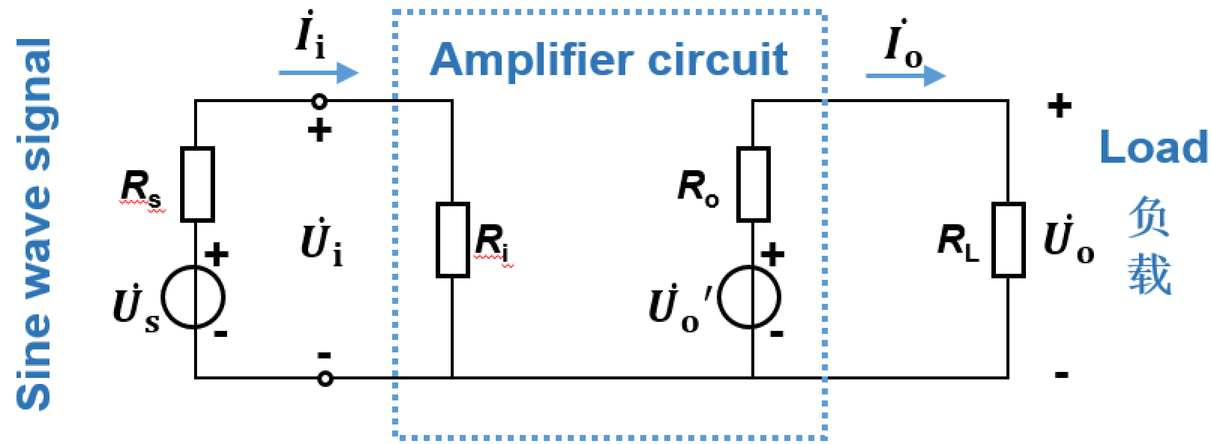
Output resistance of amplifier circuit R_o



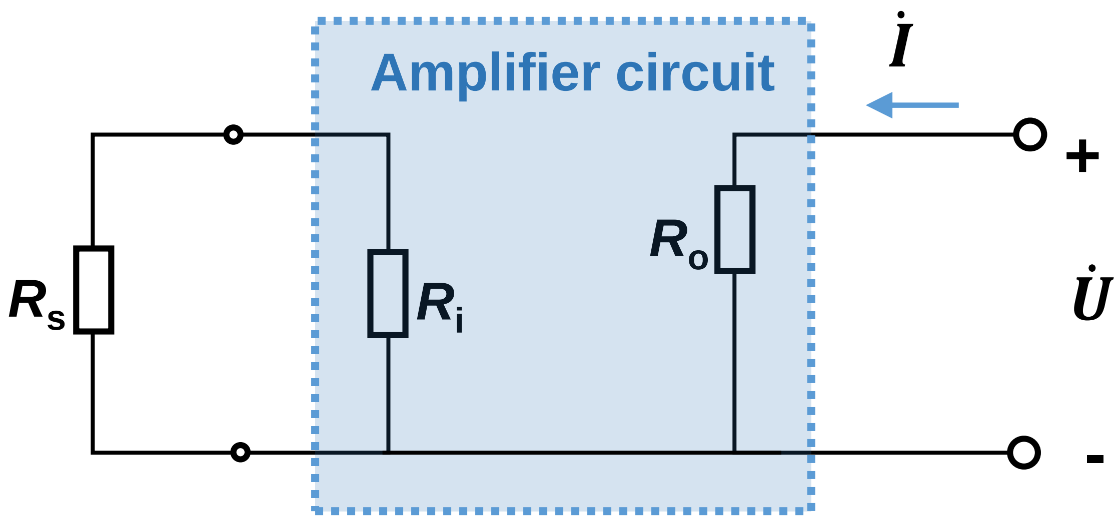
$$R_o = (\frac{U_o'}{U_o} - 1)R_L$$

$$U_o = \frac{R_L}{R_o + R_L} U_o'$$

R_o is smaller, U_o is more stable, better load capacity



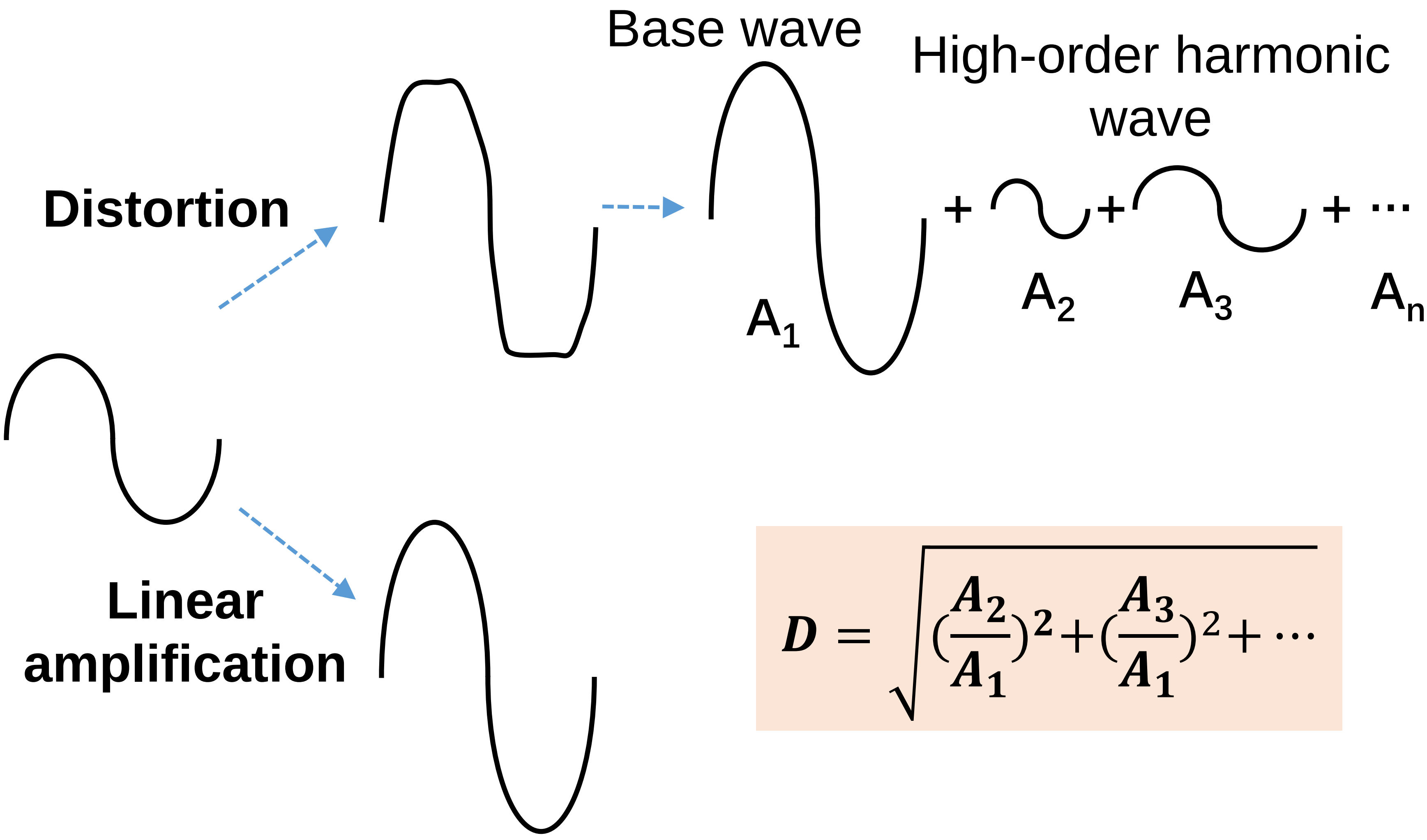
How to get output resistance?



$$R_o = \frac{\dot{U}}{\dot{I}}\Big|_{R_L=\infty, \dot{U}_i=0}$$

Nonlinear harmonic distortion coefficient

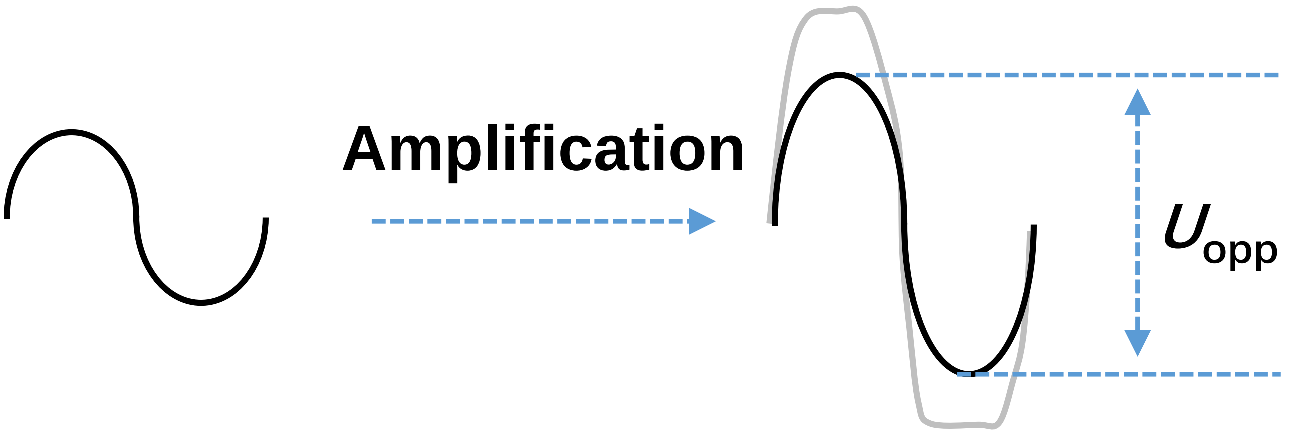
非线性(谐波)失真系数



$$D = \sqrt{(\frac{A_2}{A_1})^2 + (\frac{A_3}{A_1})^2 + \cdots}$$

The maximum non-distortion output voltage 最大不失真输出电压

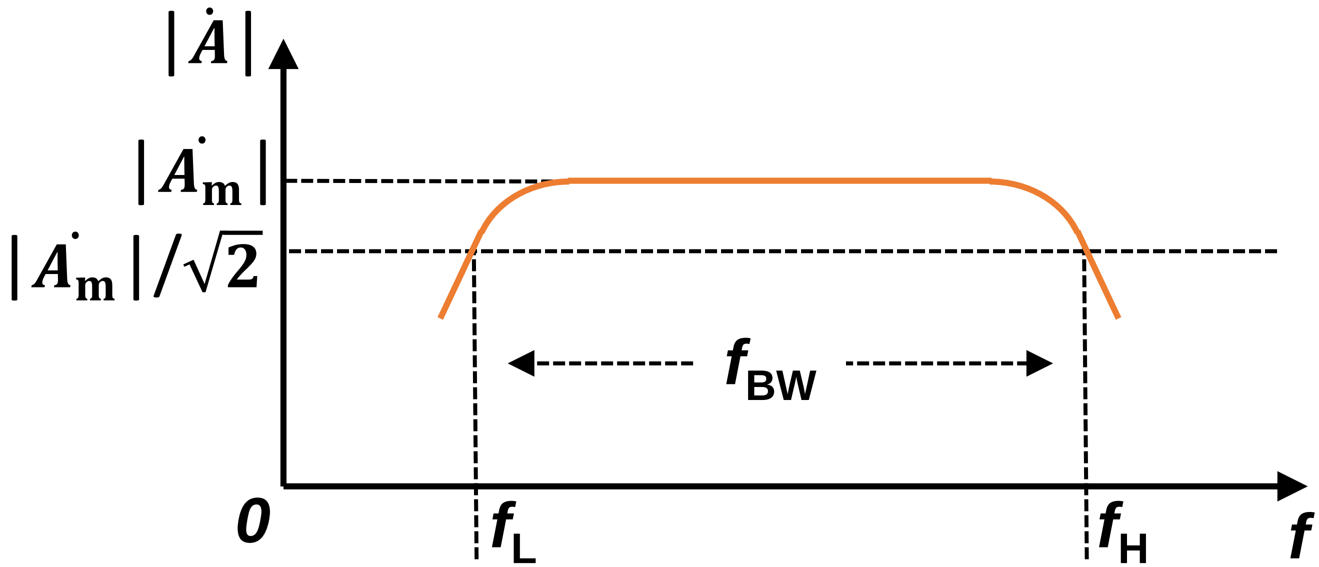
Peak-to-peak value U_{opp}, or Effective value U_{om}



$$\boldsymbol{U}_{\mathrm{opp}} = \mathbf{2}\sqrt{\mathbf{2}}\boldsymbol{U}_{\mathrm{om}}$$

Bandwidth 频带宽度/通频带

There are always capacitance and inductance in the amplifier circuit. When frequency is too high or too low, the gain will decrease. In general, an amplifier circuit can only **operate in a certain range of signal frequency**.

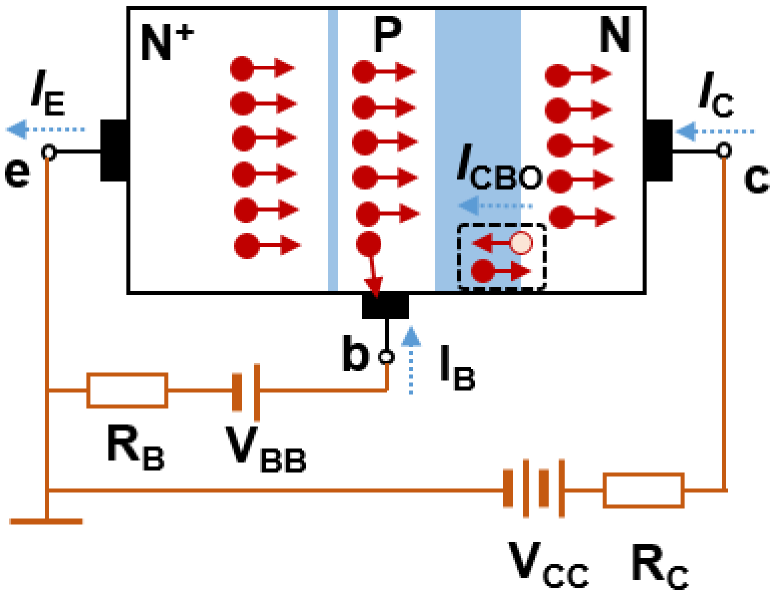


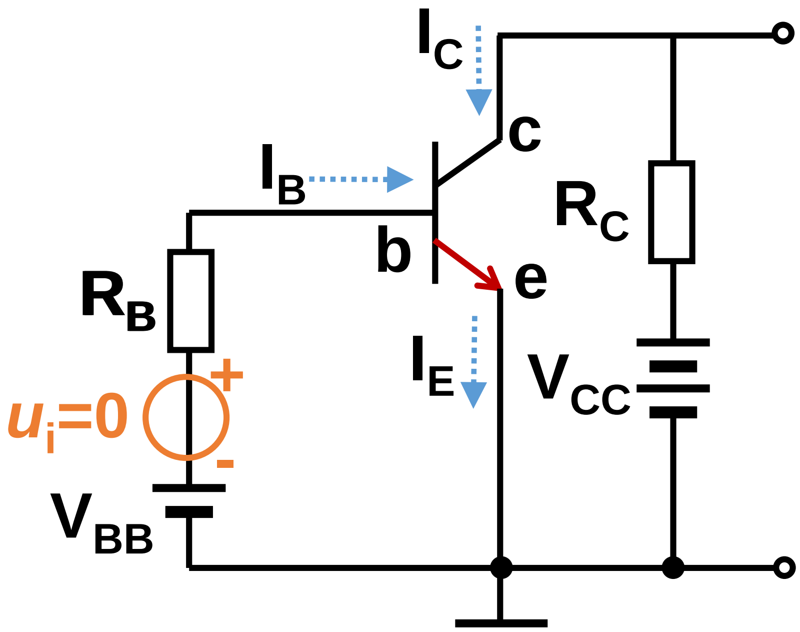
Lower cut-off frequency f_L Upper cut-off frequency f_H

Bandwidth: $f_{BW} = f_H - f_L$

2.2.3 Common-emitter amplifier circuit

Quiescent/static 静态 — no AC signal





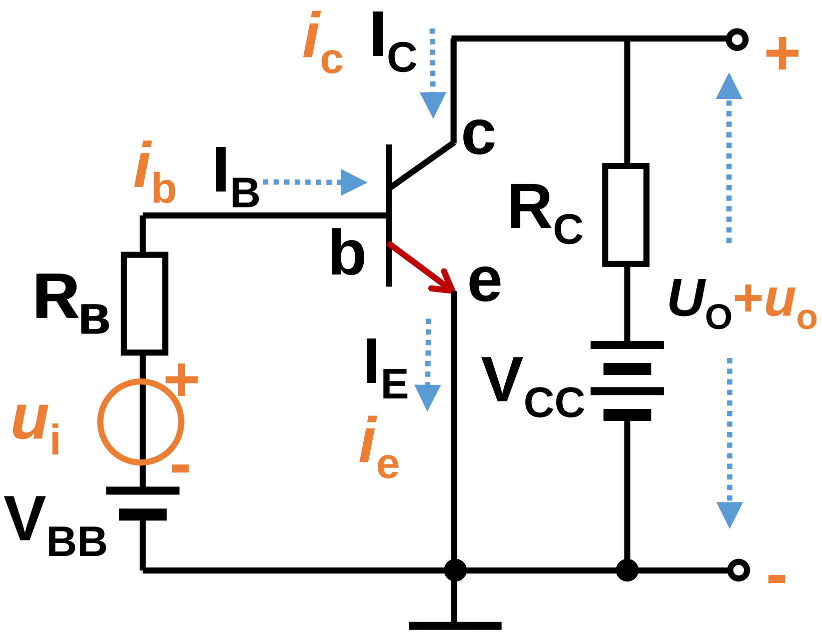
There is only DC currents in circuits

A circuit that only DC currents can pass

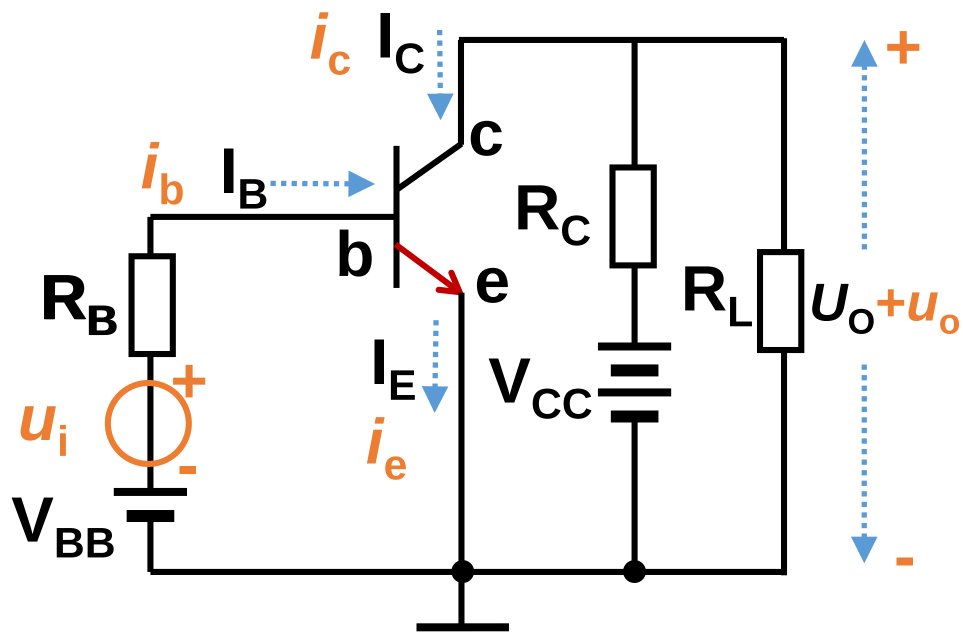
Dynamic 动态 — when there is AC input signal

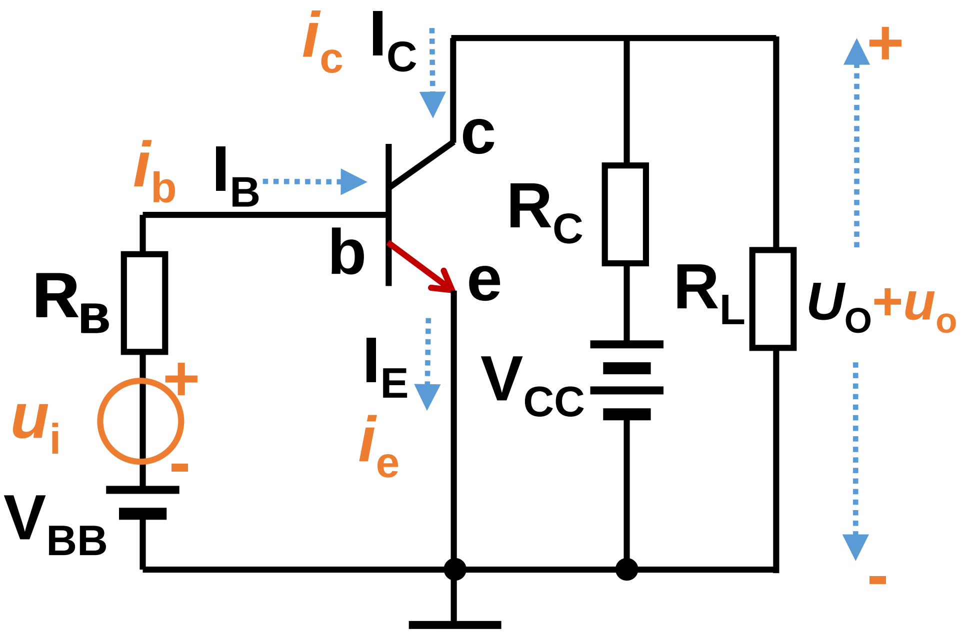
There are both DC and AC currents in circuits

Without load



With load



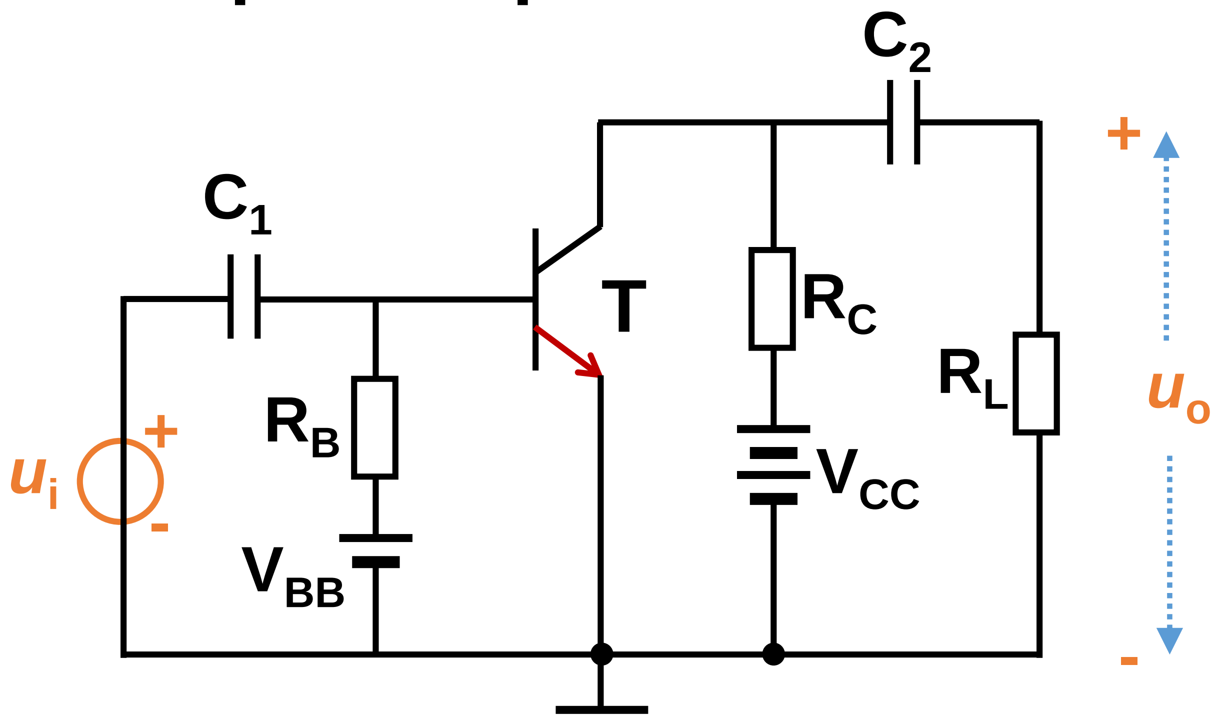


Disadvantages of the direct-coupled circuit

1) Amplification circuit and signal source will interact.

2) Amplification circuit and load resistor will interact.

Capacitor-coupled amplifier circuit:

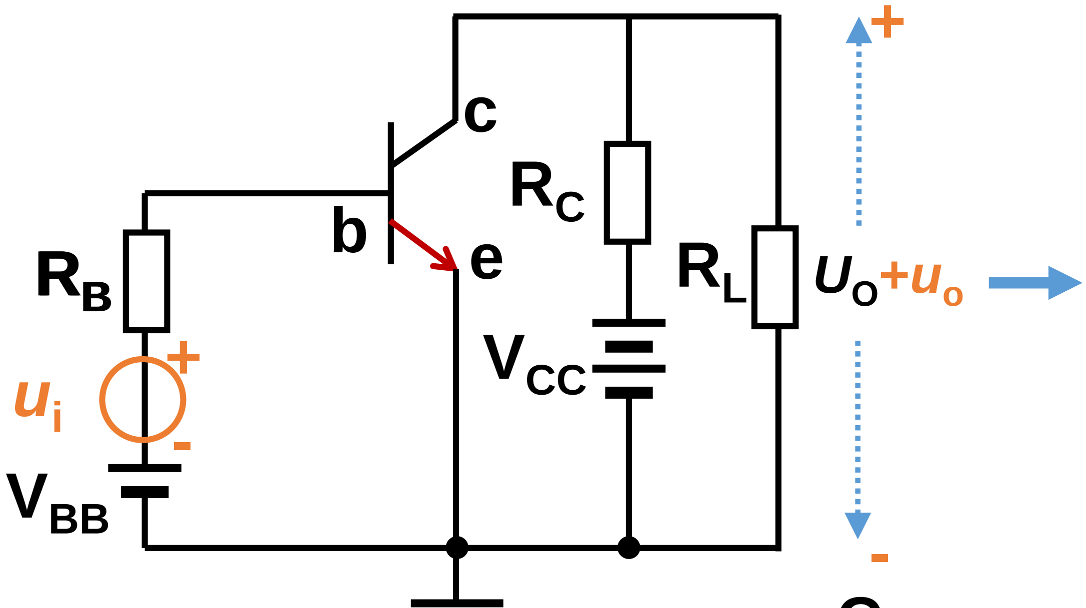


Capacitor C_1 & C_2:

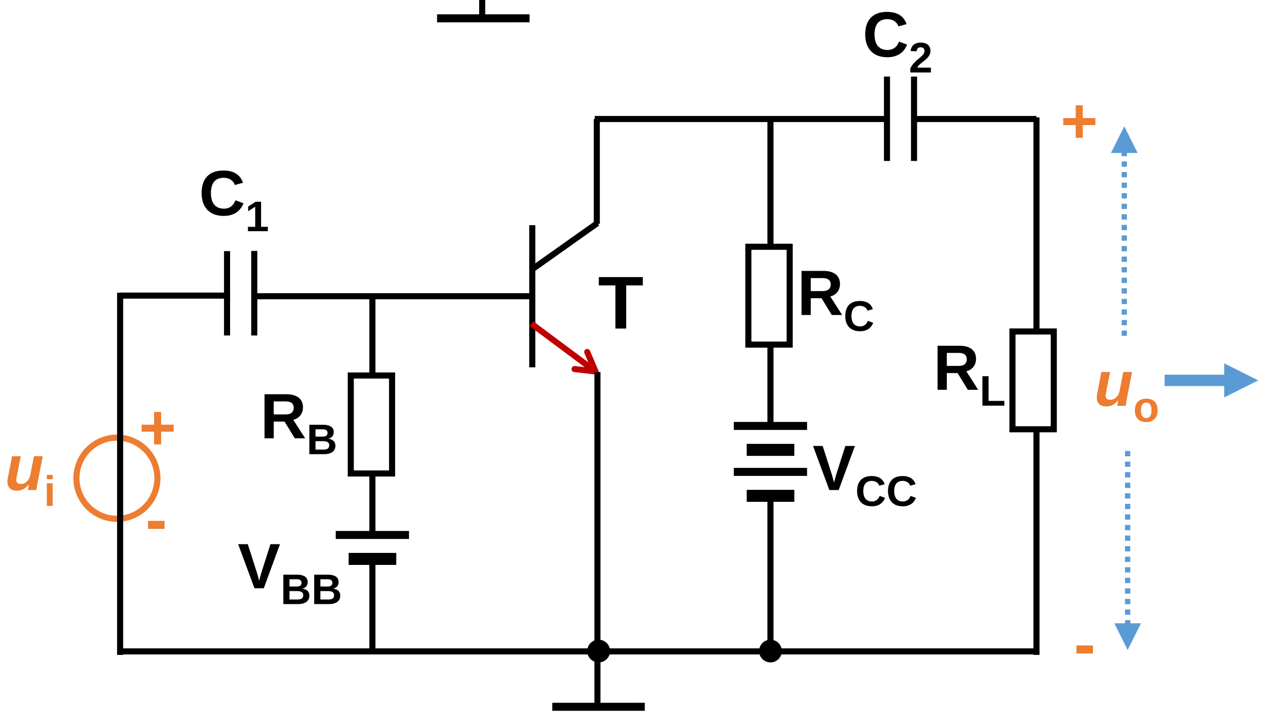
1) Separate signal and load from DC voltage and current.

2) AC signal can flow through capacitors.

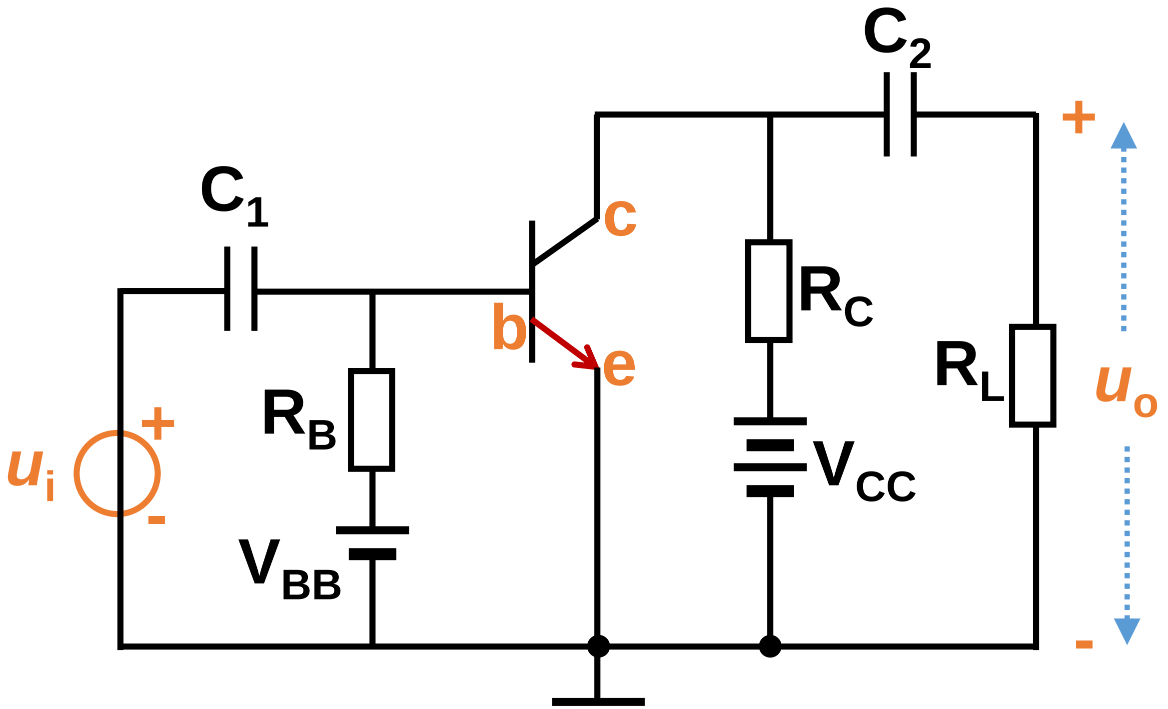
Comparison



Both DC and AC output voltages

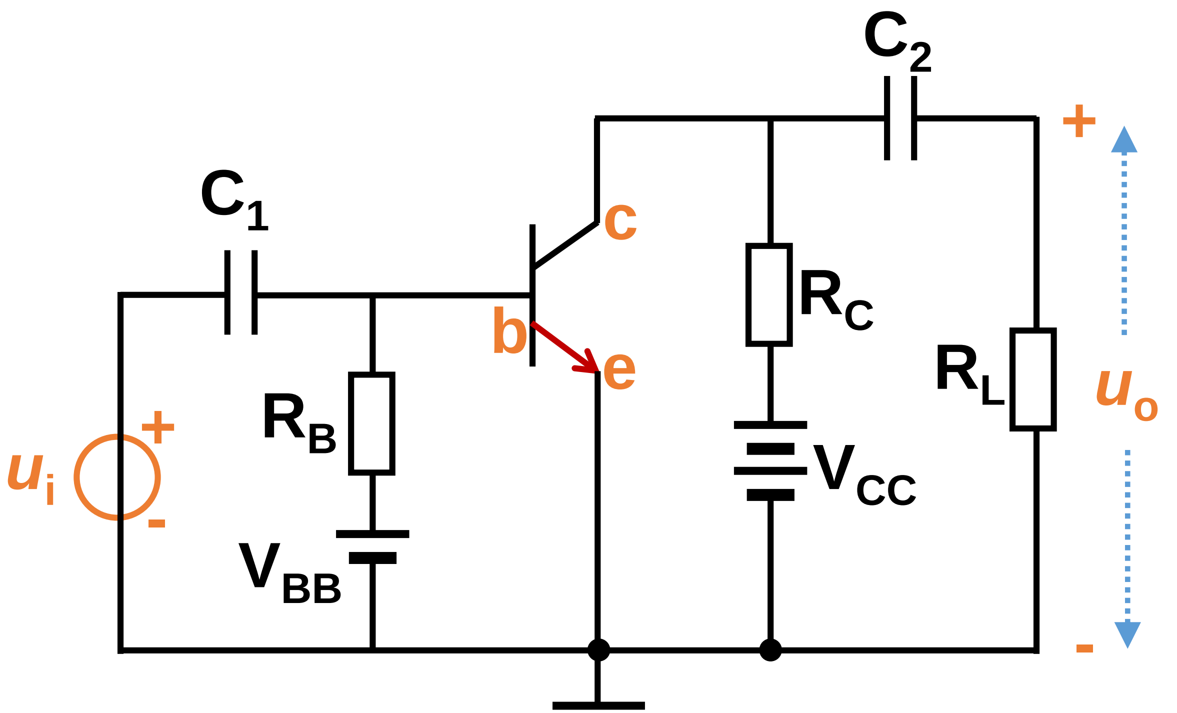


AC output voltages



V_{CC} { Provide reverse DC bias for BC junction.
Provide energy for the amplifier circuit.

V_{BB}, R_B { Provide forward DC bias U_{BE} for BE junction
Provide base DC current I_B

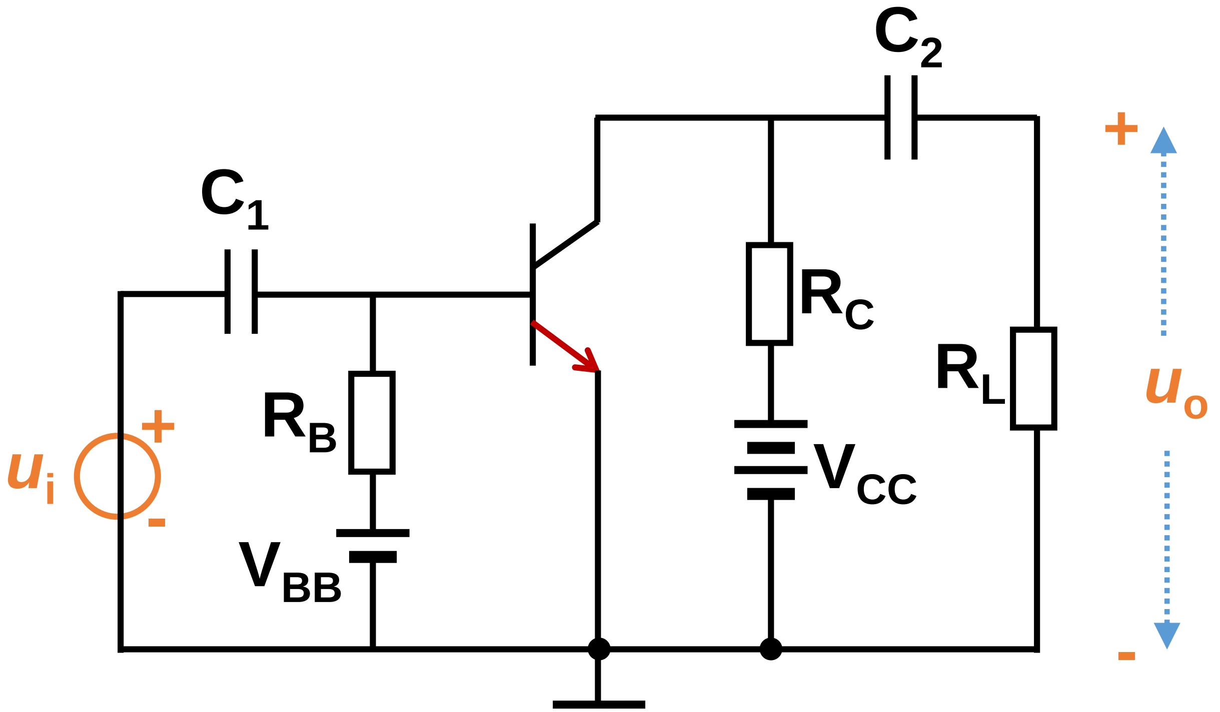


R_C {

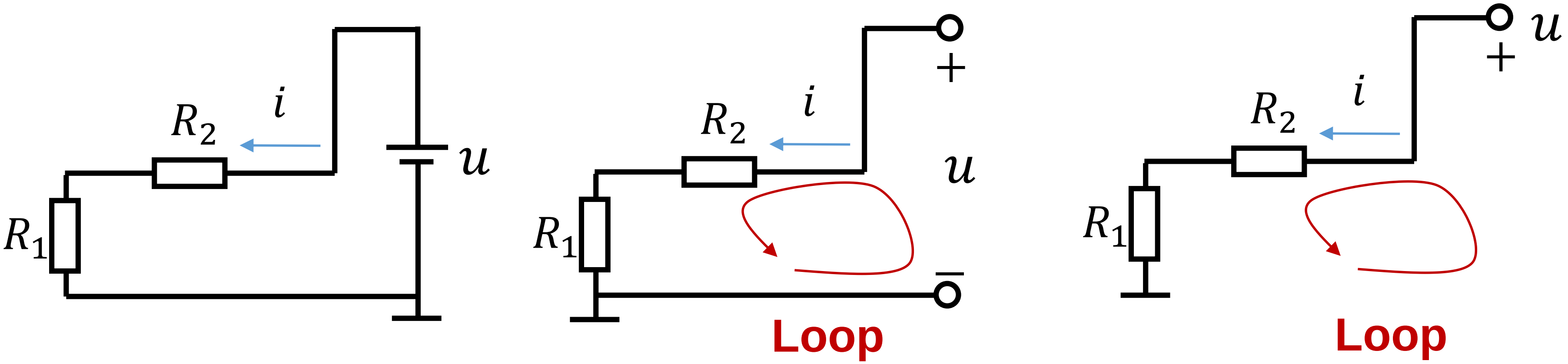
Adjust the voltage drop on BC junction.

<u>Transform current signal to voltage signal</u>, and realize the amplification of voltage.

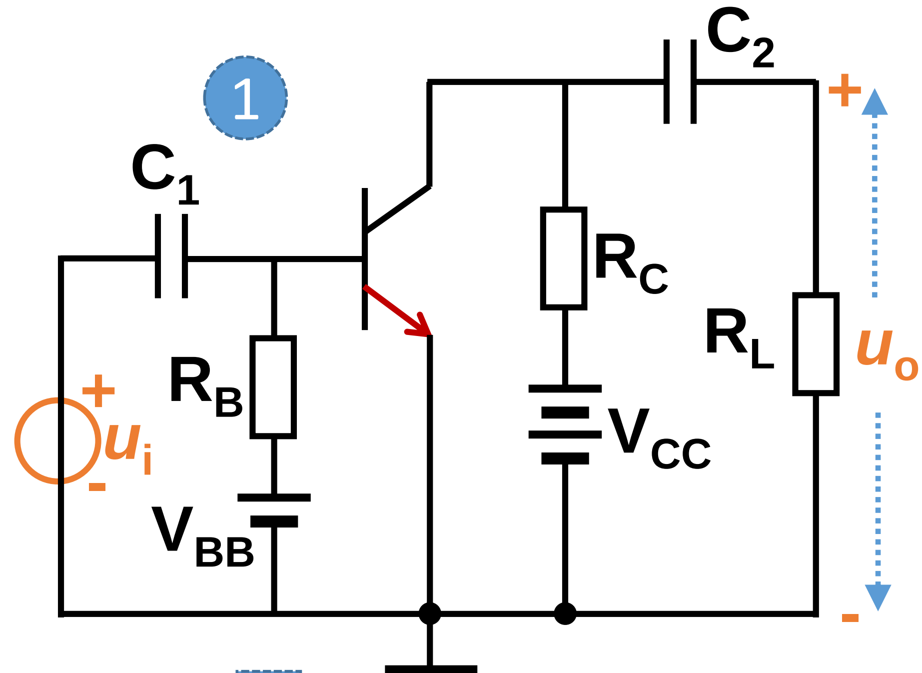
Simplify the circuit:

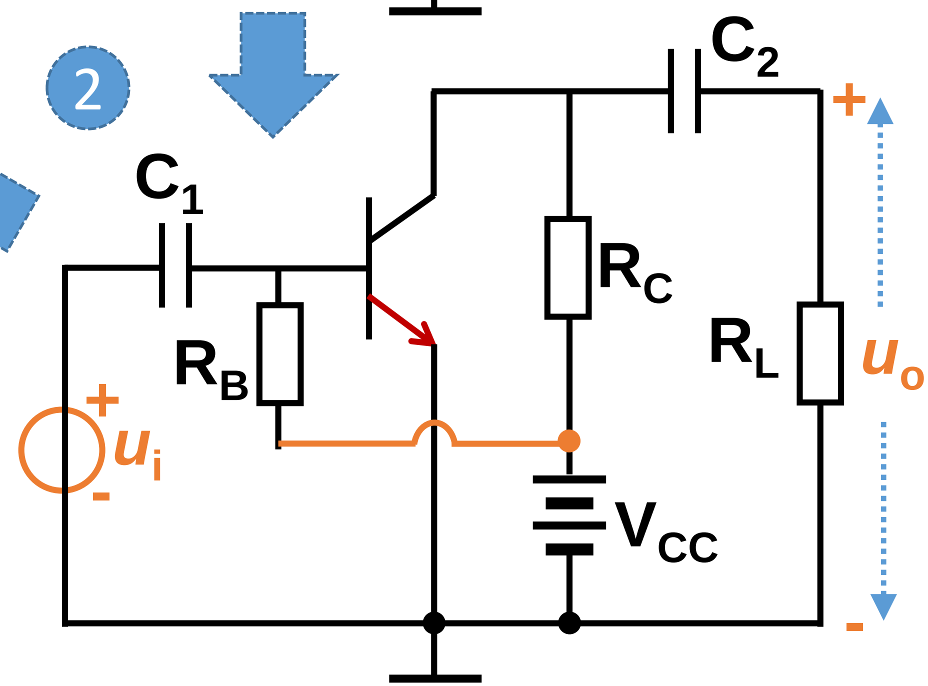


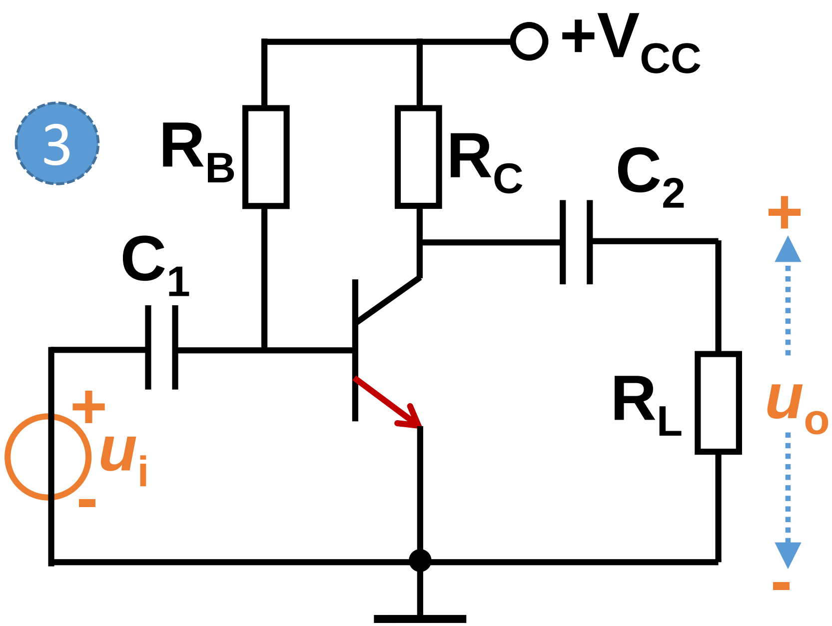
Three methods to draw a DC voltage source



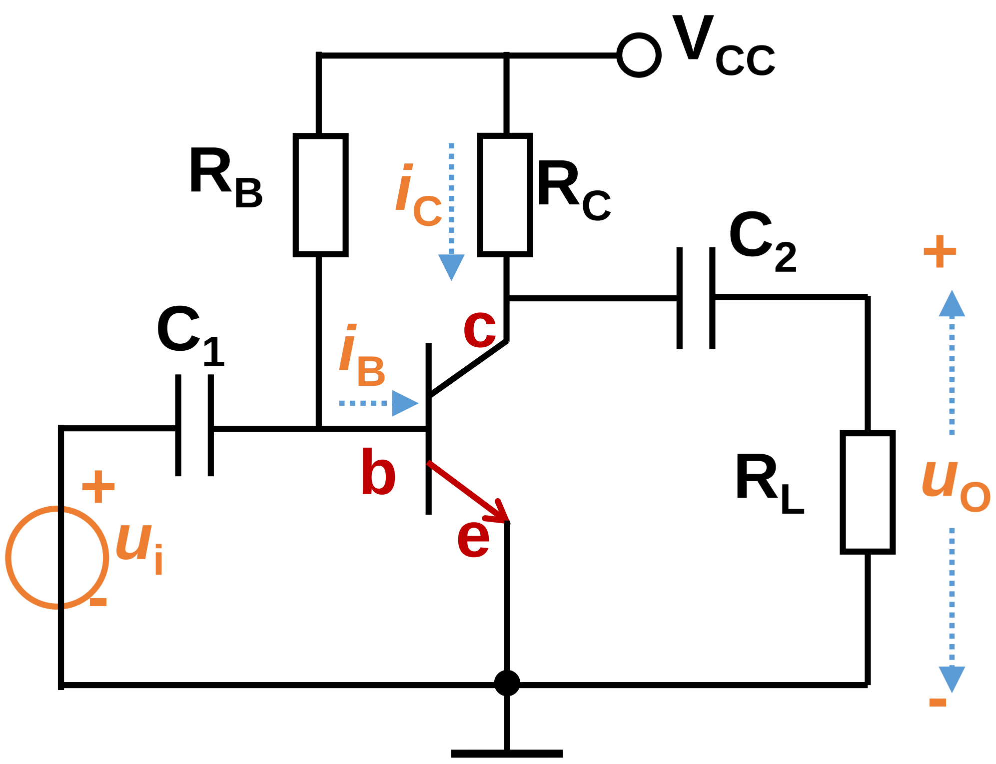
Simplify the circuit







Signal flow in the circuit



Total DC AC

$$u_{BE} = U_{BEQ} + u_{be}$$

$$i_B = I_{BQ} + i_b$$

$$i_C = I_{CQ} + i_c$$

$$u_{CE} = U_{CEQ} + u_{ce}$$

$$u_O = u_{ce}$$

Capacitor C_2

AC voltage/current：dynamic working parameter

交流电压/电流：动态工作参数

$i_{\mathrm{b}}, i_{\mathrm{c}}, u_{\mathrm{be}}, u_{\mathrm{ce}}$

Signal 信号.

DC voltage/current：quiescent working parameter

直流电压/电流：静态工作参数

$I_{\mathrm{BQ}}, I_{\mathrm{CQ}}, U_{\mathrm{BEQ}}, U_{\mathrm{CEQ}}$

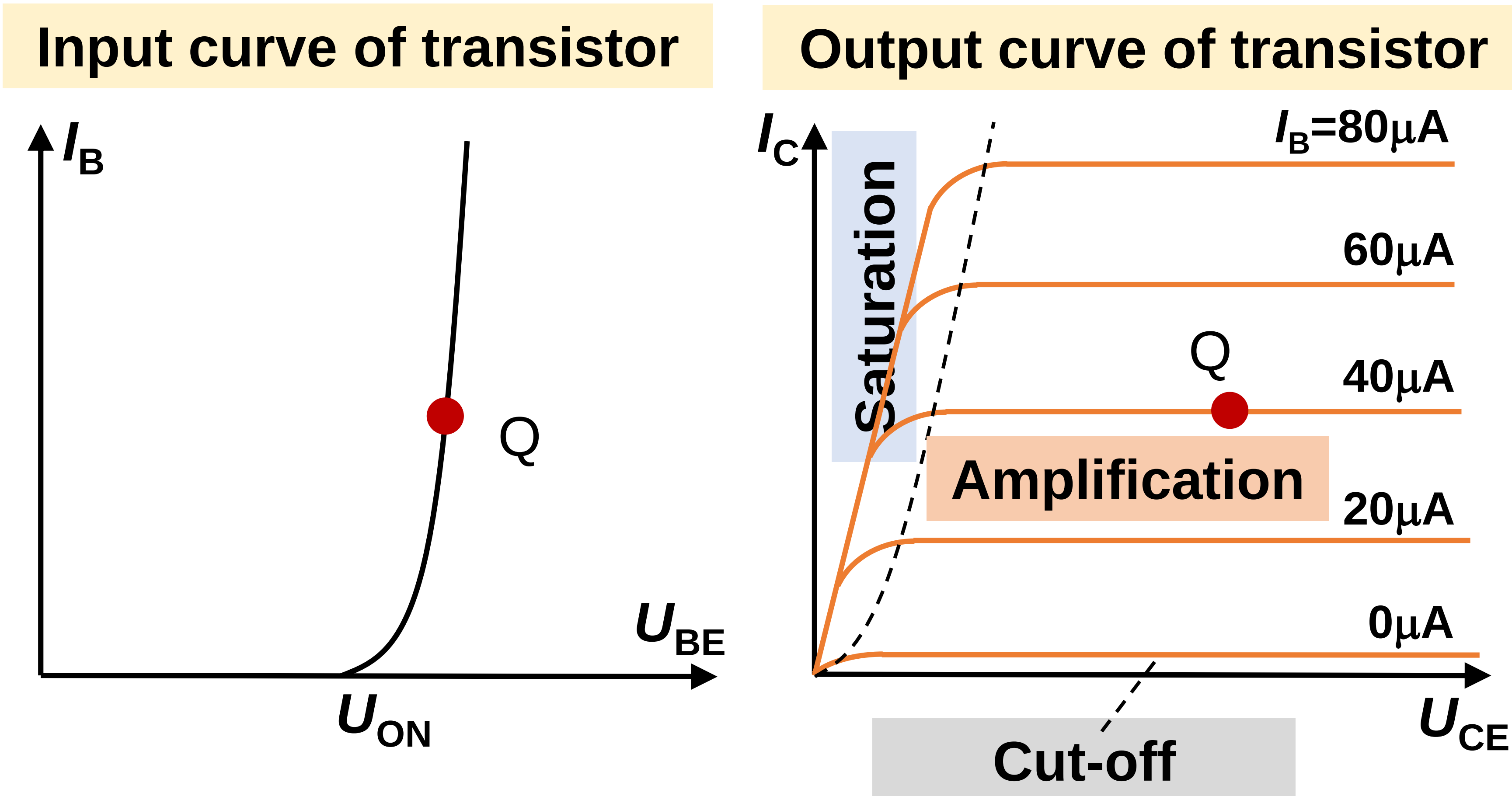
Make transistor work in amplification region.

To amplify AC signal, transistor much work in amplification region.

To work in amplification region, transistor must have proper quiescent working parameters: $I_{\mathrm{BQ}}, I_{\mathrm{CQ}}, U_{\mathrm{BEQ}}, U_{\mathrm{CEQ}}$**.**

Q is short for Quiescent

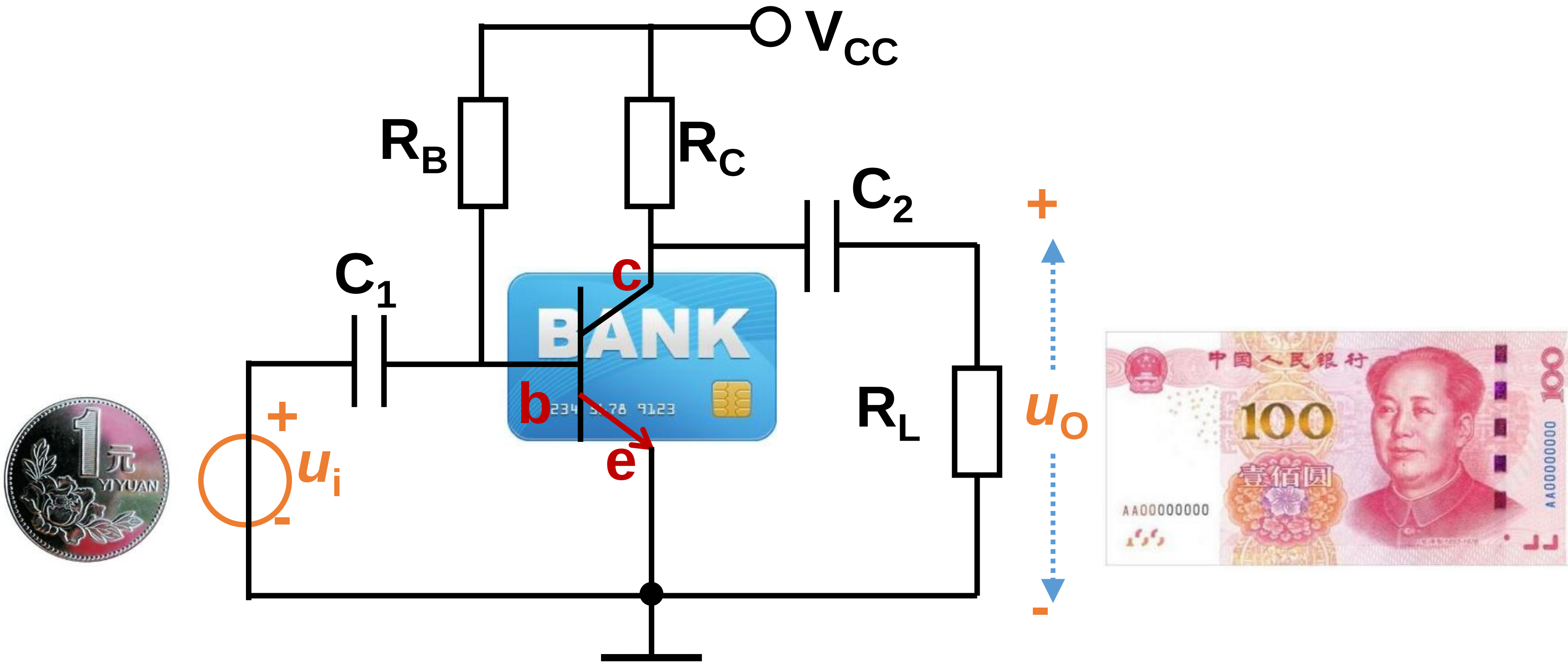
It is the same to write as: $I_{\mathrm{B}}, I_{\mathrm{C}}, U_{\mathrm{BE}}, U_{\mathrm{CE}}$



Choose V_{CC}, R_B and R_C, set a proper U_{BE}, U_{CE}, I_C and I_B, to make transistor work in the amplification region ($U_{BE} > U_{ON}$, $U_{CE} > U_{BE}$).


高利贷

Usury 高利贷

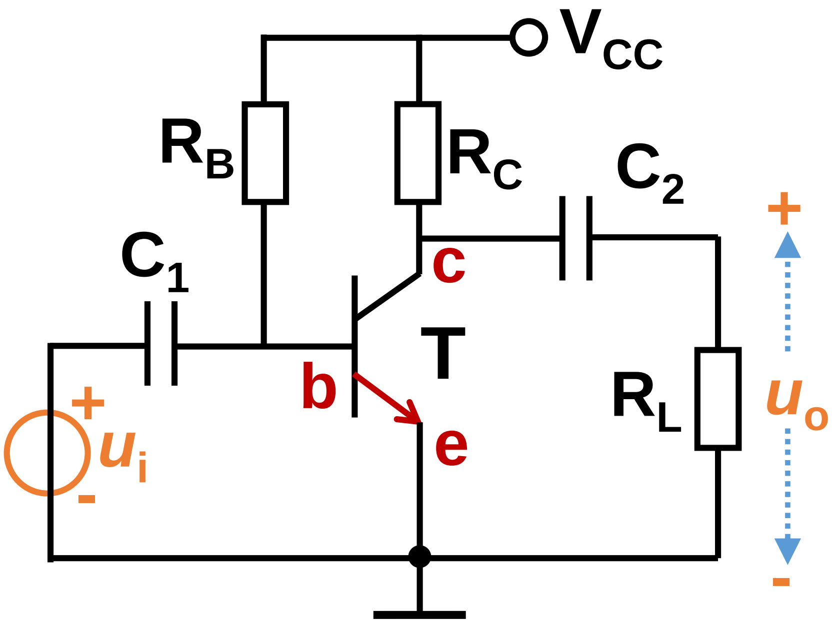

VCC
RB
RC
C2
C1
c
b
e
BANK
RL
+
uO
-
+
ui
-
1元
YI YUAN
100
壹佰圆
中国人民银行
AA00000000

Q: How to get the quiescent and dynamic working parameters?

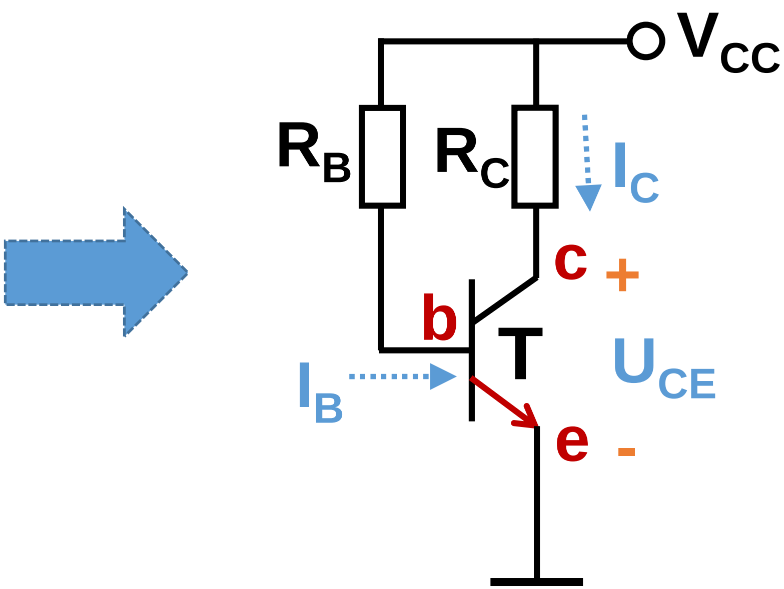
Separate DC and AC signals!

2.2.4 Analyze DC circuit of amplifiers

To get the quiescent working parameters $I_{\mathrm{BQ}}, I_{\mathrm{CQ}}, U_{\mathrm{BEQ}}, U_{\mathrm{CEQ}}$ at Q point.



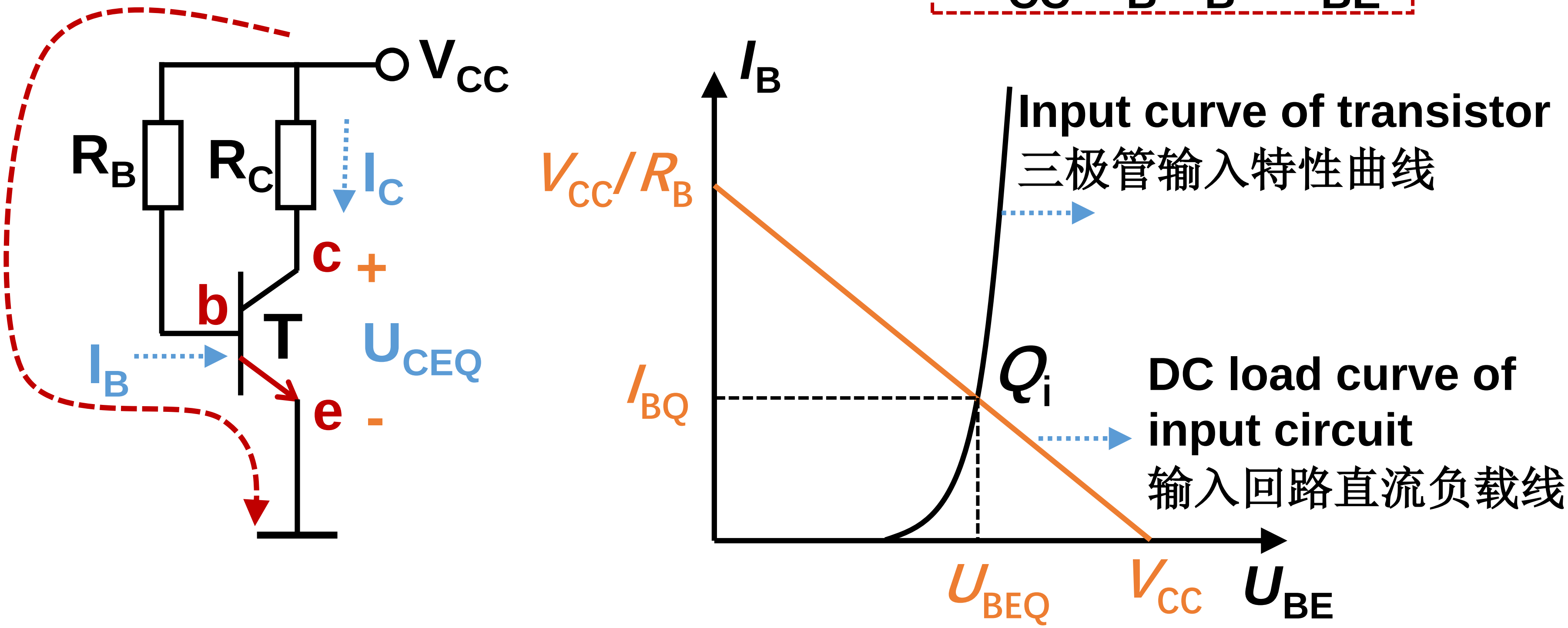
DC circuit



Graphical method 图解法

1) Input circuit 输入回路   Input circuit equation

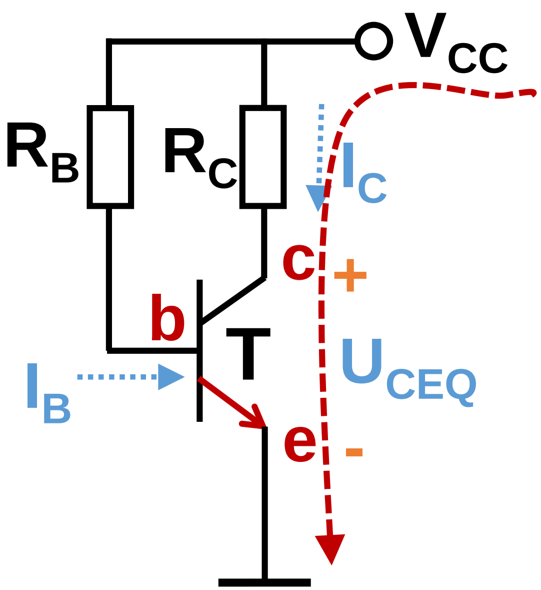
$$\boxed{V_{CC}=I_B R_B+U_{BE}}$$

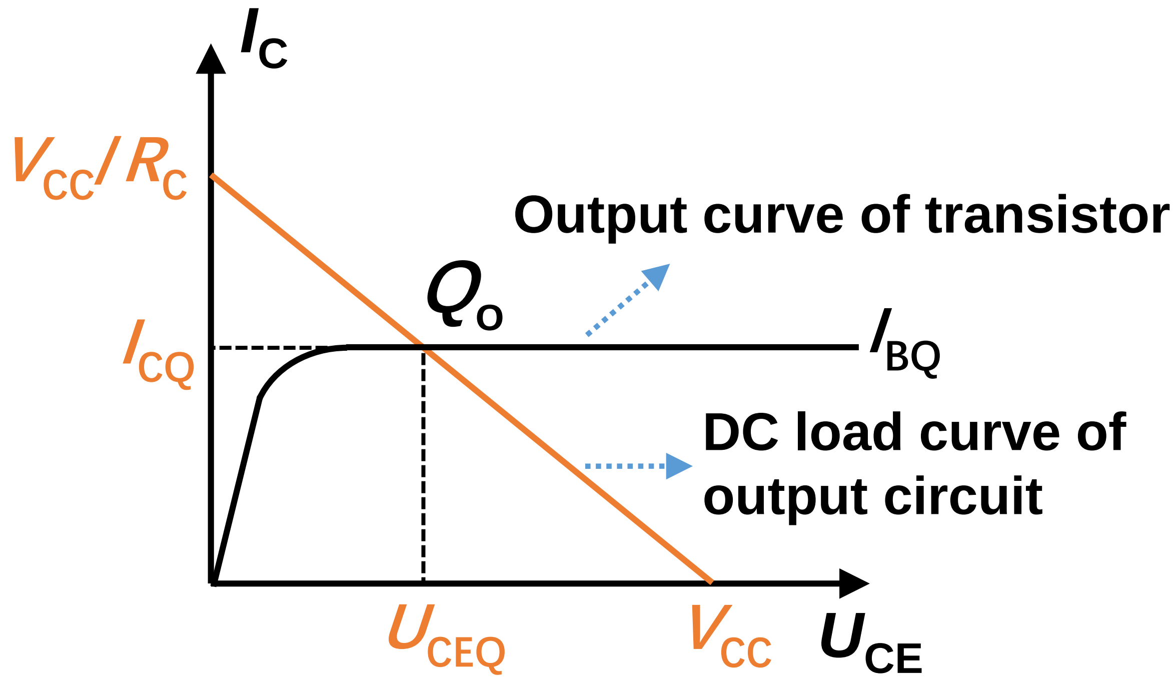


The intersection point is called the input quiescent/static working point Q_i. 交点叫做输入回路静态工作点。

2) Output circuit 输出回路

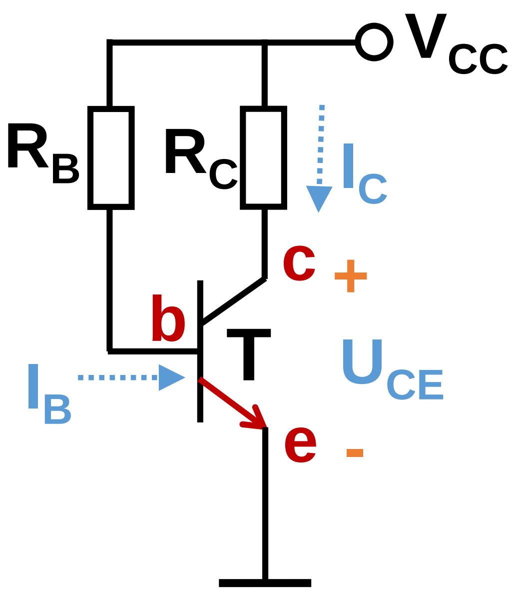
Output circuit equation: $V_{CC}=I_C R_C+U_{CE}$





The intersection point is called the output quiescent working point Q_o.

Parametric estimating method 估算法（重要）



1. Input circuit: $V_{CC}=I_{BQ}R_B+U_{BEQ}$

$$I_{BQ}=\frac{V_{CC}-U_{BEQ}}{R_B}$$

U_{BEQ}=0.7 V for Si, 0.3V for Ge

2.

$$I_{CQ}=\bar{\beta}I_{BQ}$$

3. Output circuit

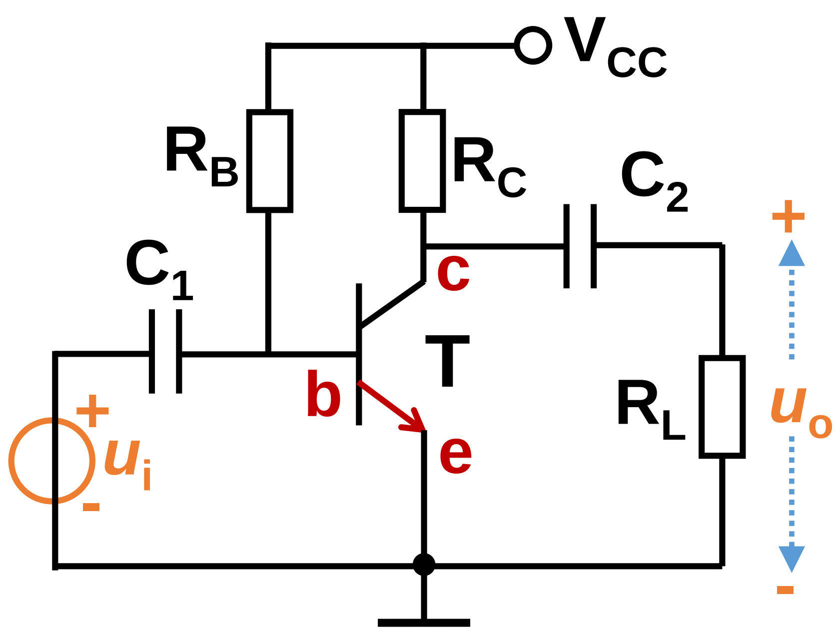
$$U_{CEQ}=V_{CC}-I_{CQ}R_C$$

The estimating method is very important and convenient.

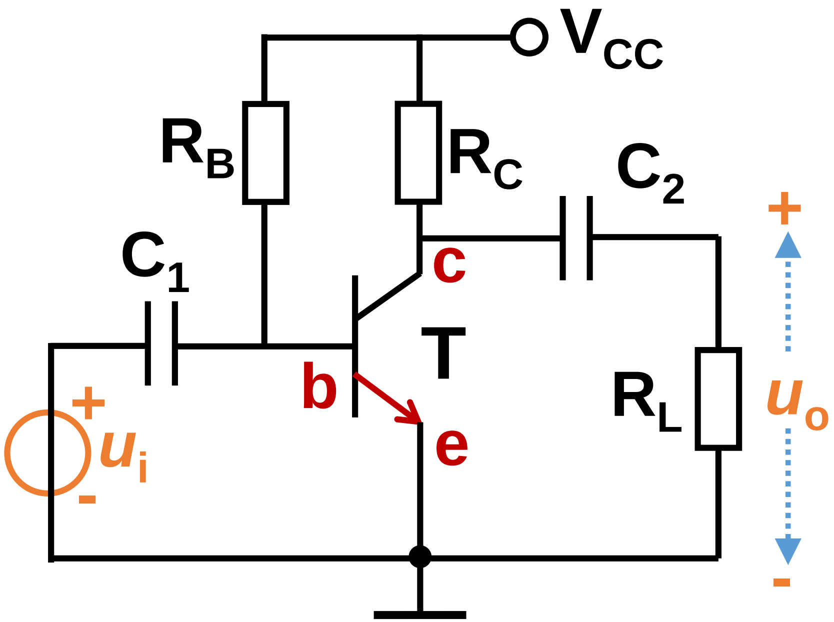
2.2.5 Dynamic analysis of amplifier circuits

Dynamic analysis: analyze the AC signal based on the DC circuit.

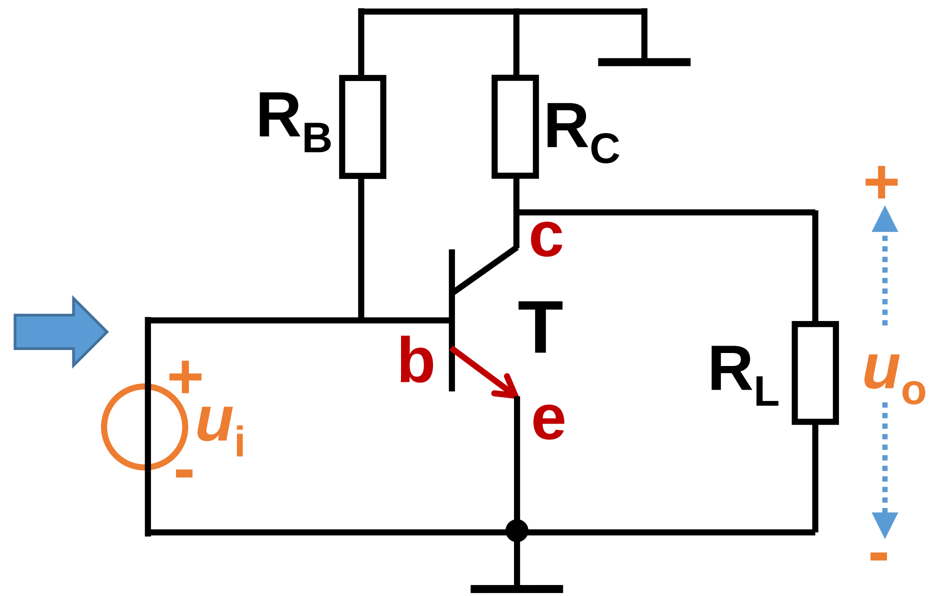
动态分析：在静态电路的基础上分析各交流信号的关系



How to obtain the AC circuit

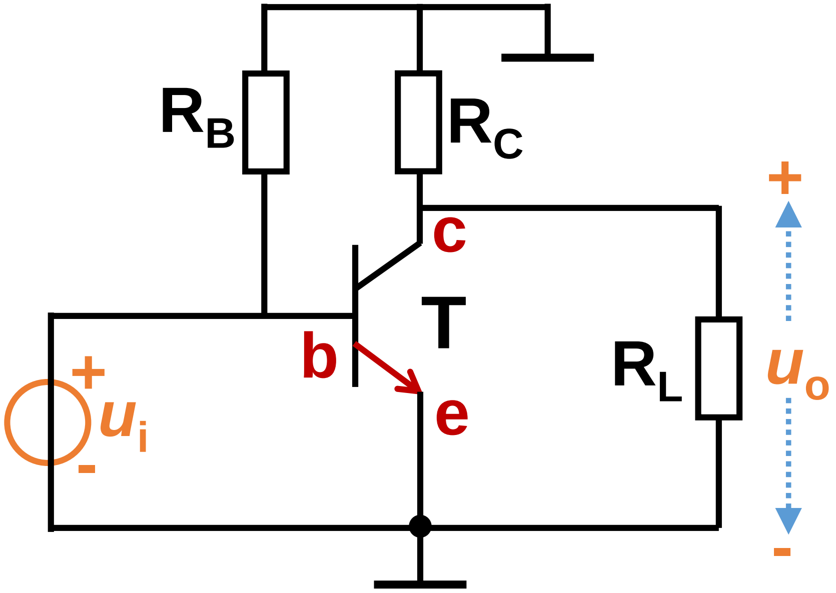


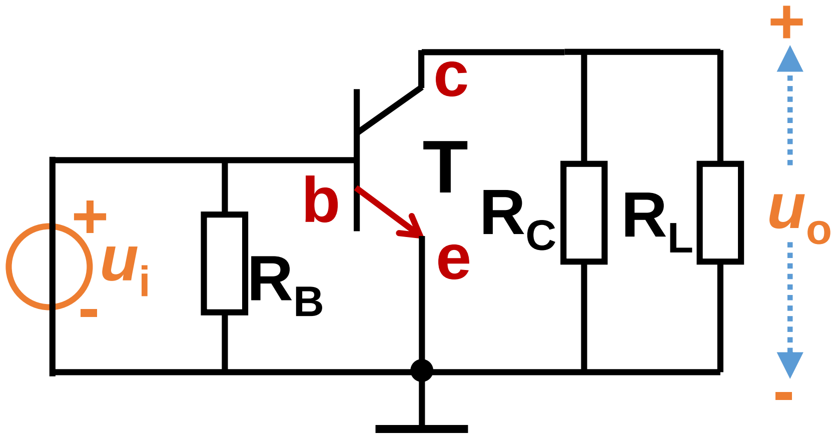
AC circuit 交流通路



1. Short the DC voltage source (Connect DC voltage source to ground) 直流电压源接地.
2. Short the capacitor 电容短路.

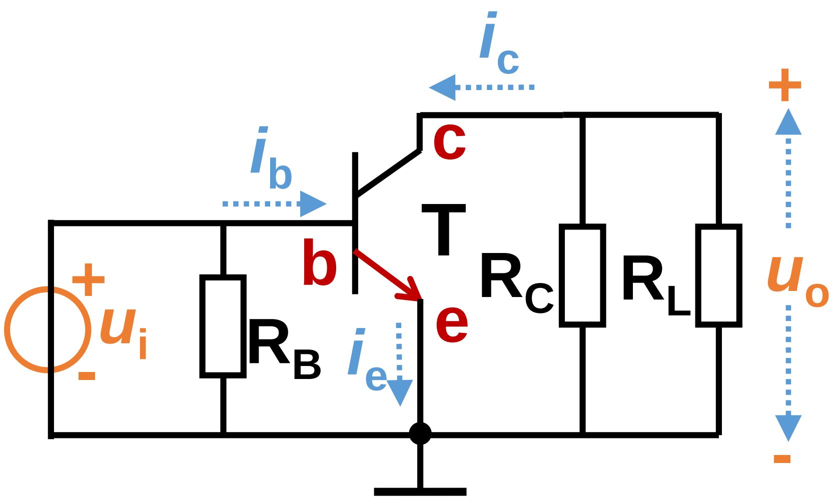
AC circuit 交流通路





Graphical analysis for AC circuit
图解法

Input and output circuit equation of AC circuit
交流通路的输入与输出方程



Input circuit equation

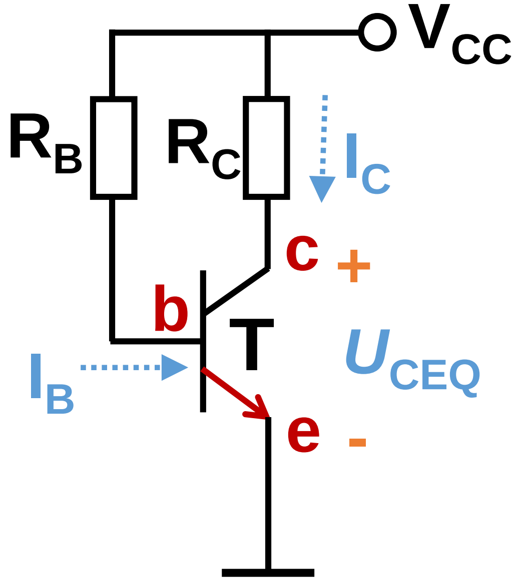
$$u_{be} = u_i$$

Output circuit equation

$$u_o = u_{ce} = -i_c(R_c \| R_L)$$

Input circuit equation: DC+AC

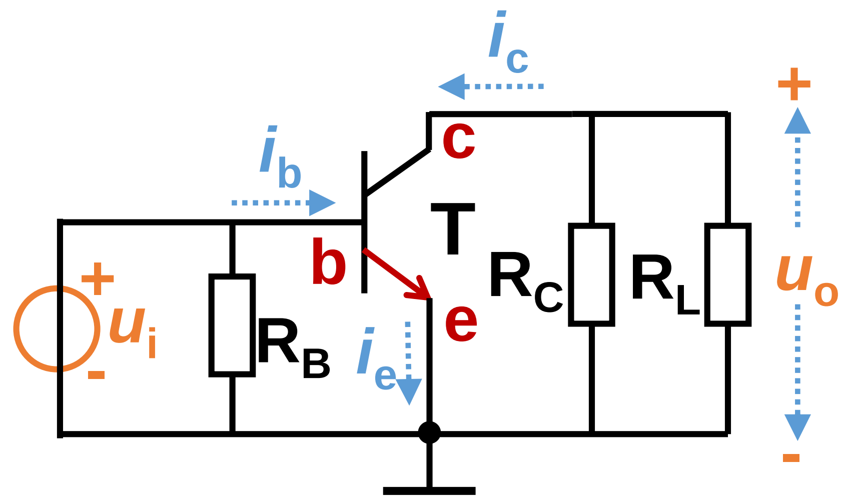
DC circuit



Input circuit equation:

$$V_{CC}=I_B R_B+U_{BE}$$

AC circuit



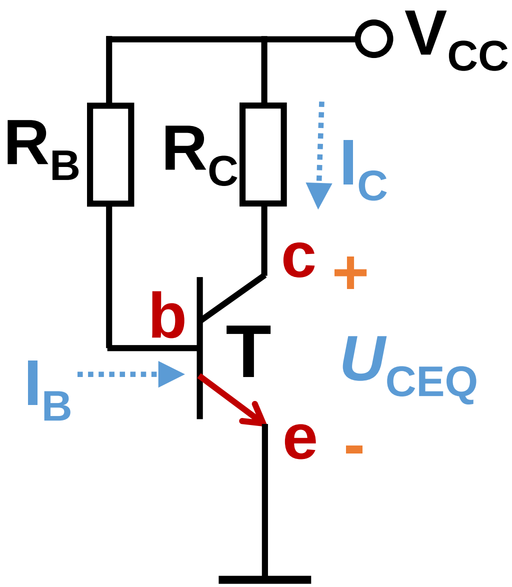
Input circuit equation:

$$u_{be}=u_i$$

Total BE voltage: $u_{BE}=U_{BE}+u_{be}$ $\boldsymbol{u_{BE}=U_{BEQ}+u_i}$

Output circuit equation: DC+AC

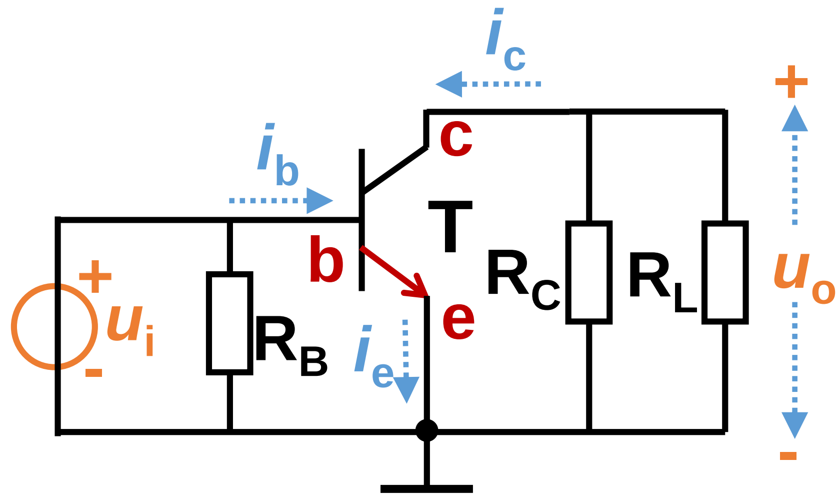
DC circuit



Output circuit:

$$U_{CE}=V_{CC}-I_C R_C$$

AC circuit



Output circuit:

$$u_o=u_{ce}=-i_c(R_c \| R_L)$$

Output circuit equation: DC+AC circuits

$$u_{\mathrm{CE}} = U_{\mathrm{CE}} + u_{\mathrm{ce}}$$

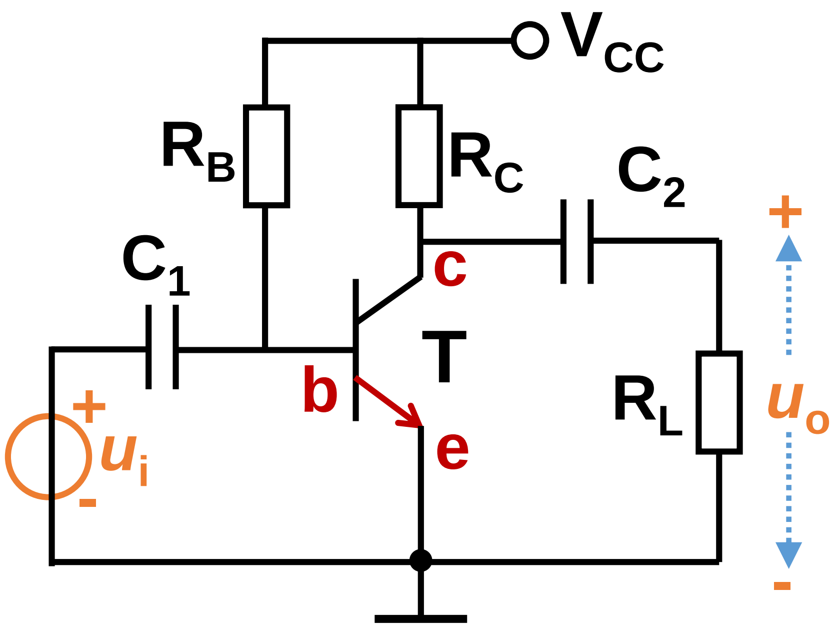
$$u_{\mathrm{CE}} = V_{\mathrm{CC}} - I_{\mathrm{C}}R_{\mathrm{C}} - i_{\mathrm{c}}(R_{\mathrm{c}} // R_{\mathrm{L}})$$

Case 1: ***R*$_{\mathrm{L}}$=∞, without load**

$$u_{\mathrm{CE}} = V_{\mathrm{CC}} - I_{\mathrm{C}}R_{\mathrm{C}} - i_{\mathrm{c}}R_{\mathrm{c}} = V_{\mathrm{CC}} - (I_{\mathrm{CQ}} + i_{\mathrm{c}})R_{\mathrm{C}}$$

$$u_{\mathrm{CE}} = V_{\mathrm{CC}} - i_{\mathrm{C}}R_{\mathrm{C}}$$

Case 1: $R_L=\infty$



Input circuit equation: DC+AC

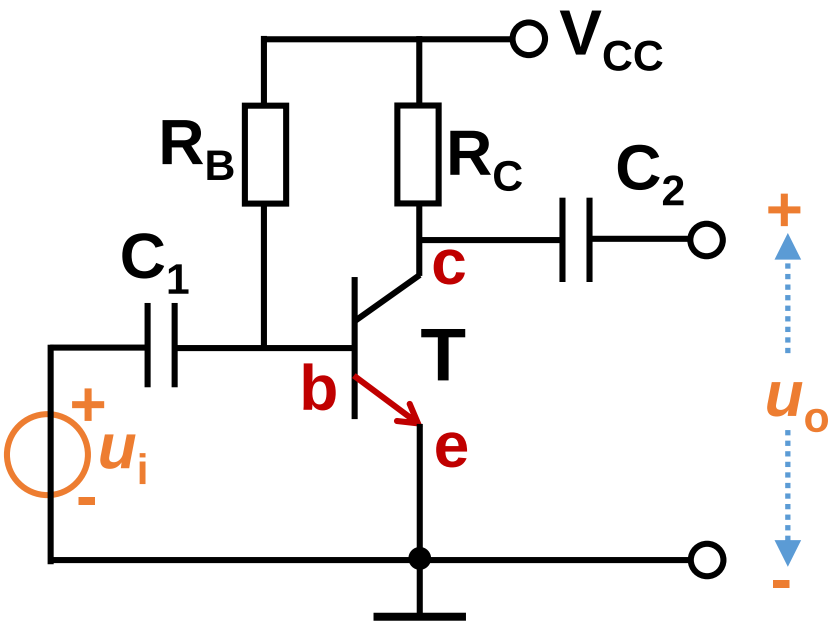
$$u_{BE}=U_{BEQ}+u_i$$

Output circuit equation: DC+AC

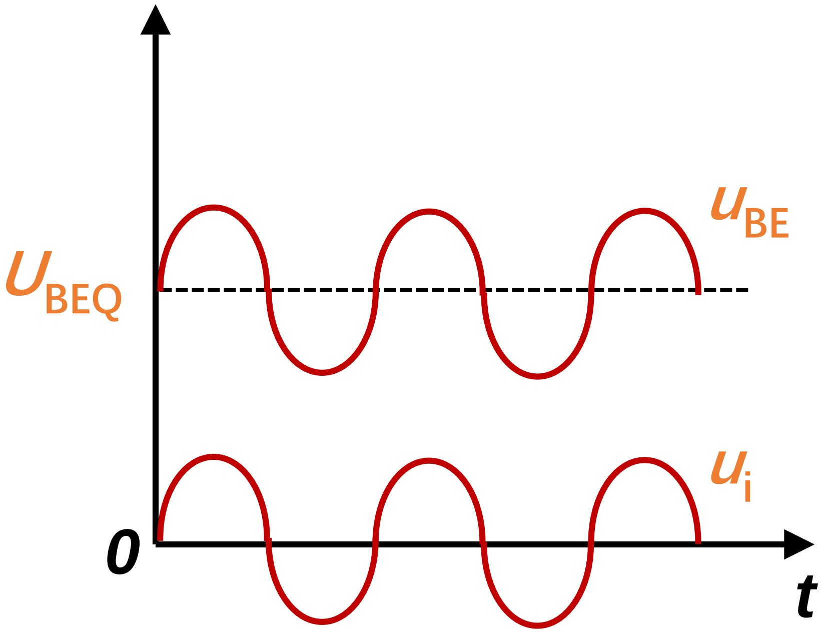
$$u_{CE}=V_{CC}-i_C R_C$$

Case 1: $R_L=\infty$

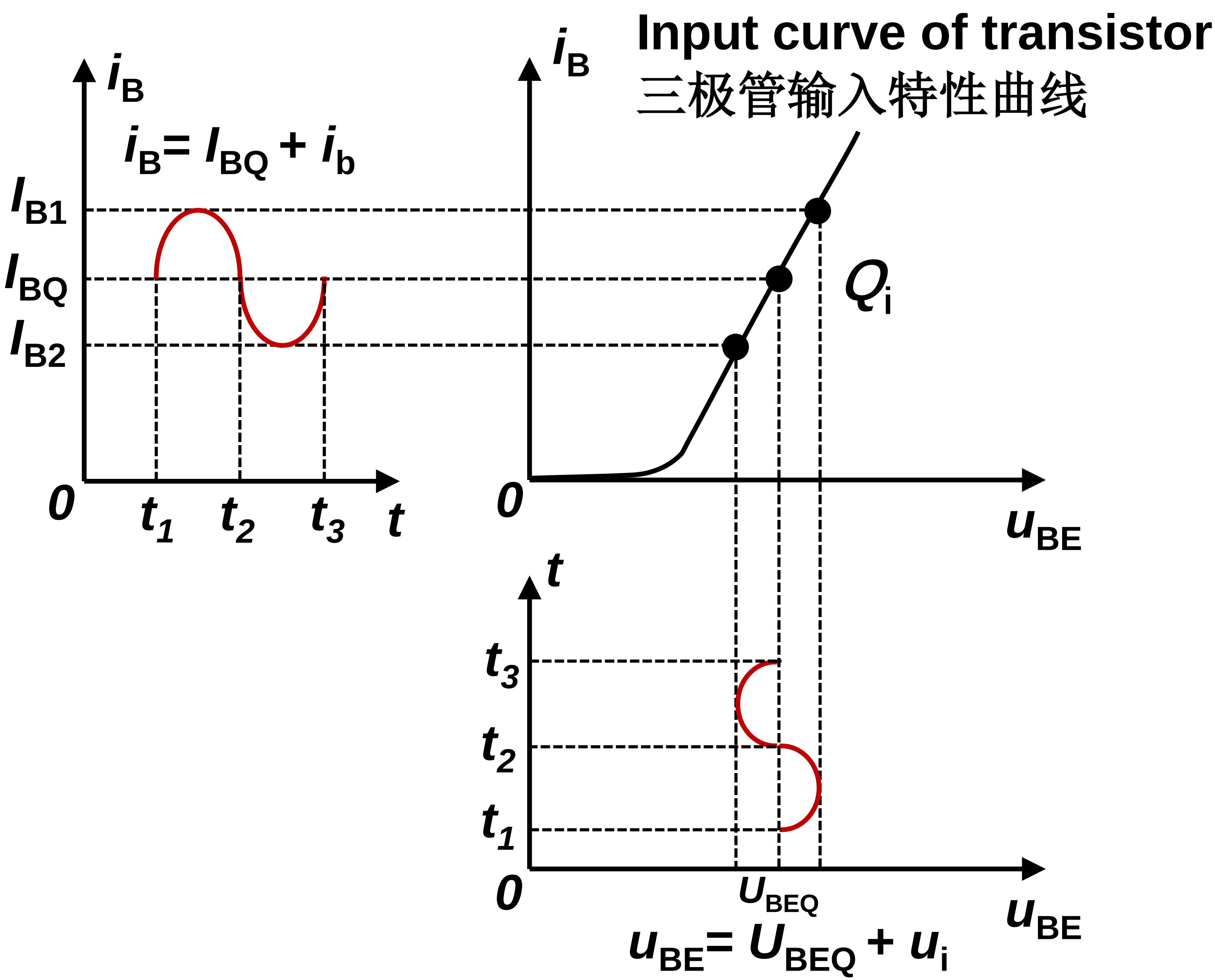
1) Input circuit



$$u_{BE}= U_{BEQ} + u_i$$

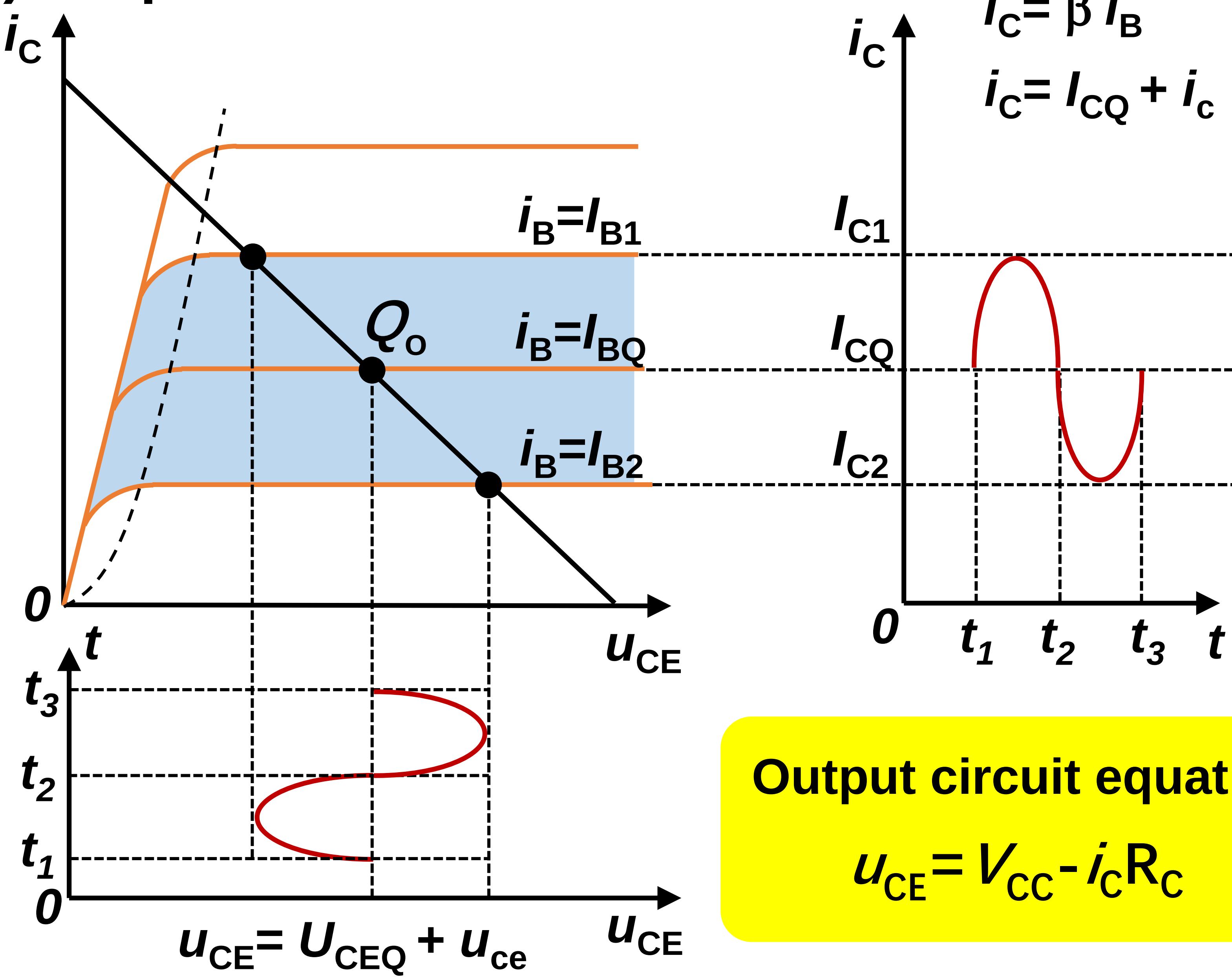


$$u_i = A\sin(\omega t)$$



Input curve of transistor
三极管输入特性曲线
i_B
$i_B = I_{BQ} + i_b$
I_{B1}
I_{BQ}
I_{B2}
0
t_1
t_2
t_3
t
i_B
Q_i
0
u_{BE}
t
t_3
t_2
t_1
0
U_{BEQ}
u_{BE}
$u_{BE} = U_{BEQ} + u_i$

2) Output circuit

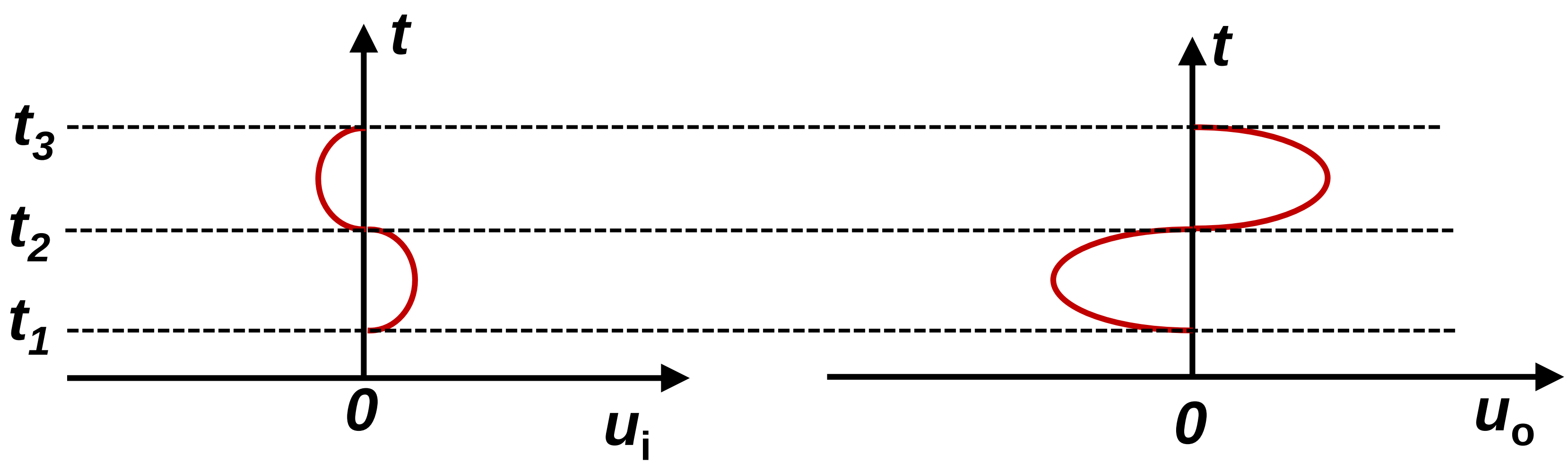


Output circuit equation:

$$u_{CE} = V_{CC} - i_C R_C$$

Input AC signal

Output AC signal



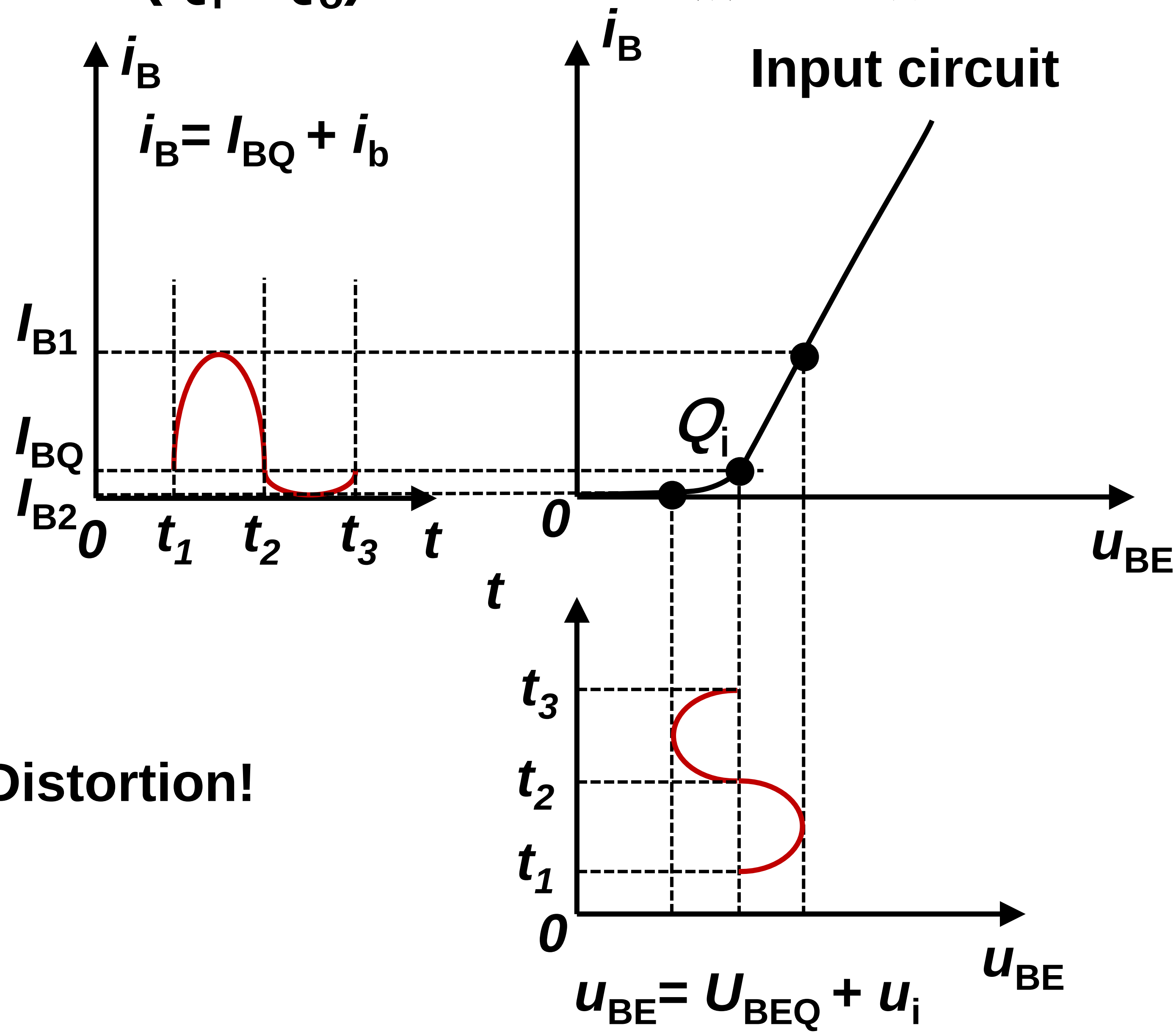
$u_i = U_{IPP} \sin(\omega t)/2$

$u_o = -U_{OPP} \sin(\omega t)/2$

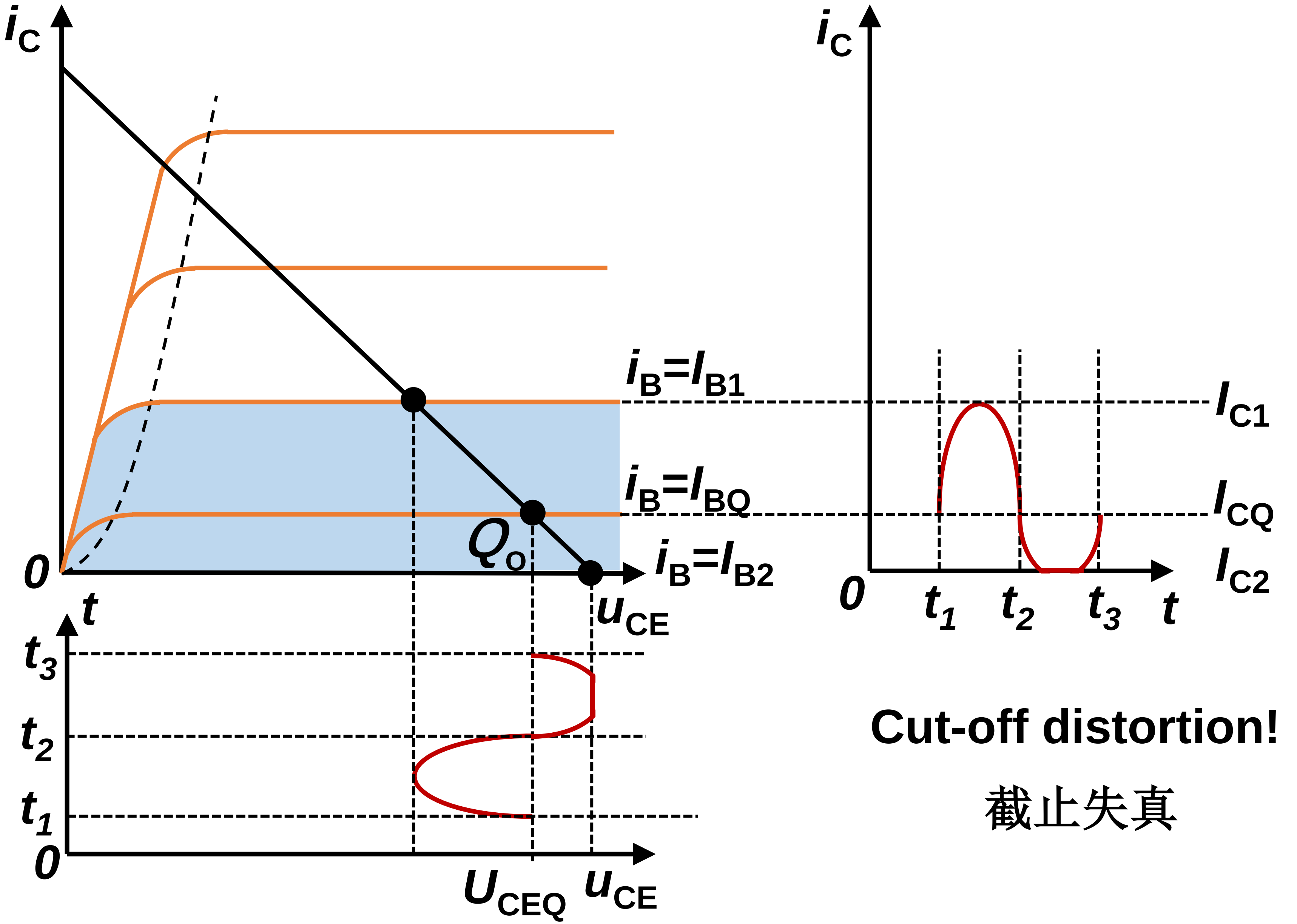
There is a 180° phase shift between input and output signal

Voltage gain 电压放大倍数：$\dot{A}_u = -\dfrac{U_{OPP}}{U_{IPP}}$

3) Q point (Q_i&Q_o) is too low 静态工作点太低



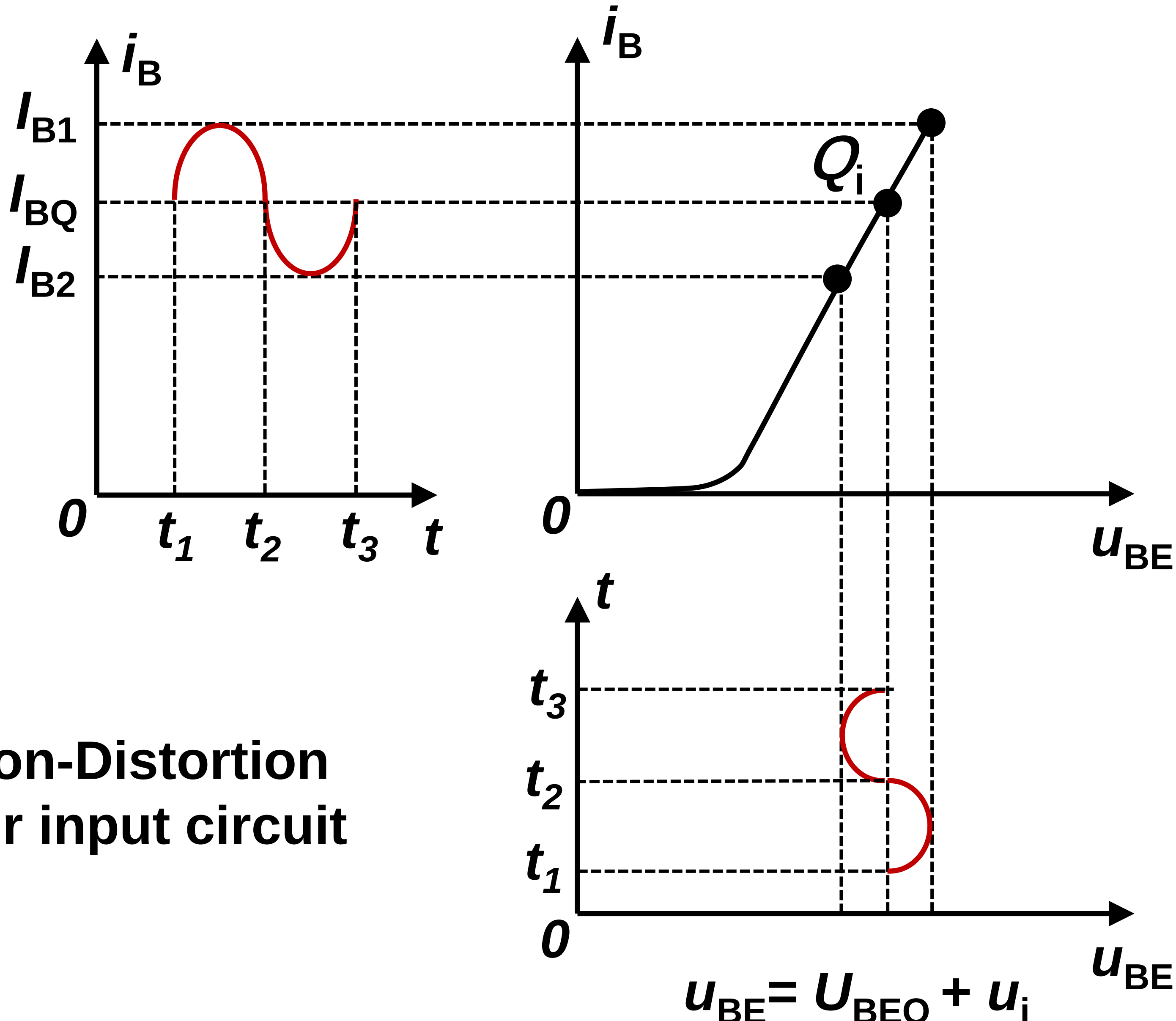
Distortion!



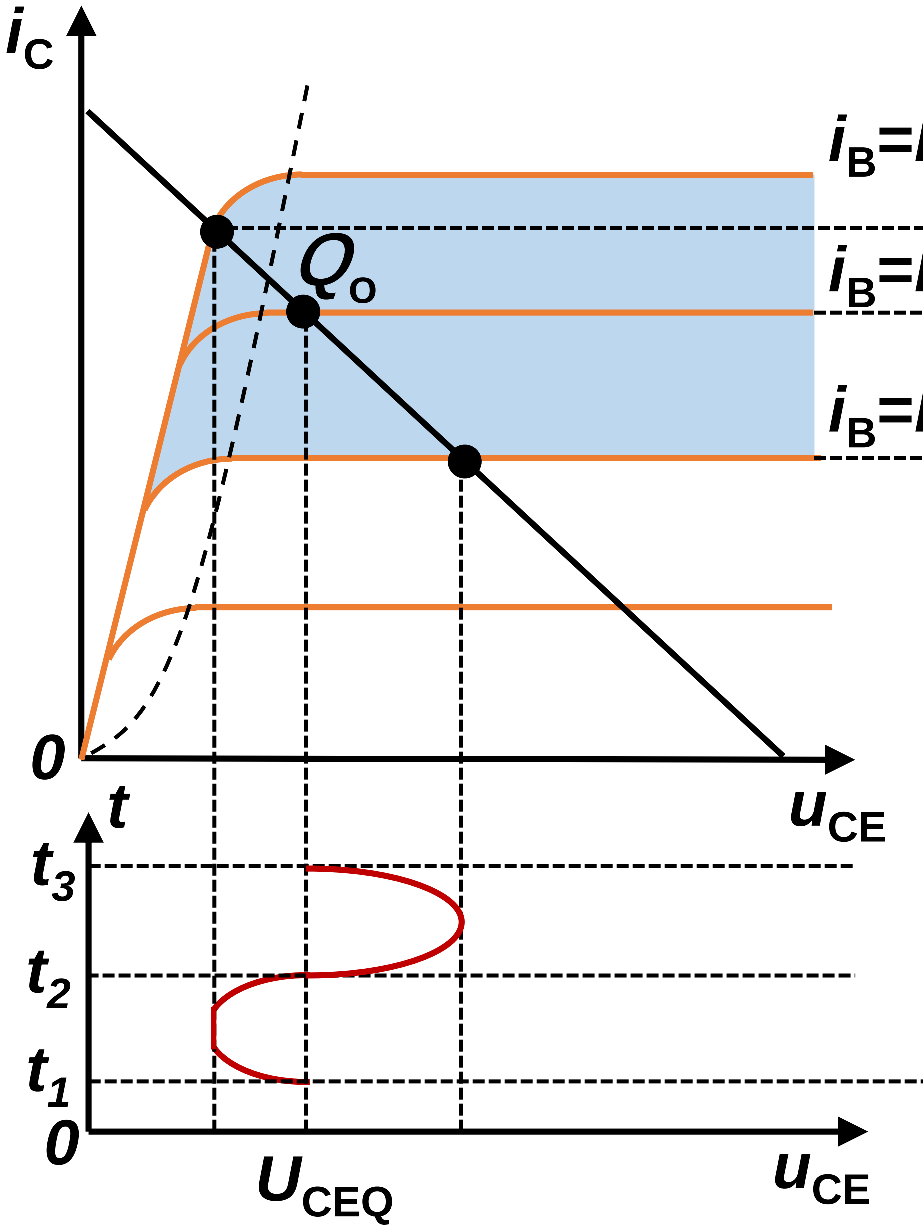
Cut-off distortion!

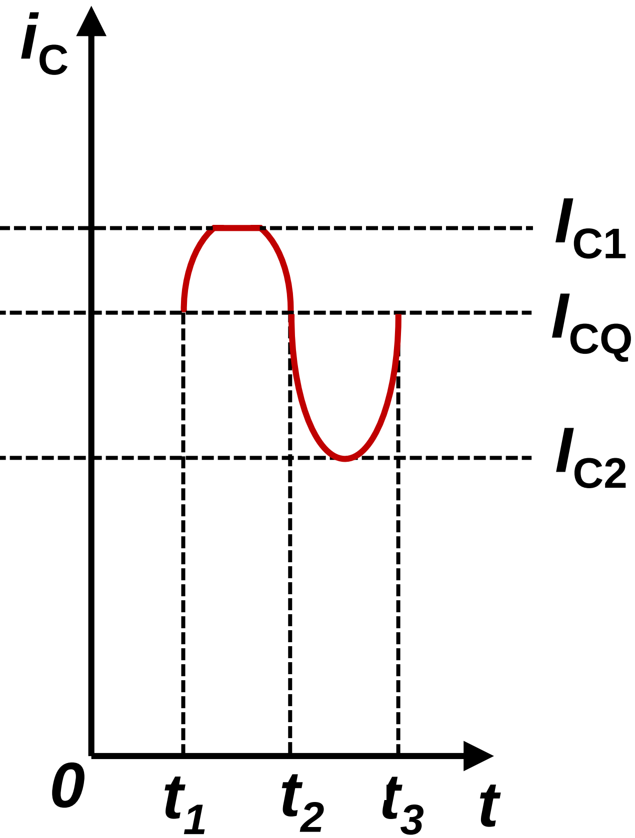
截止失真

4) Q point (Q_i&Q_o) is too high 静态工作点太高



Non-Distortion for input circuit

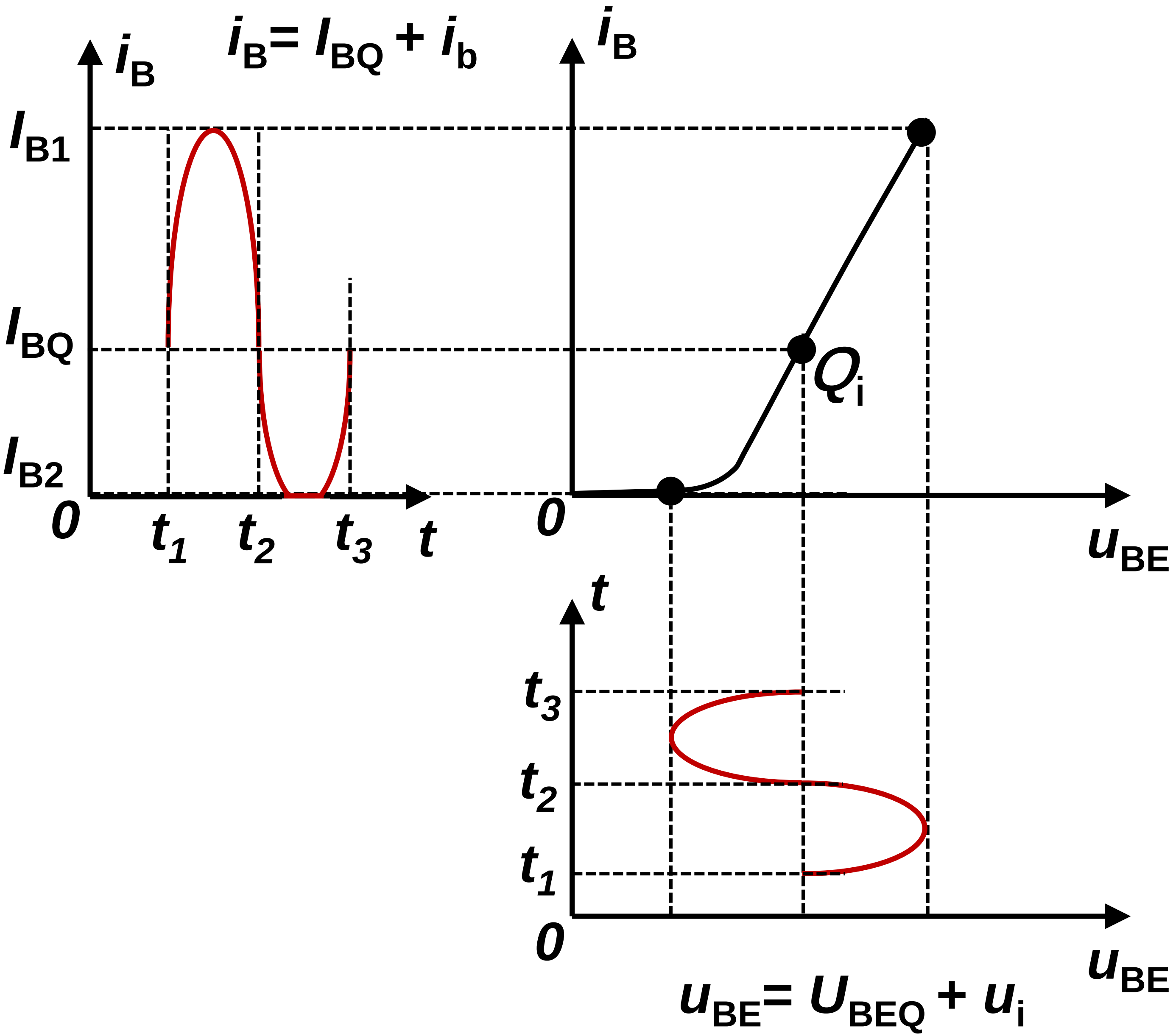


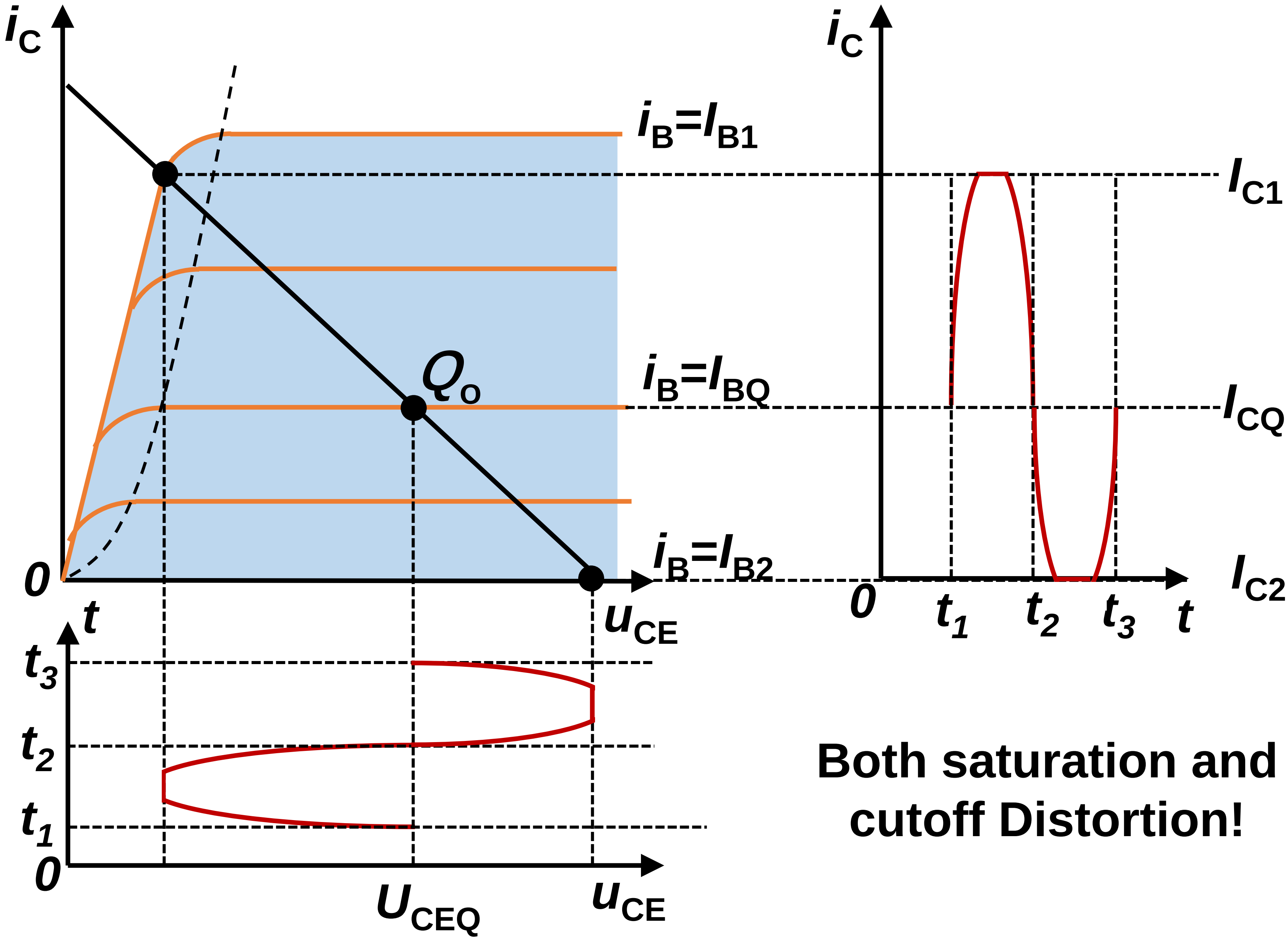


Saturation distortion for output circuit!

饱和失真

5) Amplitude of input signal is too large

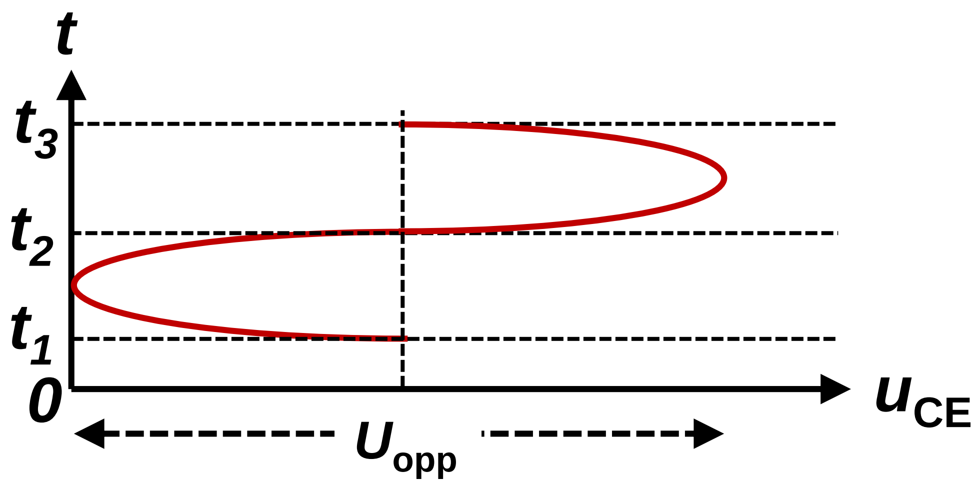




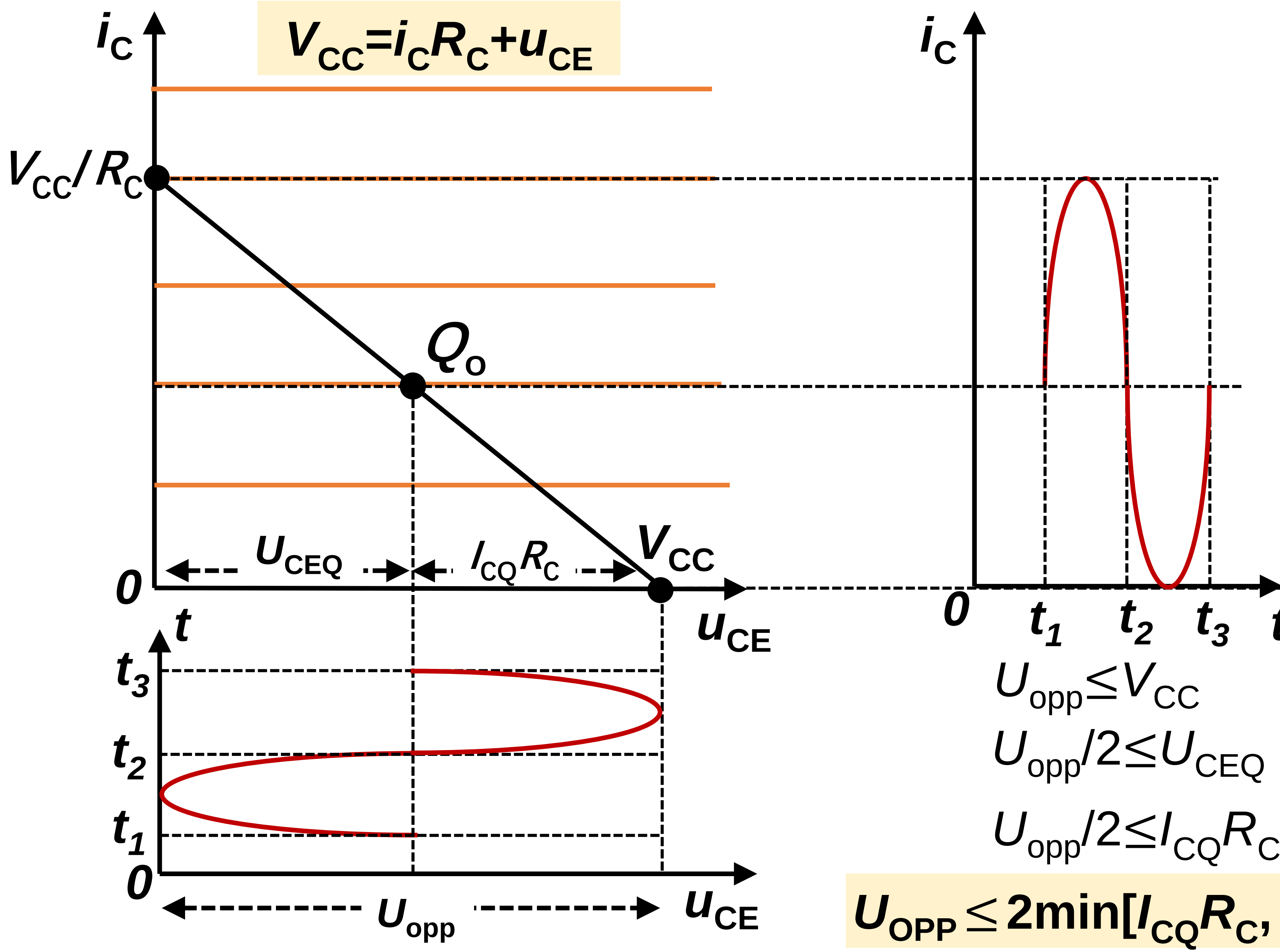
Both saturation and cutoff Distortion!

6) Dynamic range 动态范围

The maximum non-distorted output voltage (peak-to-peak value U_{opp})



Dynamic range (ignore the saturation region)



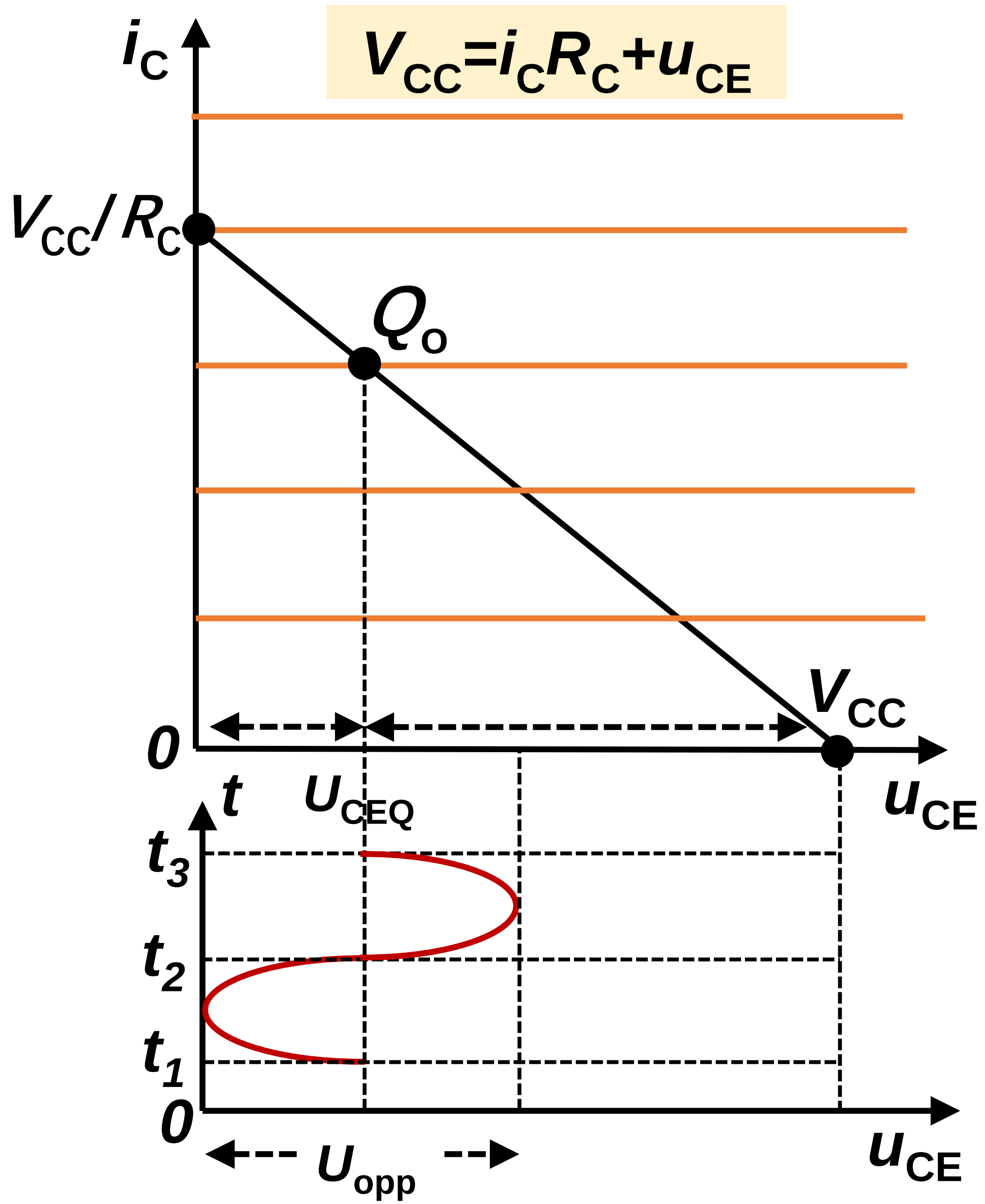
$U_{opp} \leq V_{CC}$

$U_{opp}/2 \leq U_{CEQ}$

$U_{opp}/2 \leq I_{CQ}R_C$

$$U_{OPP} \leq 2\min[I_{CQ}R_C, U_{CEQ}]$$

Dynamic range (ignore the saturation region)



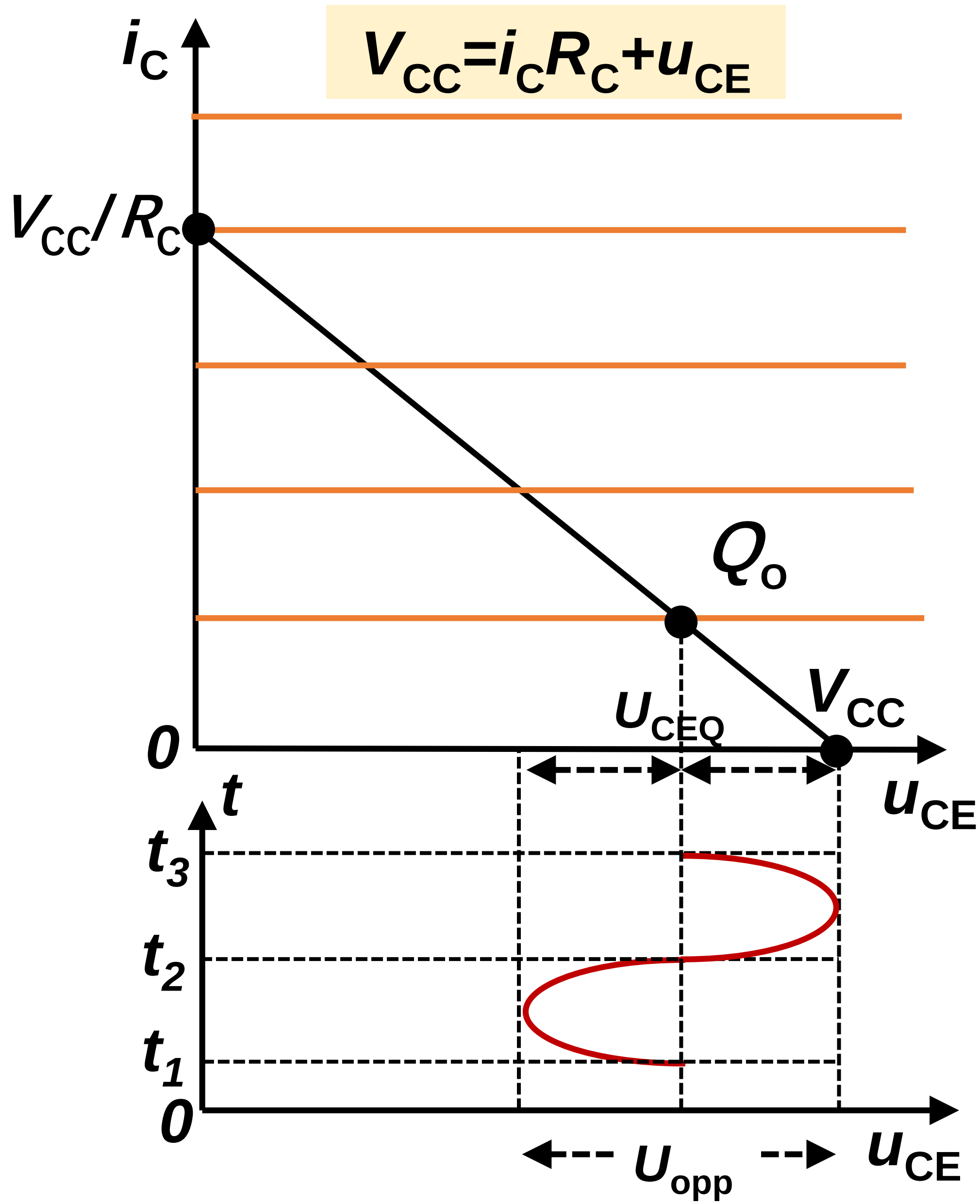
When $U_{CEQ} < V_{CC}/2$

$$U_{opp}/2 \le U_{CEQ}$$

$$U_{opp}/2 \le V_{CC}-U_{CEQ}$$

$$U_{OPP} \le 2U_{CEQ}$$

Dynamic range (ignore the saturation region)



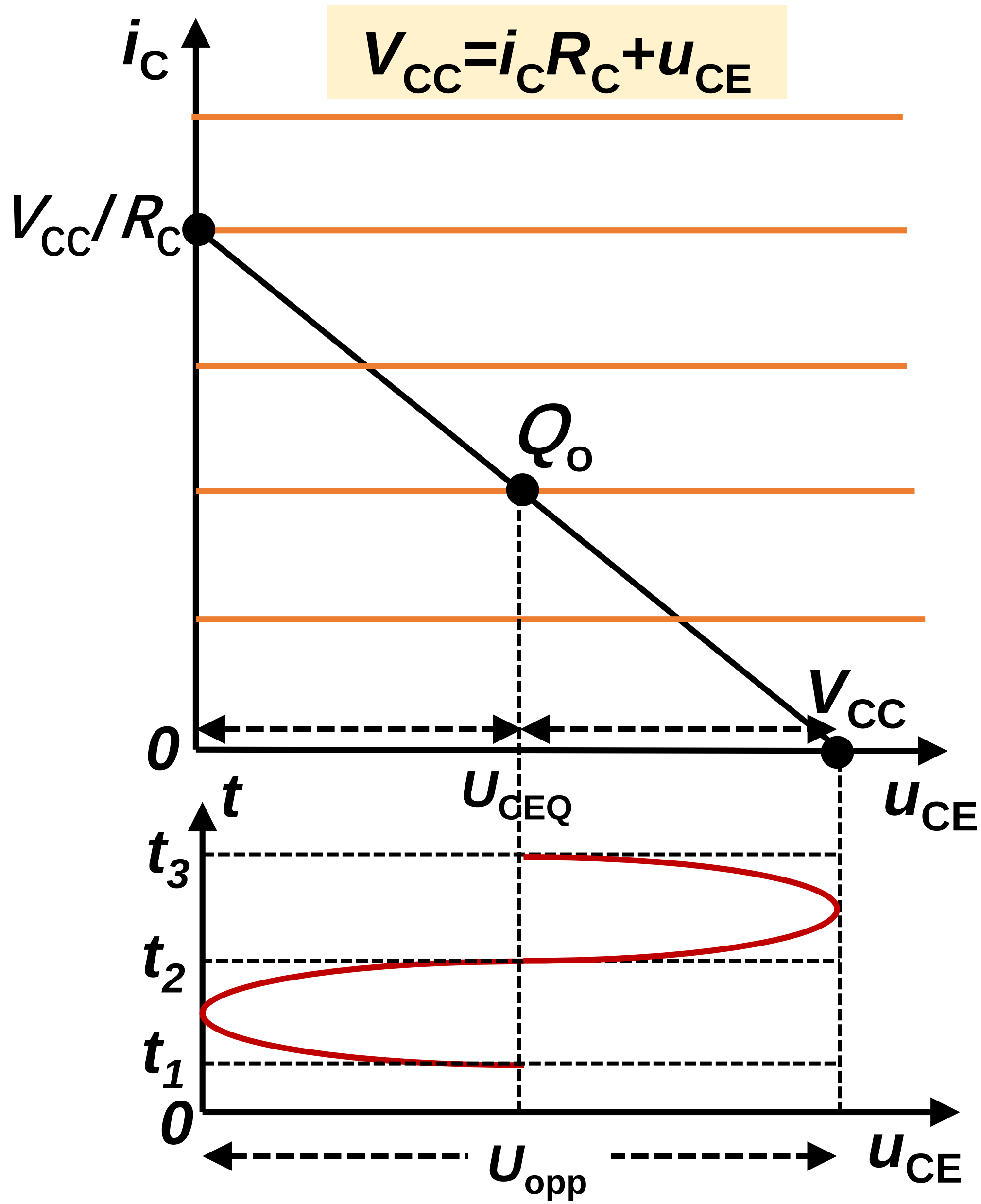
When U_{CEQ} > $V_{CC}/2$

$U_{opp}/2 \le U_{CEQ}$

$U_{opp}/2 \le V_{CC}\text{-}U_{CEQ} = I_{CQ}R_C$

$$U_{OPP} \le 2I_{CQ}R_C$$

Dynamic range (ignore the saturation region)



When $U_{CEQ} = V_{CC}/2$

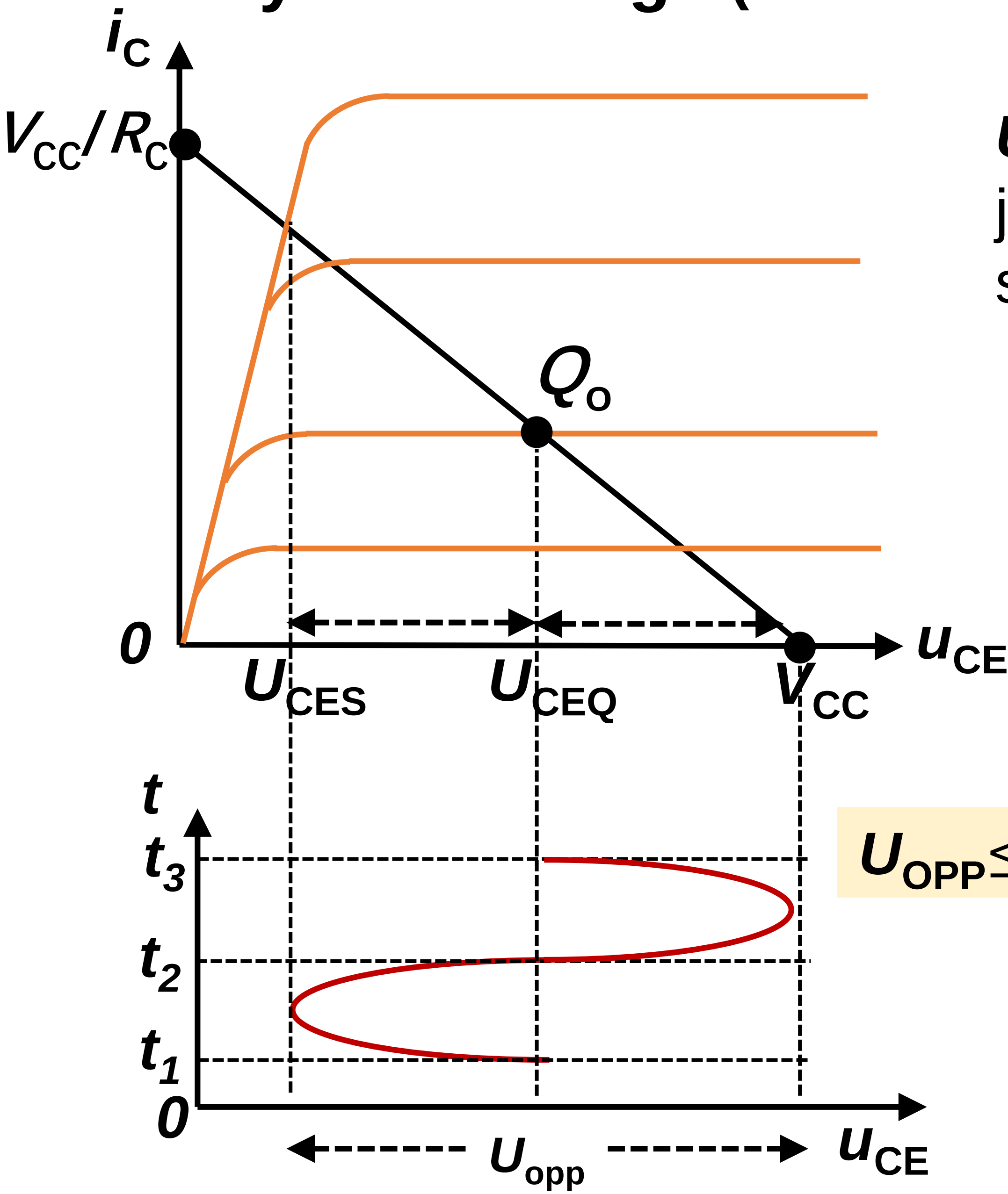
$U_{opp}/2 \leq U_{CEQ}$

$U_{opp}/2 \leq V_{CC}\text{-}U_{CEQ} = U_{CEQ}$

$U_{OPP} \leq V_{CC}$

Dynamic range reaches the maximum value V_{CC}

Dynamic range (consider the saturation region)



U_{CES}: voltage drop on CE junction when enters into saturation region

$$U_{opp} \le V_{CC} - U_{CES}$$

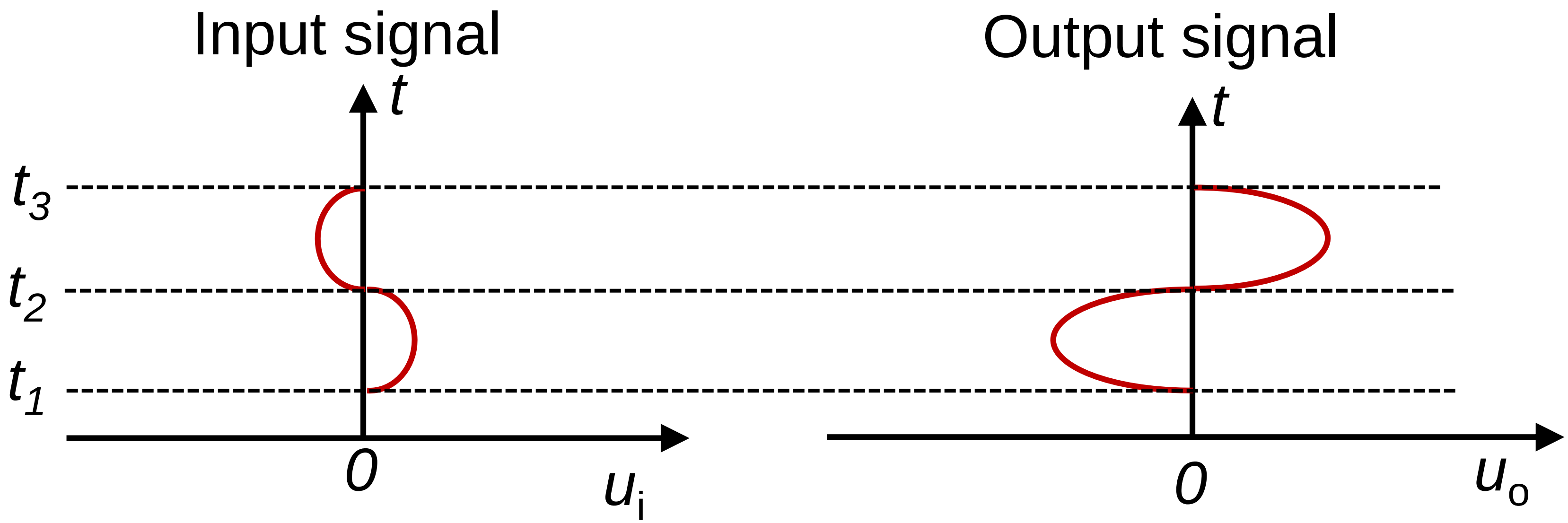
$$U_{opp}/2 \le U_{CEQ} - U_{CES}$$

$$U_{opp}/2 \le V_{CC} - U_{CEQ}$$

$$U_{OPP} \le 2\min[V_{CC} - U_{CEQ},\ U_{CEQ} - U_{CES}]$$

Conclusion: Case1: $R_L=\infty$ Common-e amplifier circuit

1. There is a 180º phase shift between input and output signal

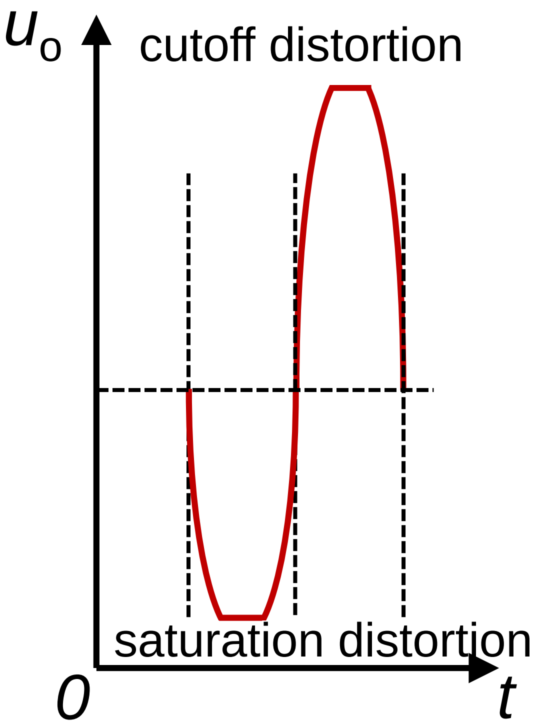


$u_{\mathrm{i}} = U_{\mathrm{IPP}}\sin(\omega t)/2$ $\quad$ $u_{\mathrm{o}} = -U_{\mathrm{OPP}}\sin(\omega t)/2$

Voltage gain: $\dot{A}_{\mathbf{u}} = -\dfrac{\boldsymbol{U}_{\mathrm{OPP}}}{\boldsymbol{U}_{\mathrm{IPP}}}$

2. Choose proper quiescent working point Q_i and Q_o

Otherwise there is nonlinear distortion effect (saturation distortion, cutoff distortion)



3. The amplitude of output signal

$U_{opp} \leq 2\min[I_C R_C, U_{CEQ}]$ (Ignore saturation region)

$U_{opp} \leq 2\min[I_C R_C, U_{CEQ} - U_{CES}]$ (include saturation region)

Case 2: R_L is finite   The output equation

$$u_{CE} = U_{CEQ} + u_{ce}$$

$$= V_{CC} - I_{CQ}R_C - i_c(R_C || R_L)$$

$$= V_{CC} - I_{CQ}R_C - i_c R_L'$$

$$R_L' = R_C || R_L$$

$$= V_{CC} - I_{CQ}R_C + I_{CQ}R_L' - (I_{CQ} + i_c)R_L'$$

$$= V_{CC} - I_{CQ}R_C + I_{CQ}R_L' - i_C R_L'$$

$$V_{CC}' = V_{CC} - I_{CQ}R_C + I_{CQ}R_L' = U_{CEQ} + I_{CQ}R_L'$$

$$u_{CE} = V_{CC}' - i_C R_L'$$

$$R_L' < R_C \qquad V_{CC}' < V_{CC}$$

Case1: $R_L=\infty$

$u_{CE}=V_{CC}-i_C R_C$

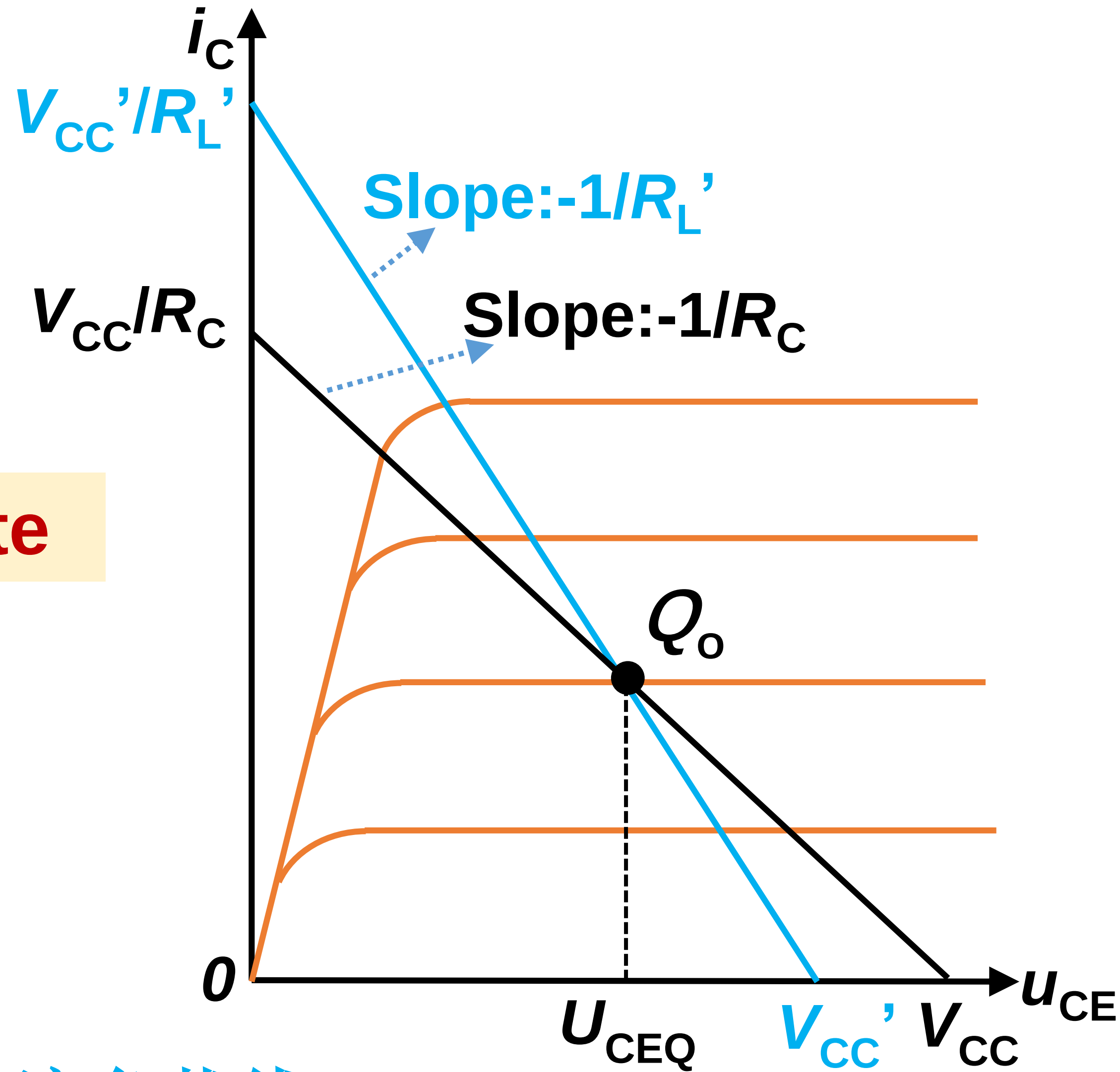
$V_{CC}=U_{CEQ}+I_{CQ}R_C$

Case2: R_L is finite

$u_{CE}=V_{CC}'-i_C R_L'$

$V_{CC}'=U_{CEQ}+I_{CQ}R_L'$

$R_L'=R_c||R_L$



AC load curve 交流负载线

R_L is finite

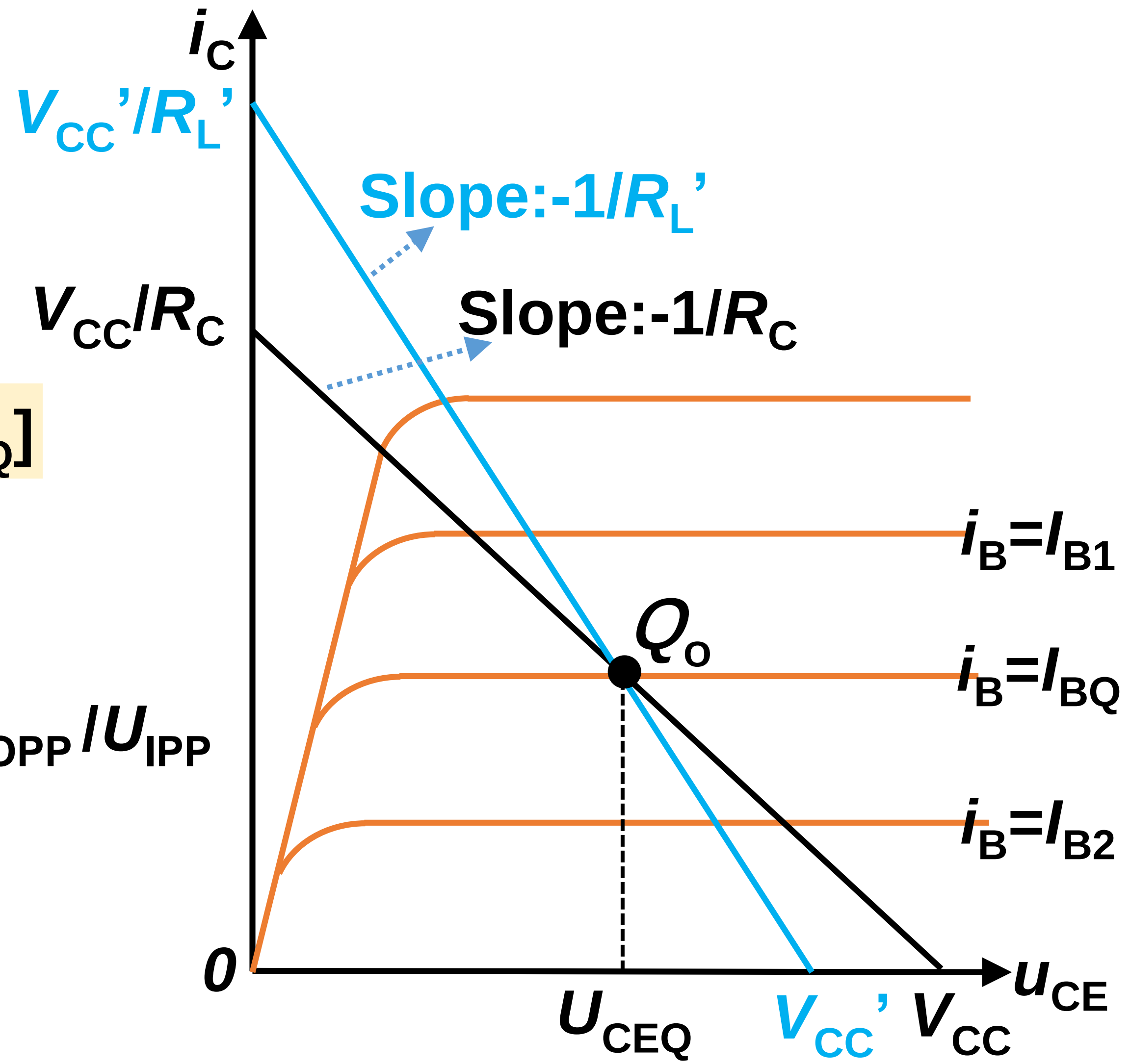
Dynamic range:

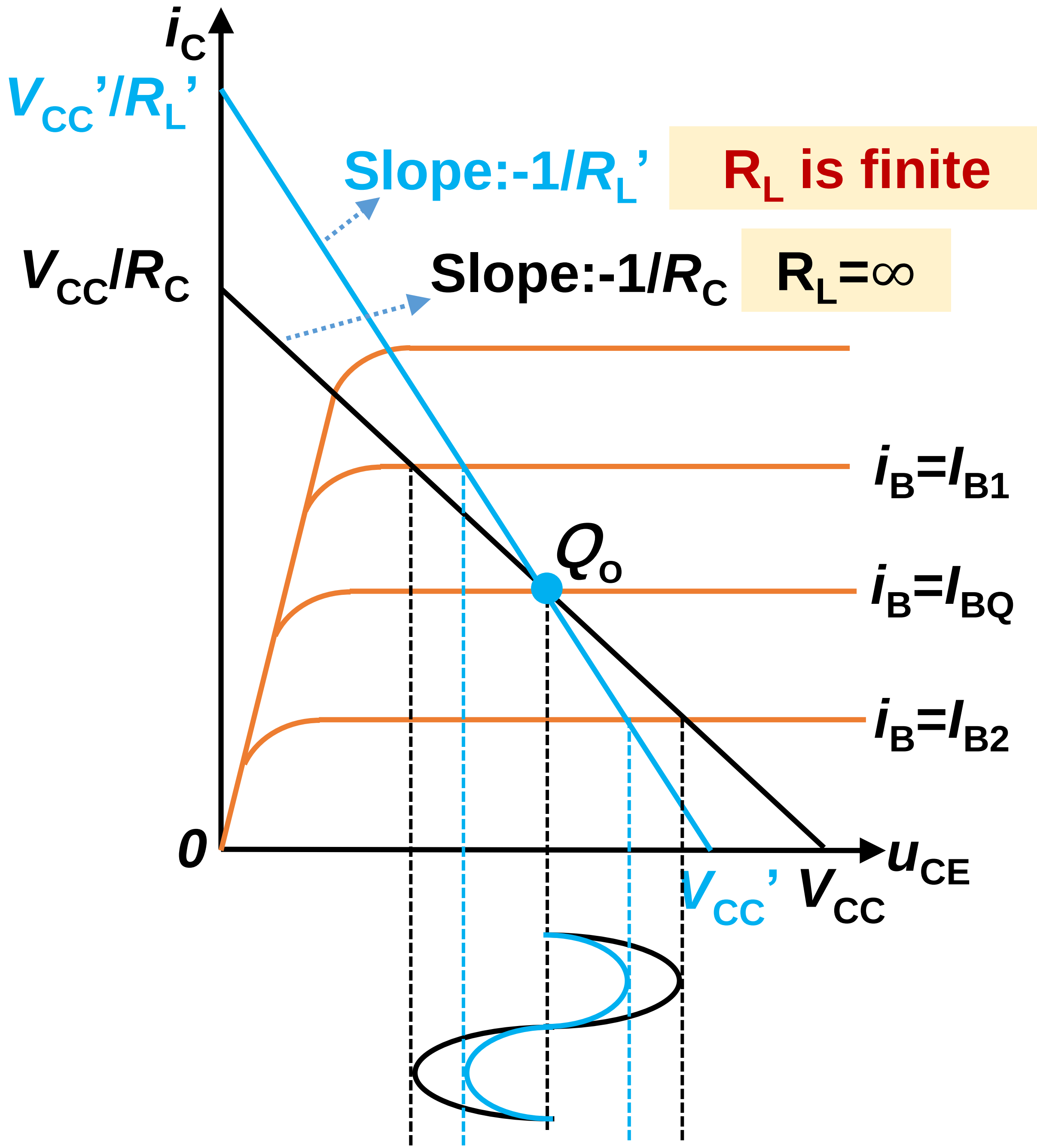
$U_{OPP} \leq 2\min[I_C R_L', U_{CEQ}]$

Decrease

Voltage gain: $|A_u| = U_{OPP} / U_{IPP}$

Decrease





i_C
V_{CC}'/R_L'
Slope:-1/R_L'
R_L is finite
V_{CC}/R_C
Slope:-1/R_C
$R_L=\infty$
$i_B=I_{B1}$
Q_o
$i_B=I_{BQ}$
$i_B=I_{B2}$
0
u_{CE}
V_{CC}'
V_{CC}

Homework 2-1：

Refer to blackboard and QQ group

暂不上交，和下次作业**2-2**一起。

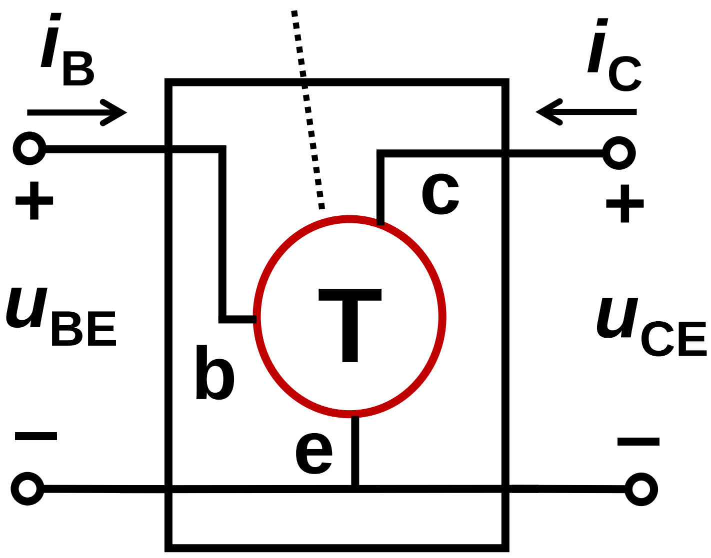
2.2.6 Equivalent circuit method (*h*-model)

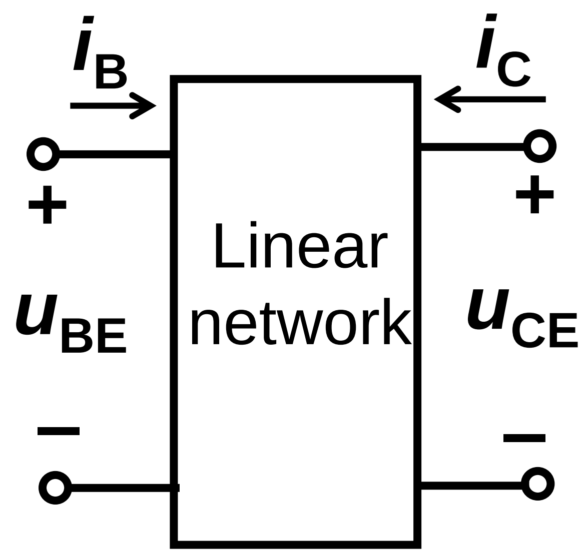
To replace transistor with a linear network

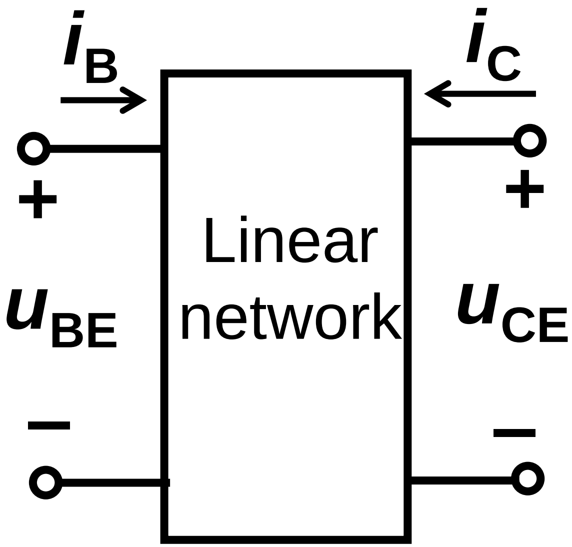
Transistor

Equivalent circuit

NPN or PNP







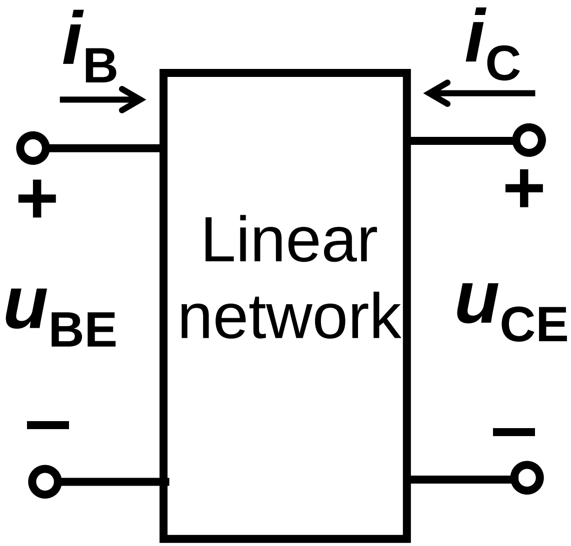
1 $\begin{cases} u_{BE} = f(i_B, u_{CE}) \\ i_C = g(i_B, u_{CE}) \end{cases}$

2 $$\begin{cases} du_{BE} = \dfrac{\partial u_{BE}}{\partial i_B}|_{U_{CEQ}} di_B + \dfrac{\partial u_{BE}}{\partial u_{CE}}|_{I_{BQ}} du_{CE} \\ di_C = \dfrac{\partial i_C}{\partial i_B}|_{U_{CEQ}} di_B + \dfrac{\partial i_C}{\partial u_{CE}}|_{I_{BQ}} du_{CE} \end{cases}$$

Input AC signal u_{be} has a small amplitude

du_{BE} can be replaced by a small quantity: u_{be} or $\dot{U}_{be}$

$$\begin{pmatrix} h_{11e} = \dfrac{\partial u_{BE}}{\partial i_B}|_{U_{CEQ}} = \dfrac{\dot{U}_{be}}{\dot{I}_b}|_{U_{CEQ}} & h_{12e} = \dfrac{\partial u_{BE}}{\partial u_{CE}}|_{I_{BQ}} = \dfrac{\dot{U}_{be}}{\dot{U}_{ce}}|_{I_{BQ}} \\ h_{21e} = \dfrac{\partial i_C}{\partial i_B}|_{U_{CEQ}} = \dfrac{\dot{I}_c}{\dot{I}_b}|_{U_{CEQ}} & h_{22e} = \dfrac{\partial i_C}{\partial u_{CE}}|_{I_{BQ}} = \dfrac{\dot{I}_c}{\dot{U}_{ce}}|_{I_{BQ}} \end{pmatrix}$$

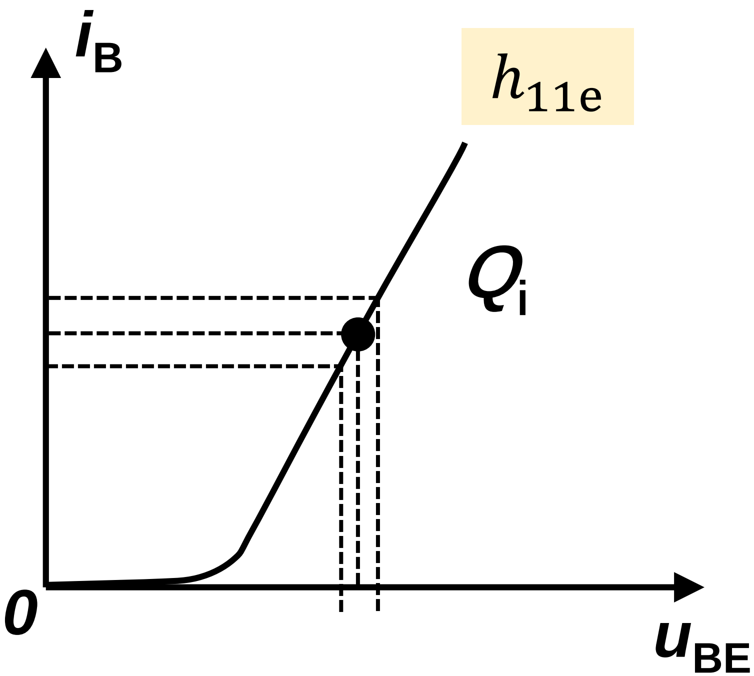


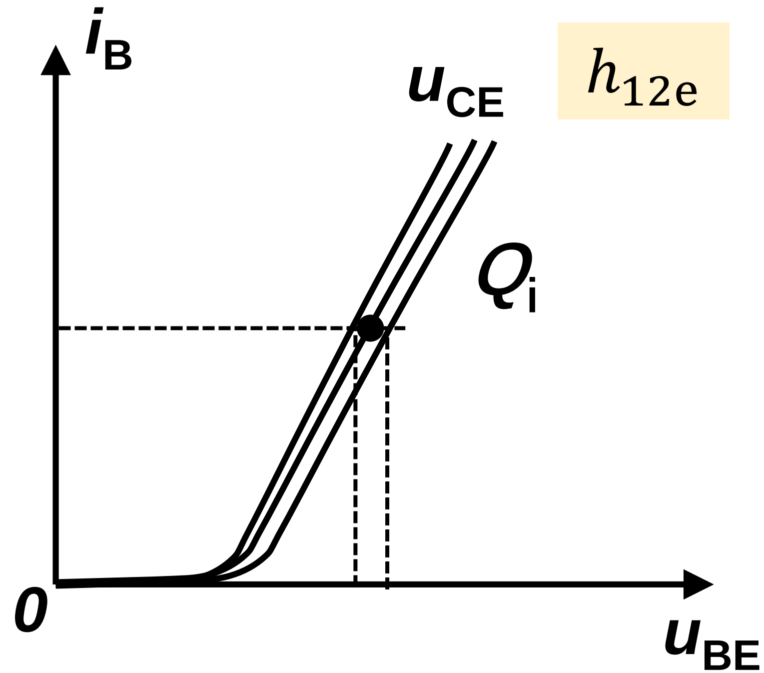
$$\begin{cases} \dot{U}_{\mathrm{be}} = h_{11\mathrm{e}}\dot{I}_{\mathrm{b}} + h_{12\mathrm{e}}\dot{U}_{\mathrm{ce}} \\ \dot{I}_{\mathrm{c}} = h_{21\mathrm{e}}\dot{I}_{\mathrm{b}} + h_{22\mathrm{e}}\dot{U}_{\mathrm{ce}} \end{cases}$$

$$\begin{cases} u_{\mathrm{be}} = h_{11\mathrm{e}}i_{\mathrm{b}} + h_{12\mathrm{e}}u_{\mathrm{ce}} \\ i_{\mathrm{c}} = h_{21\mathrm{e}}i_{\mathrm{b}} + h_{22\mathrm{e}}u_{\mathrm{ce}} \end{cases}$$

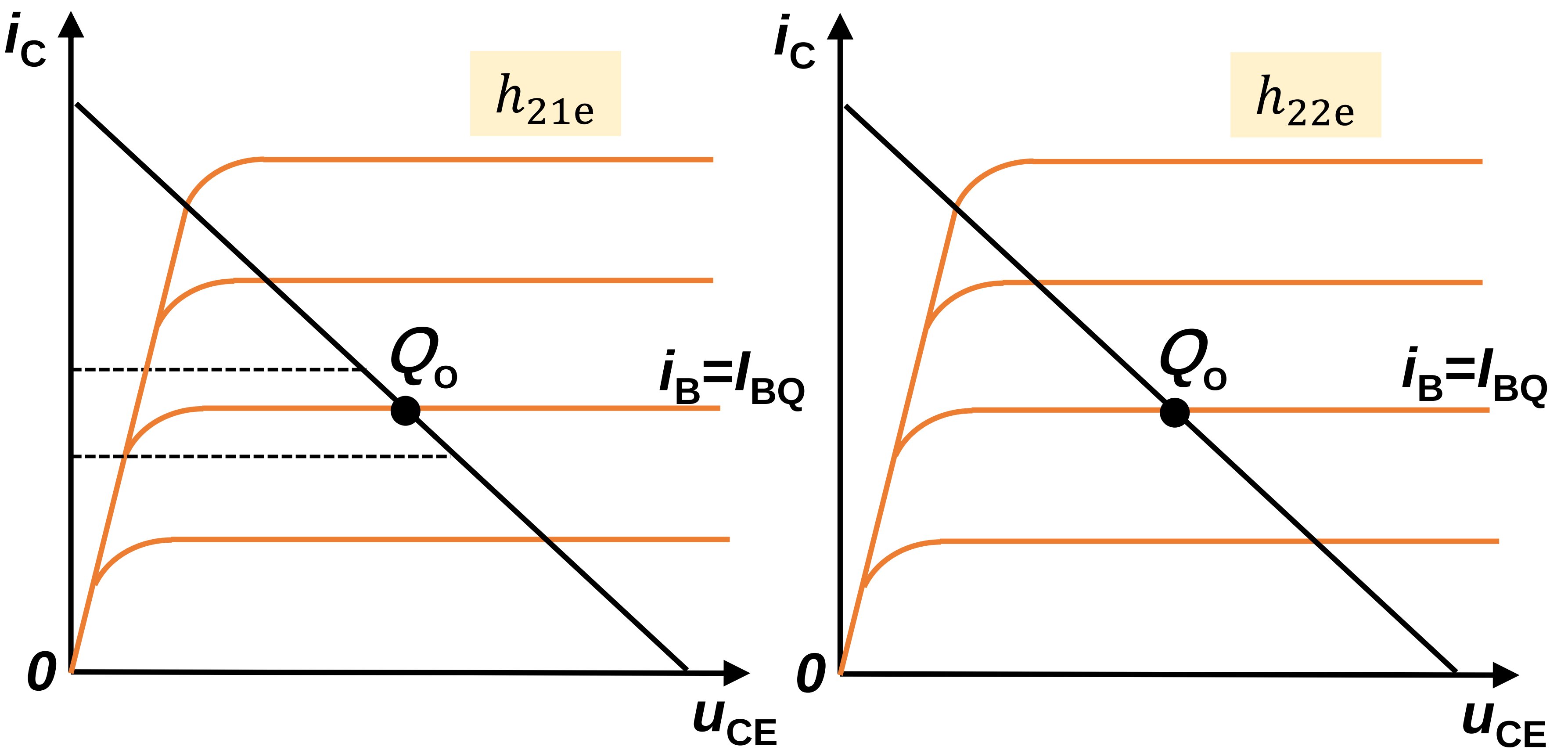
$$\left(\begin{array}{ll} h_{11\mathrm{e}} = \dfrac{\partial u_{\mathrm{BE}}}{\partial i_{\mathrm{B}}}\Big|_{U_{\mathrm{CEQ}}} = \dfrac{\dot{U}_{\mathrm{be}}}{\dot{I}_{\mathrm{b}}}\Big|_{U_{\mathrm{CEQ}}} & h_{12\mathrm{e}} = \dfrac{\partial u_{\mathrm{BE}}}{\partial u_{\mathrm{CE}}}\Big|_{I_{\mathrm{BQ}}} = \dfrac{\dot{U}_{\mathrm{be}}}{\dot{U}_{\mathrm{ce}}}\Big|_{I_{\mathrm{BQ}}} \\ h_{21\mathrm{e}} = \dfrac{\partial i_{\mathrm{C}}}{\partial i_{\mathrm{B}}}\Big|_{U_{\mathrm{CEQ}}} = \dfrac{\dot{I}_{\mathrm{c}}}{\dot{I}_{\mathrm{b}}}\Big|_{U_{\mathrm{CEQ}}} & h_{22\mathrm{e}} = \dfrac{\partial i_{\mathrm{C}}}{\partial u_{\mathrm{CE}}}\Big|_{I_{\mathrm{BQ}}} = \dfrac{\dot{I}_{\mathrm{c}}}{\dot{U}_{\mathrm{ce}}}\Big|_{I_{\mathrm{BQ}}} \end{array} \right)$$

The physical interpretation of *h*

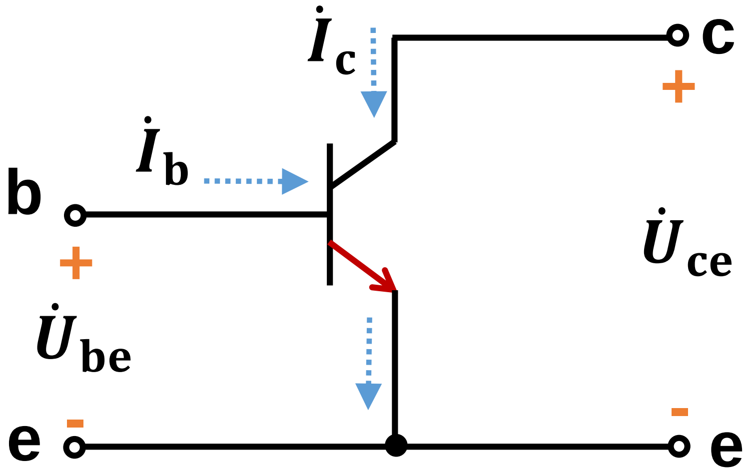




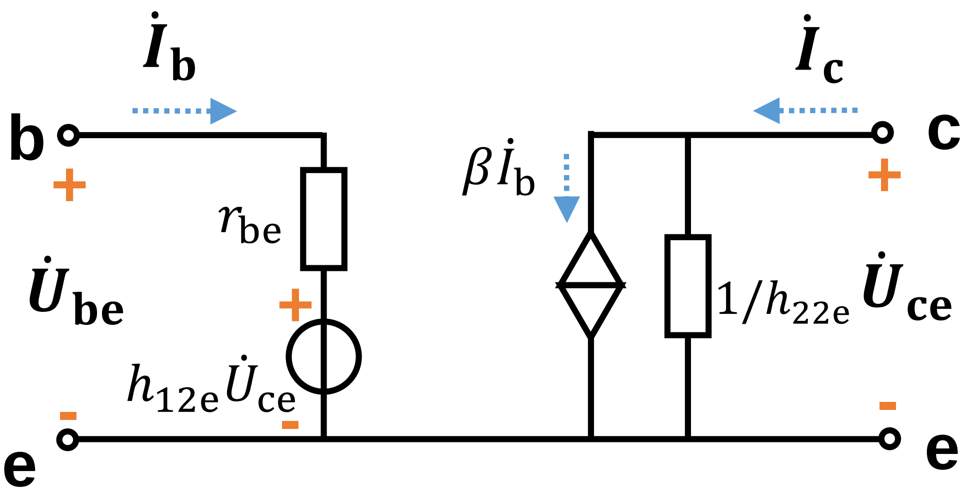
$$\left(\begin{array}{ll} h_{11e} = \dfrac{\partial u_{BE}}{\partial i_B}\Big|_{U_{CEQ}} = \dfrac{\dot{U}_{be}}{\dot{I}_b}\Big|_{U_{CEQ}} & h_{12e} = \dfrac{\partial u_{BE}}{\partial u_{CE}}\Big|_{I_{BQ}} = \dfrac{\dot{U}_{be}}{\dot{U}_{ce}}\Big|_{I_{BQ}} \\ h_{21e} = \dfrac{\partial i_C}{\partial i_B}\Big|_{U_{CEQ}} = \dfrac{\dot{I}_c}{\dot{I}_b}\Big|_{U_{CEQ}} & h_{22e} = \dfrac{\partial i_C}{\partial u_{CE}}\Big|_{I_{BQ}} = \dfrac{\dot{I}_c}{\dot{U}_{ce}}\Big|_{I_{BQ}} \end{array} \right)$$



$$
\left\{\begin{array}{ll}
h_{11\mathrm{e}} = \dfrac{\partial u_{\mathrm{BE}}}{\partial i_{\mathrm{B}}}\Big|_{U_{\mathrm{CEQ}}} = \dfrac{\dot{U}_{\mathrm{be}}}{\dot{I}_{\mathrm{b}}}\Big|_{U_{\mathrm{CEQ}}} & h_{12\mathrm{e}} = \dfrac{\partial u_{\mathrm{BE}}}{\partial u_{\mathrm{CE}}}\Big|_{I_{\mathrm{BQ}}} = \dfrac{\dot{U}_{\mathrm{be}}}{\dot{U}_{\mathrm{ce}}}\Big|_{I_{\mathrm{BQ}}} \\
h_{21\mathrm{e}} = \dfrac{\partial i_{\mathrm{C}}}{\partial i_{\mathrm{B}}}\Big|_{U_{\mathrm{CEQ}}} = \dfrac{\dot{I}_{\mathrm{c}}}{\dot{I}_{\mathrm{b}}}\Big|_{U_{\mathrm{CEQ}}} & h_{22\mathrm{e}} = \dfrac{\partial i_{\mathrm{C}}}{\partial u_{\mathrm{CE}}}\Big|_{I_{\mathrm{BQ}}} = \dfrac{\dot{I}_{\mathrm{c}}}{\dot{U}_{\mathrm{ce}}}\Big|_{I_{\mathrm{BQ}}}
\end{array}\right.
$$

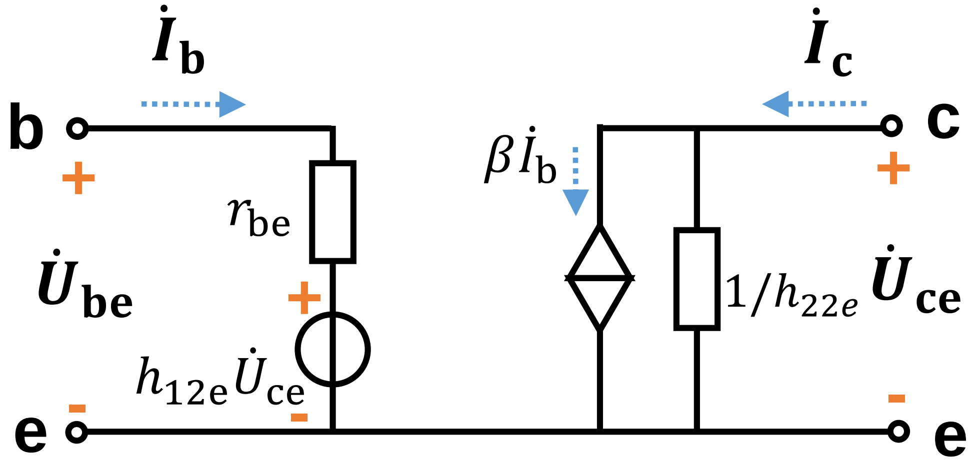


$$\begin{cases} \dot{U}_{be} = h_{11e}\dot{I}_b + h_{12e}\dot{U}_{ce} \\ \dot{I}_c = h_{21e}\dot{I}_b + h_{22e}\dot{U}_{ce} \end{cases}$$



$$r_{be} = h_{11e}$$

$$h_{21e} = \beta$$

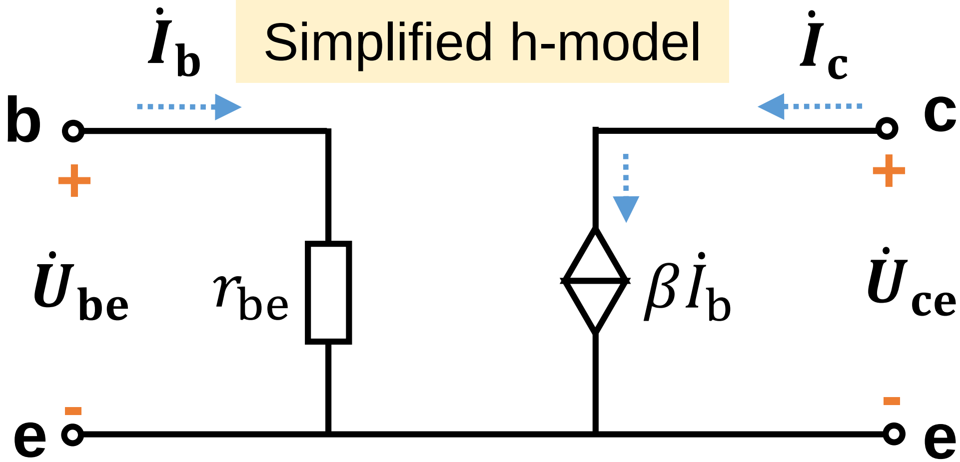


$$r_{be} = h_{11e}$$

$$h_{21e} = \beta$$

Because h_{12e} and h_{22e} is very small, we can simplify the circuit

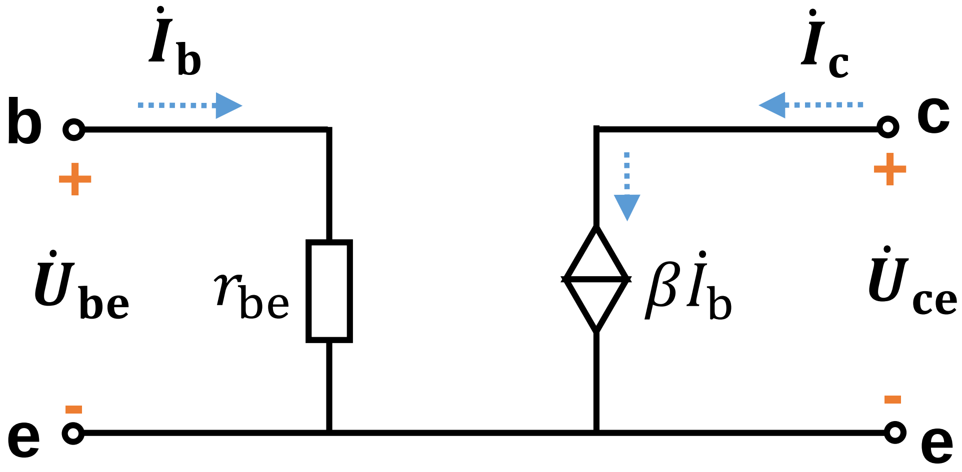
Simplified h-model



$$r_{be} = r_{bb'} + \beta\frac{U_T}{|I_{CQ}|}$$

$$U_T = 26\text{mV}$$

Question: What's the h-model of PNP transistor?

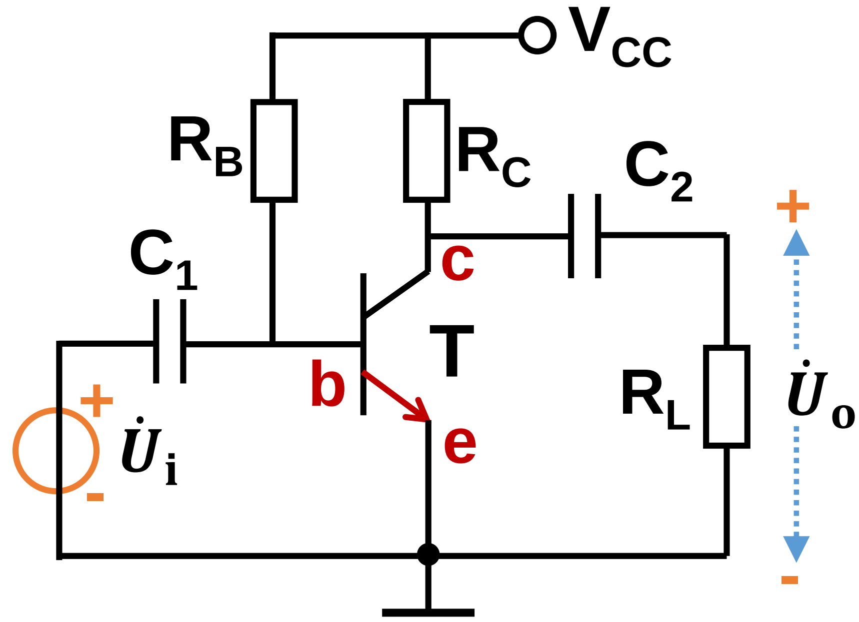


$$r_{be} = r_{bb'} + \beta \frac{U_T}{|I_{CQ}|}$$

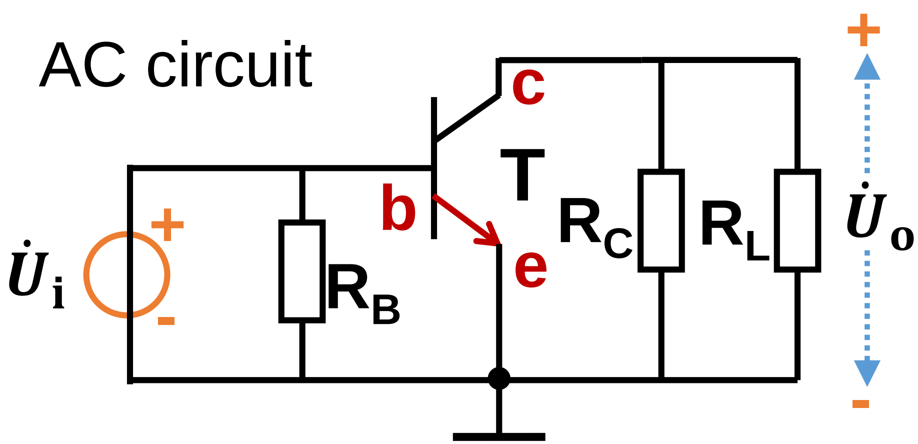
$$U_T = 26\text{mV}$$

This equivalent circuit is the same for both NPN and PNP transistors.

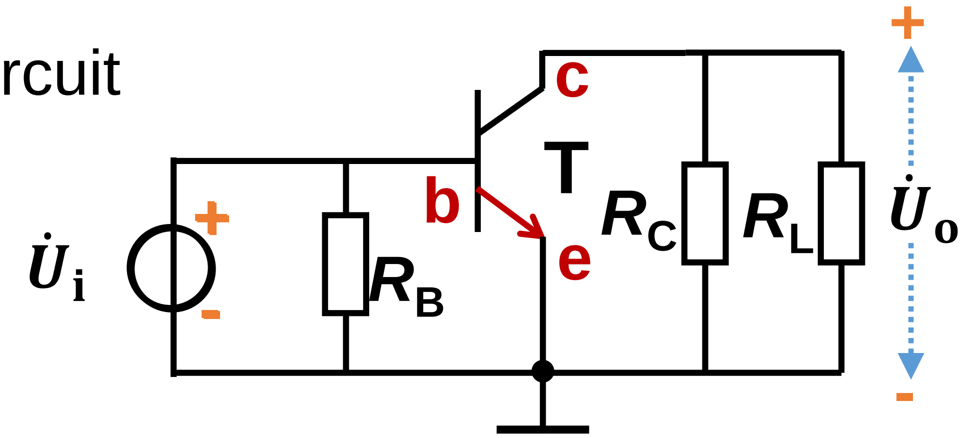
Example:



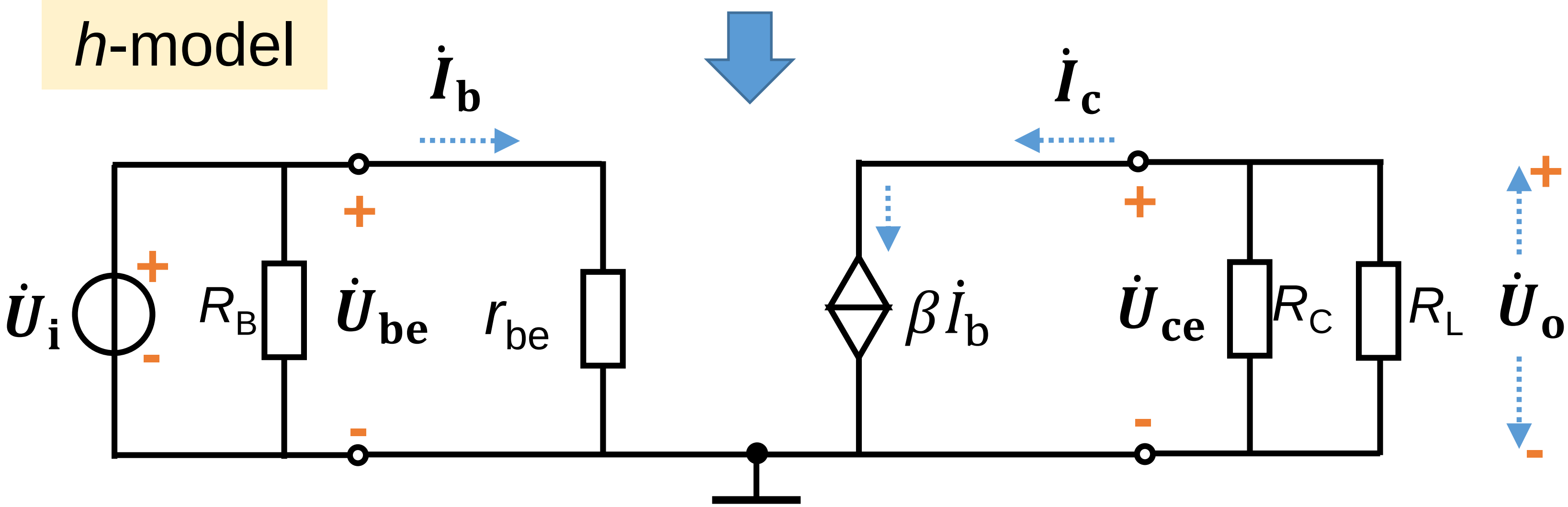
(1) What's the equivalent circuit of *h*-model 等效电路?



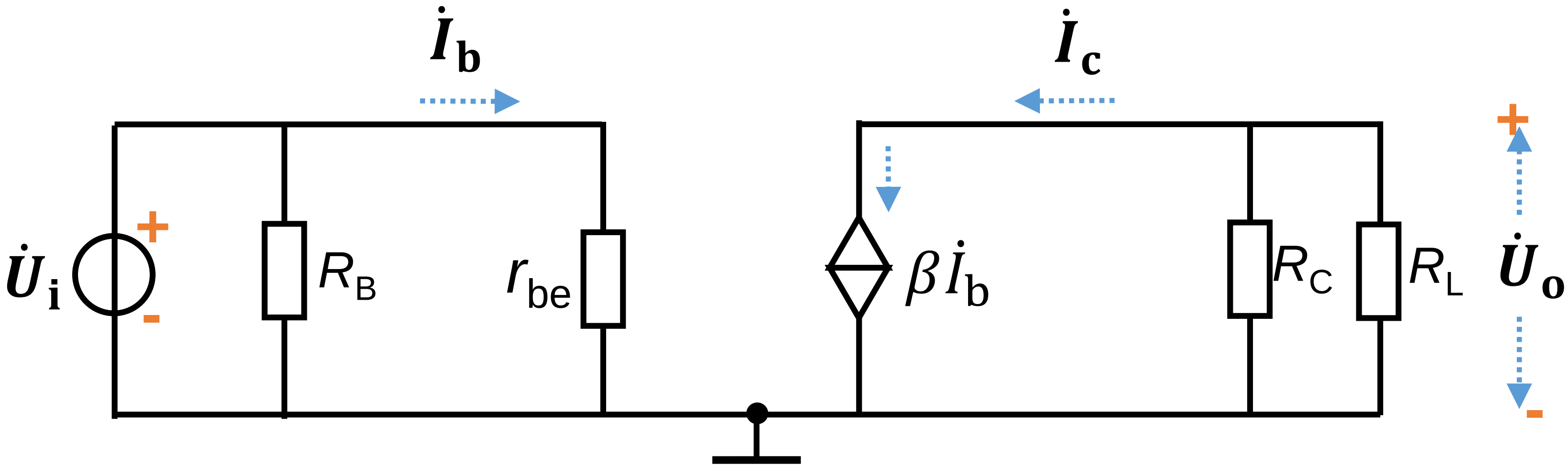
AC circuit



h-model



(2) Voltage gain?



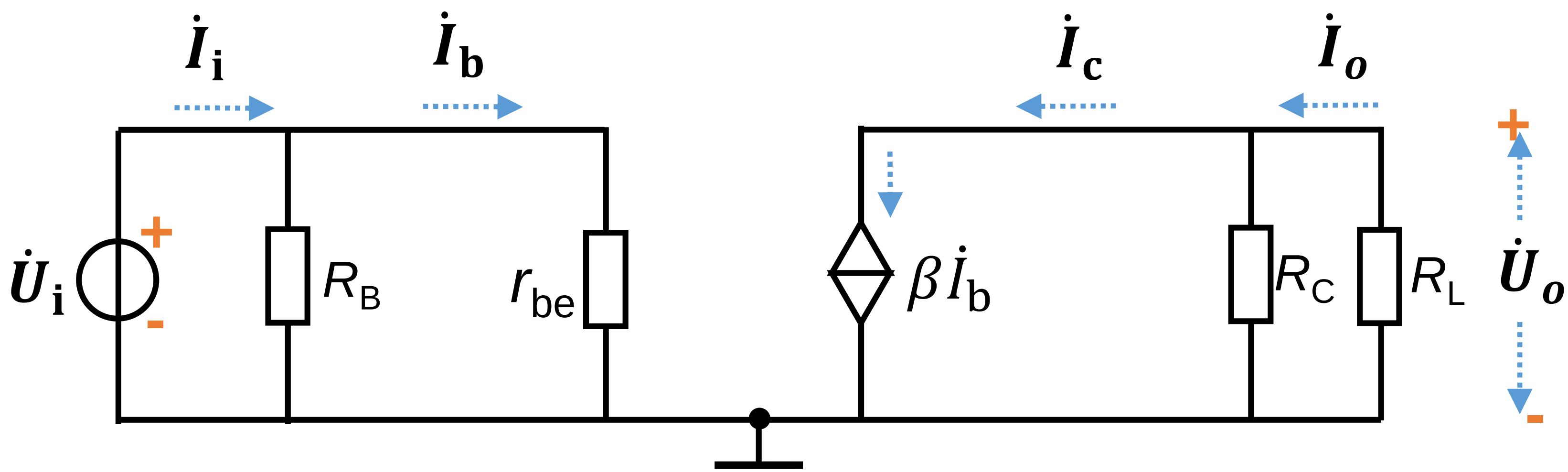
Voltage gain： $\dot{A}_u = \dfrac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_i = \dot{I}_b r_{be}$$

$$\dot{U}_o = -\dot{I}_c(R_C||R_L) = -\beta\dot{I}_b R_L{}'$$

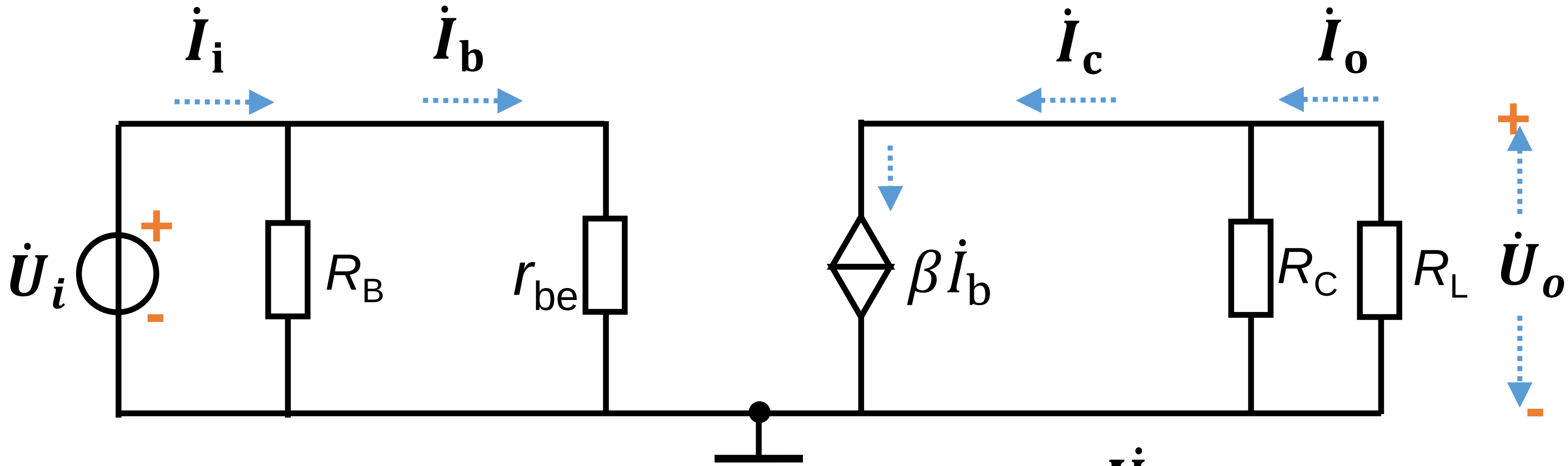
$$\boxed{\dot{A}_u = -\frac{\beta R_L{}'}{r_{be}}}$$

(3) Input resistance?

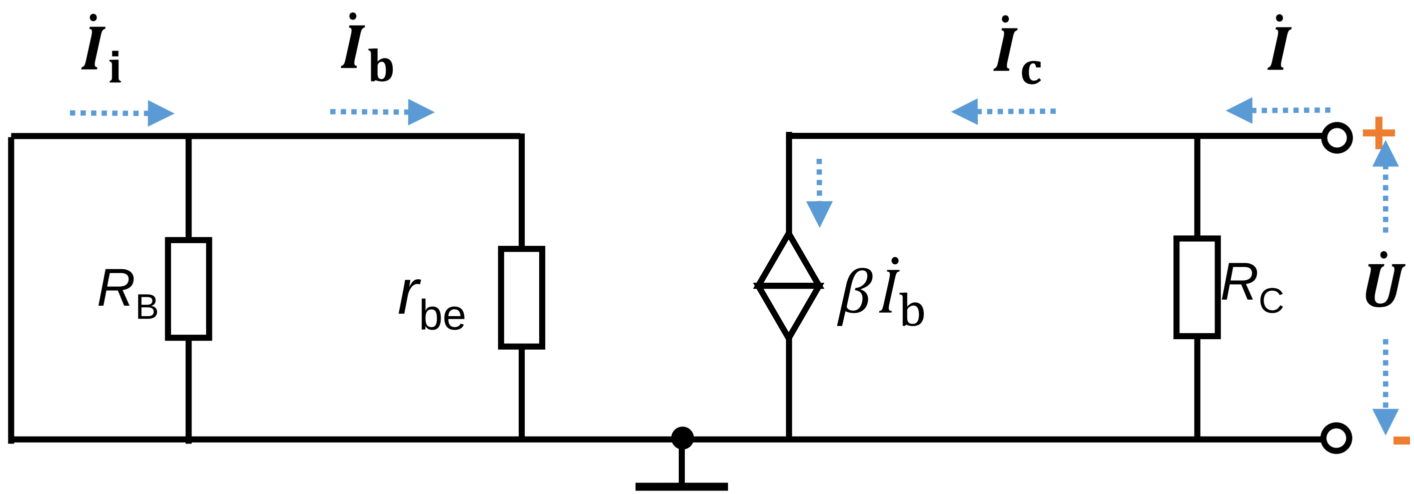


Input resistance： $R_i = \dfrac{\dot{U}_i}{\dot{I}_i} = R_B || r_{be}$

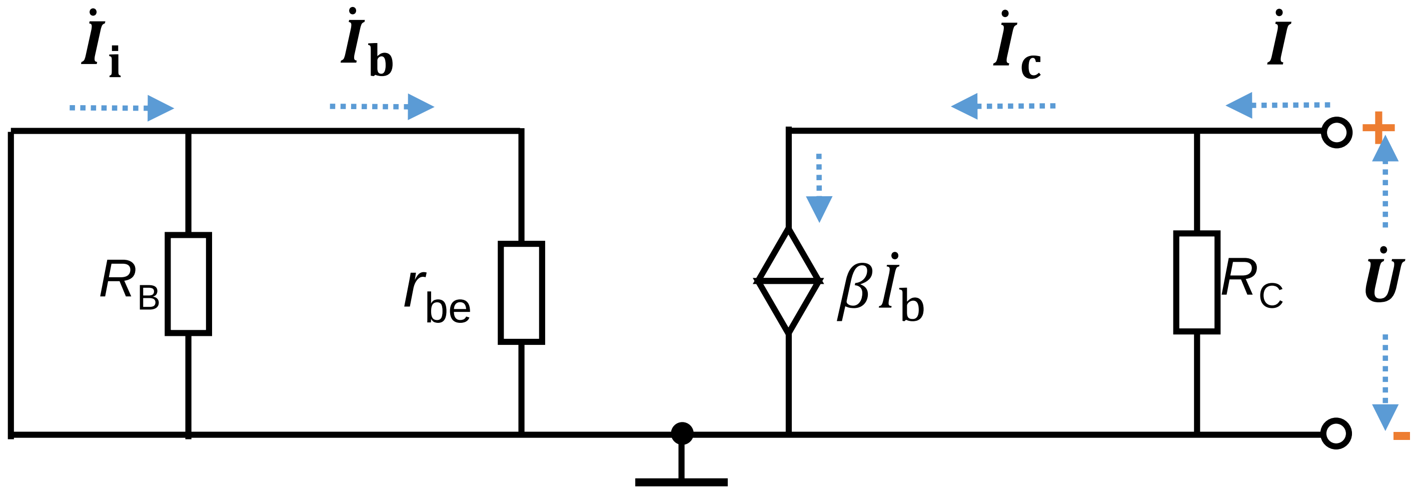
(4) Output resistance?



Output resistance: $R_o \neq \dfrac{\dot{U}_o}{\dot{I}_o}$



Output resistance: $R_o = \dfrac{\dot{U}}{\dot{I}}\Big|_{R_L=\infty,\dot{U}_i=0}$



Output resistance： $R_o = \frac{\dot{U}}{\dot{I}}\Big|_{R_L=\infty, \dot{U}_i=0}$

$$\dot{I}_b = 0$$

$$\dot{I}_c = 0$$

Current source has infinite resistance

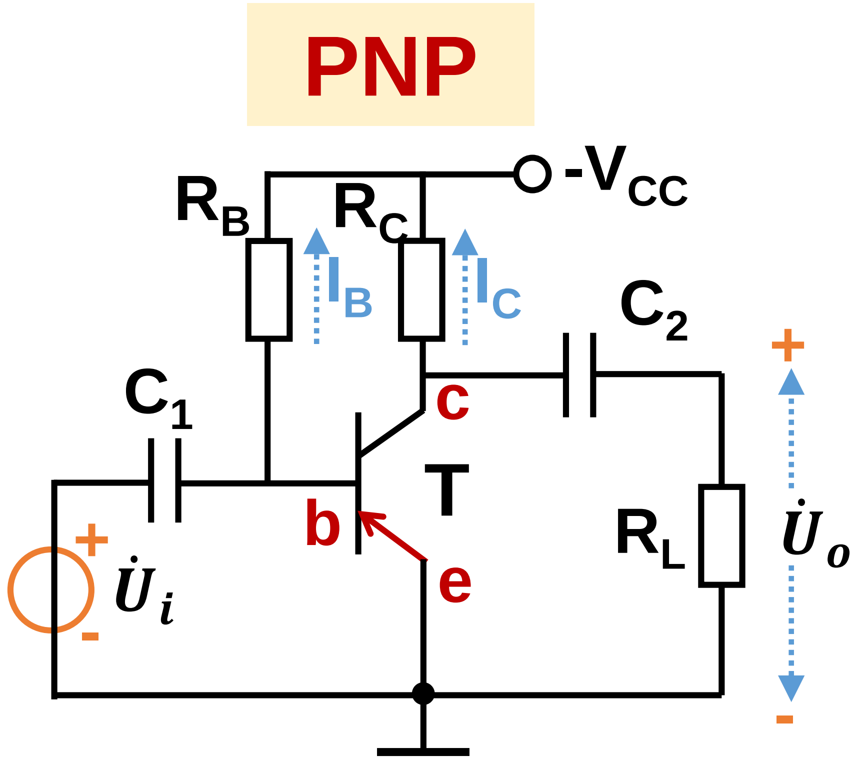
$$R_o = R_C$$

[Q1] In the figure below, V_{CC}=12V, R_C=2kΩ, and R_B=360kΩ. Transistor is made by Ge, $\beta = \overline{\beta} = 60$, $r_{bb'}$=300Ω. C_1=C_2=10μF, R_L=2kΩ. Ask:

(a) Quiescent parameters: I_{BQ}, I_{CQ} and U_{CEQ};

(b) Dynamic range U_{opp};

(c) A_u, R_i, R_o.



(a) I_{BQ}, I_{CQ} and U_{CEQ}.

$$-V_{CC}=-I_{BQ}R_B+U_{BEQ}$$

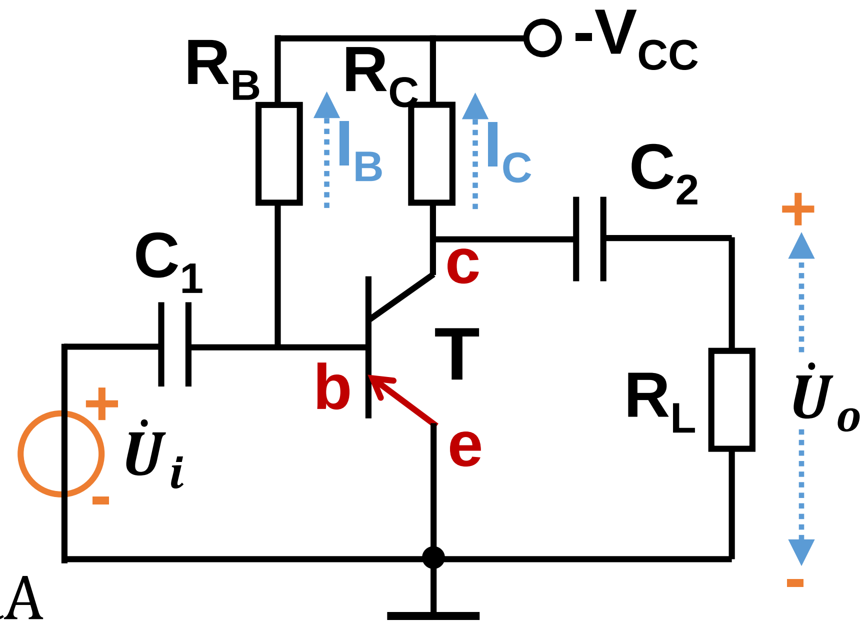
$$I_{BQ}=\frac{V_{CC}+U_{BEQ}}{R_B}$$

$$I_{BQ}=\frac{12+(-0.3)}{360\times10^3}=32.5\mu\text{A}$$

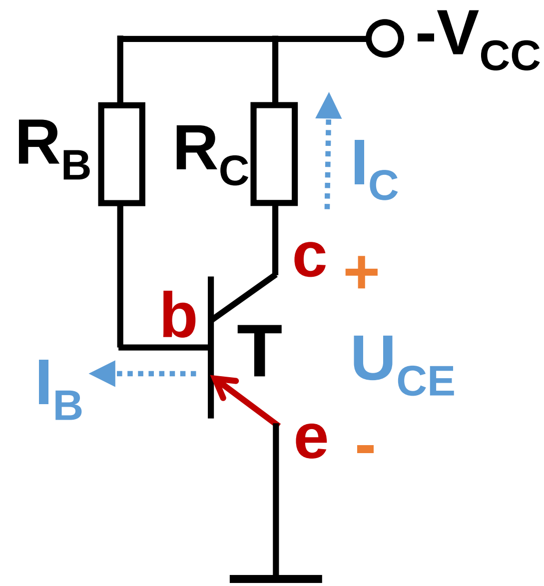
$$I_{CQ}=\beta I_{BQ}=60\times I_{BQ}$$
$$=1.95\text{mA}$$

$$-V_{CC}=-I_{CQ}R_C+U_{CEQ}$$

$$U_{CEQ}=-V_{CC}+I_{CQ}R_B$$
$$=-12+1.95\times10^{-3}\times2\times10^3$$
$$=-8.1\text{V}$$

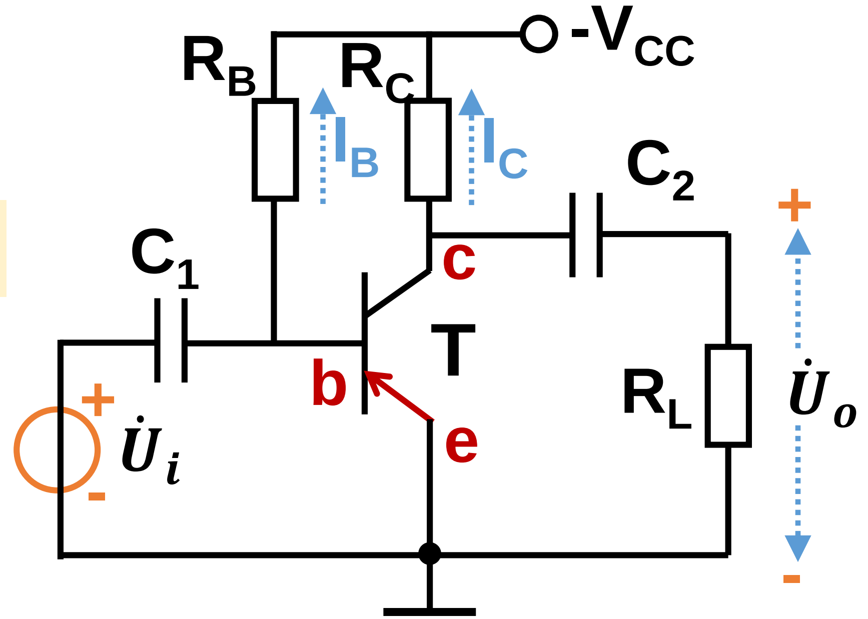


DC circuit



(b) Dynamic range U_{opp};

$U_{OPP} \leq 2\min[|I_{CQ}R_L'|, |U_{CEQ}|]$



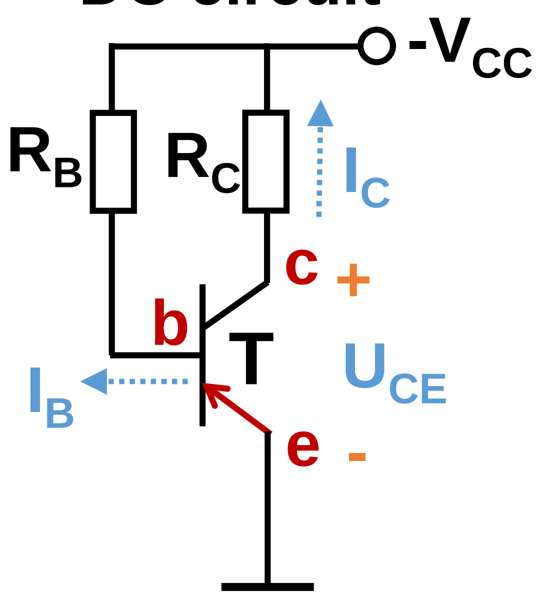
$$I_{CQ} = 1.95\text{mA}$$

$$R_L' = R_L||R_C = 1\ \text{K}\Omega$$

$$I_{CQ}R_L' = 1.95\ \text{V}$$

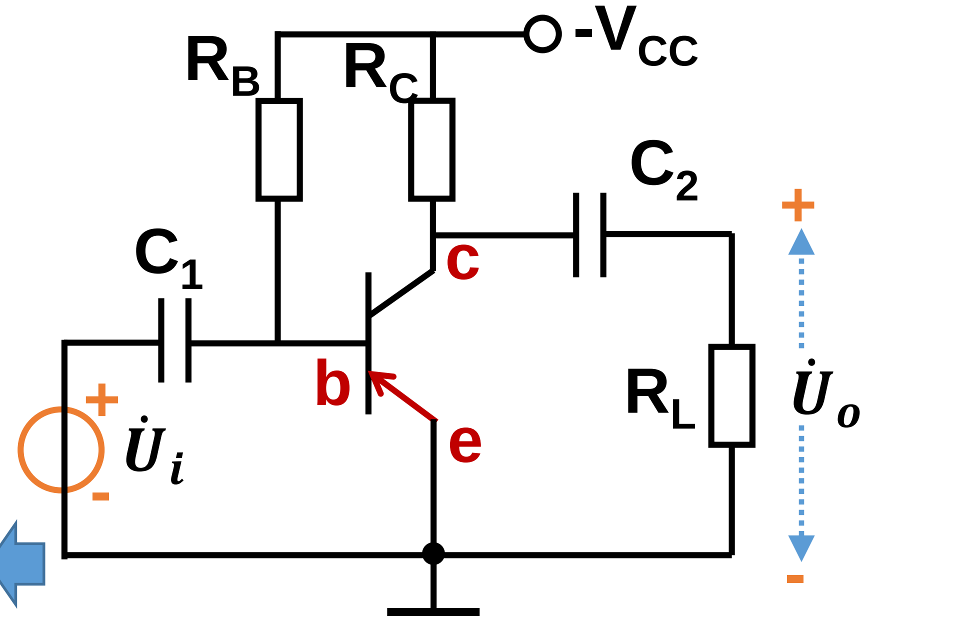
$$U_{CEQ} = -8.1\text{V}$$

DC circuit

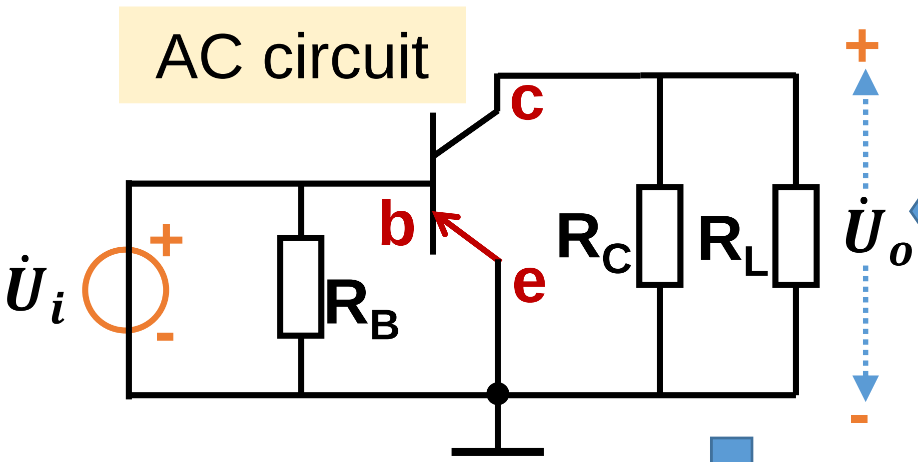


$U_{OPP} = 3.9\text{V}$

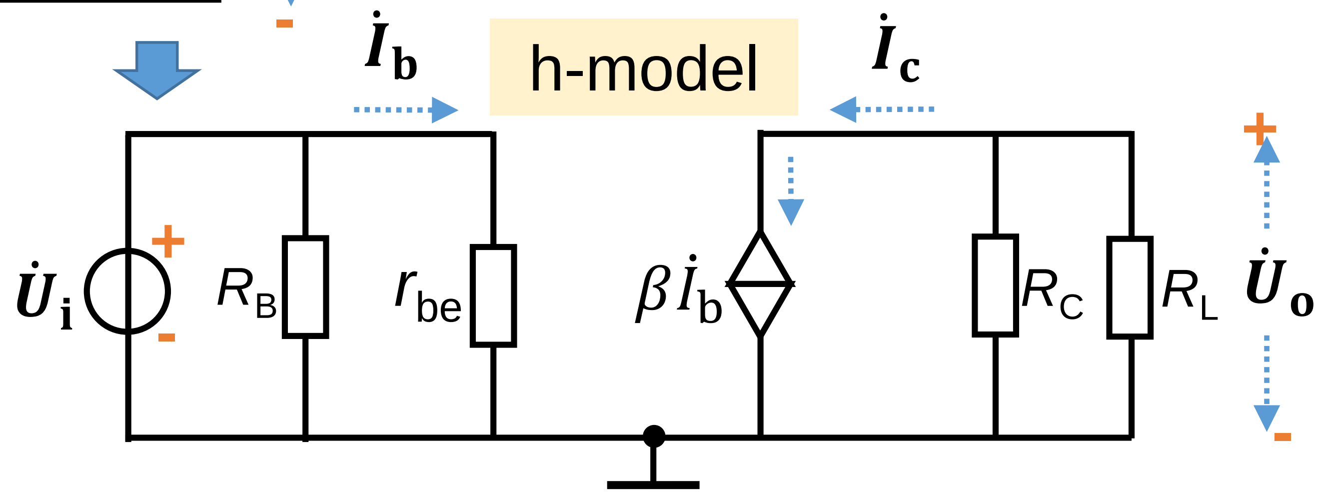
(c) A_u, R_i, R_o.



-V_CC
R_B
R_C
C_2
C_1
c
b
e
R_L
+
-
$\dot{U}_o$
$\dot{U}_i$

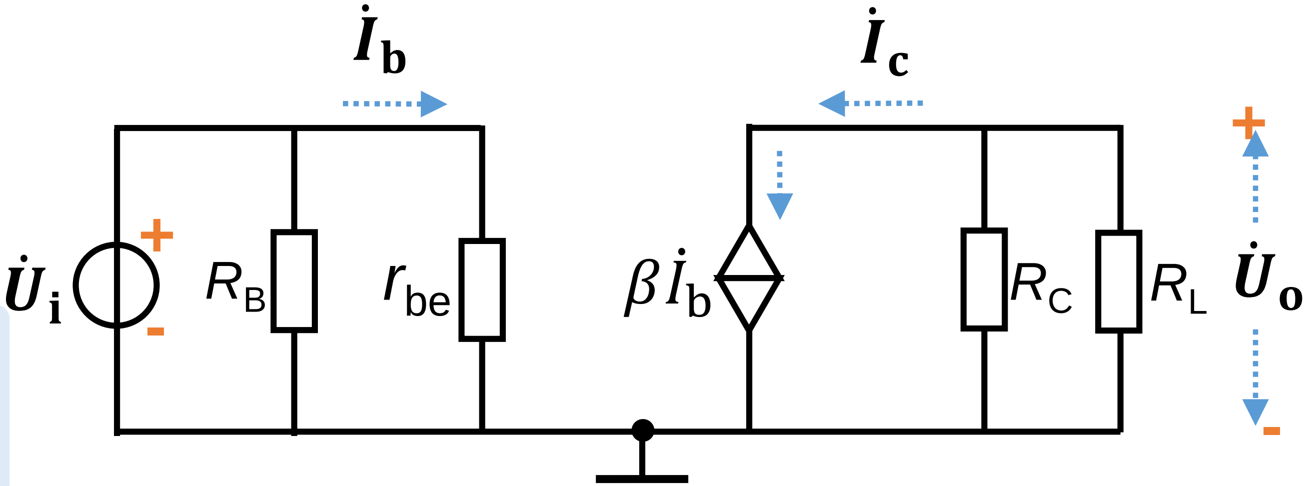


AC circuit
c
b
e
R_C
R_L
R_B
+
-
$\dot{U}_o$
$\dot{U}_i$



h-model
$\dot{I}_b$
$\dot{I}_c$
$\dot{U}_i$
R_B
r_{be}
$\beta \dot{I}_b$
R_C
R_L
$\dot{U}_o$
+
-

(c) A_u, R_i, R_o.



Voltage gain

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$$

$$\dot{U}_i = \dot{I}_b r_{be}$$

$$\dot{U}_o = -\dot{I}_c (R_C || R_L)$$

$$\dot{A}_u = -\frac{\beta (R_C || R_L)}{r_{be}}$$

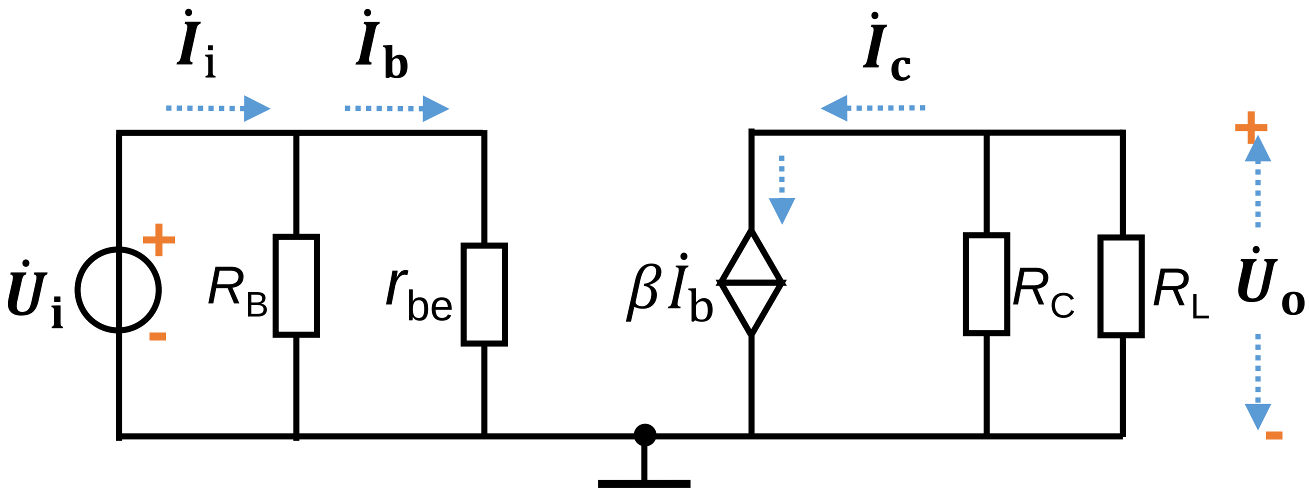
$$R_C || R_L = 1\mathrm{k\Omega}$$

$$r_{be} = r_{bb'} + \beta \frac{U_T}{|I_{CQ}|}$$

$$= 300 + 60 \frac{26 \times 10^{-3}}{1.95 \times 10^{-3}} = 1.1\mathrm{k\Omega}$$

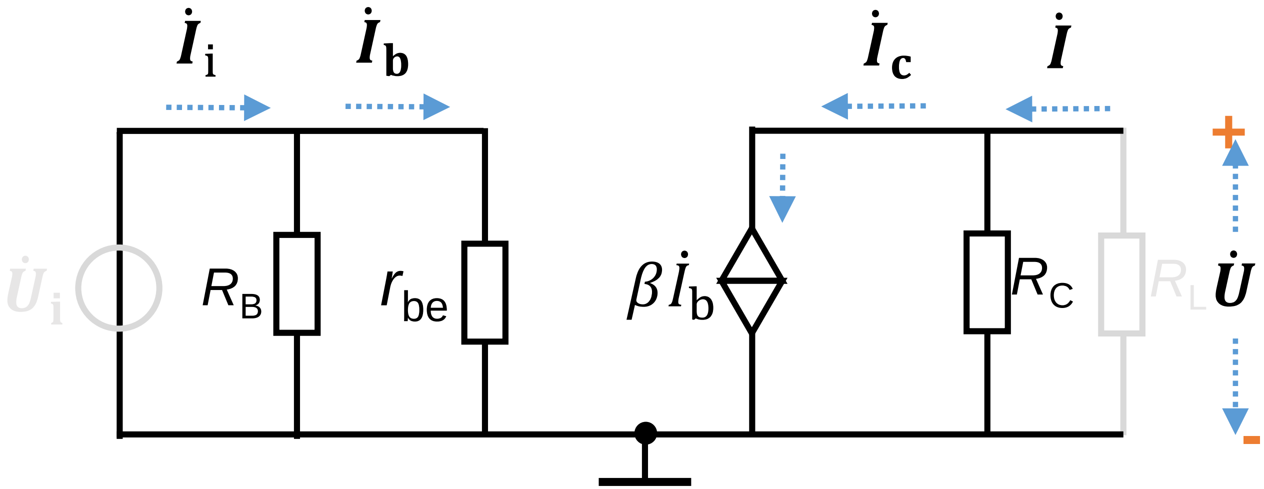
$$\dot{A}_u = -\frac{\beta (R_C || R_L)}{r_{be}} = -\frac{60 \times 1}{1.1} = -54.5$$

(c) A_u, R_i, R_o.



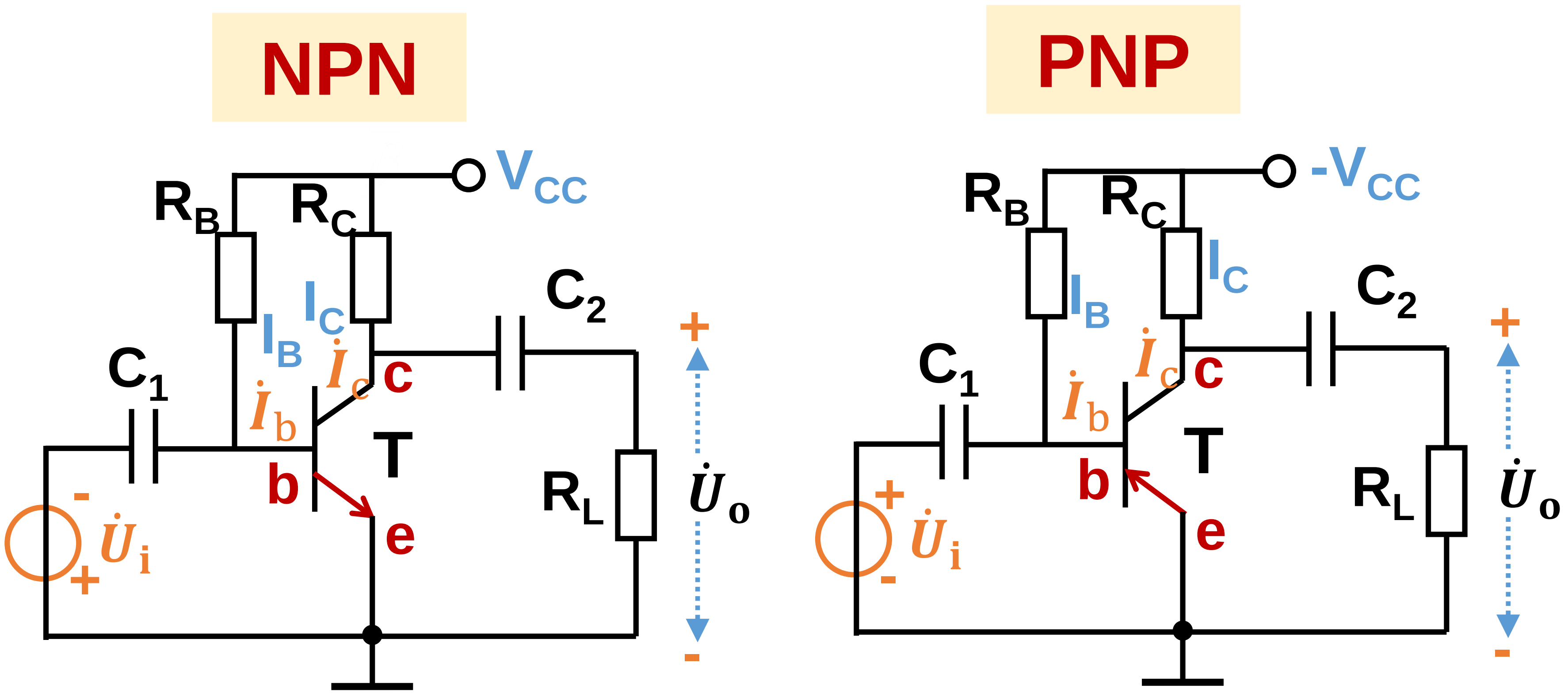
Input resistance: $$R_i = \frac{\dot{U}_i}{\dot{I}_i} = R_B || r_{be} = 1.1\text{k}\Omega$$

(c) A_u, R_i, R_o.



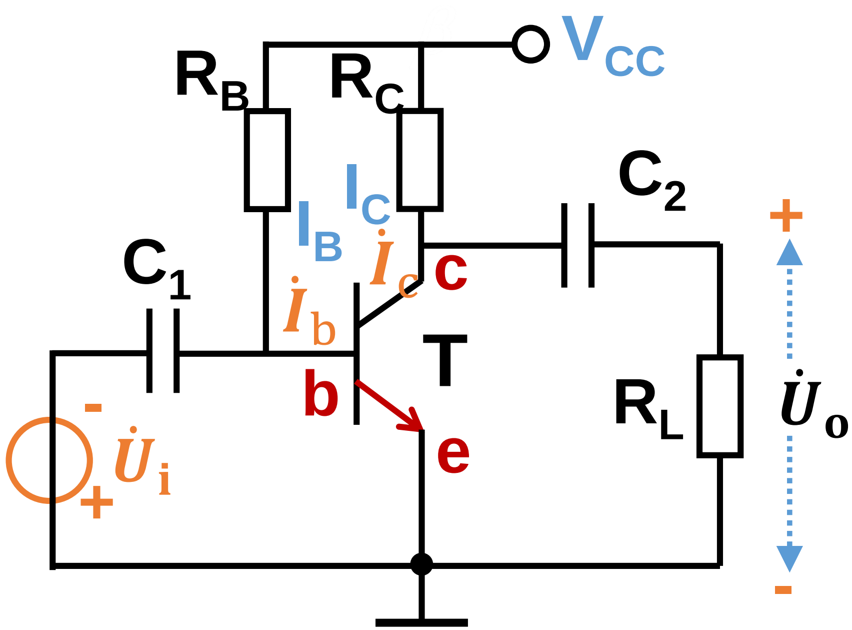
Output resistance: $\boldsymbol{R_o = \frac{\dot{U}}{\dot{I}}\Big|_{R_L=\infty,\dot{U}_i=0} = R_C = 2\mathrm{k}\Omega}$

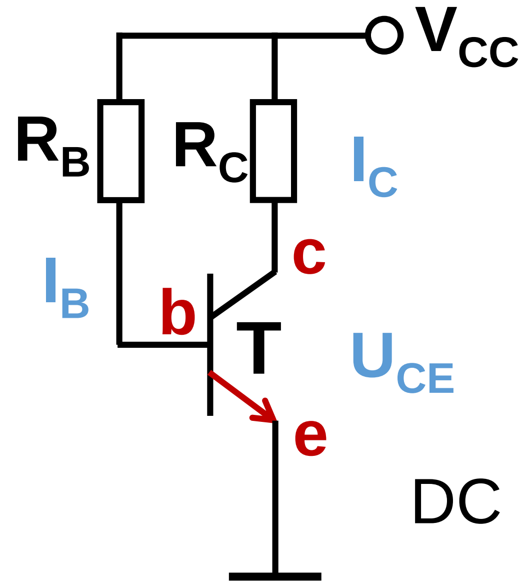
[Q2] The direction of DC current (I_B, I_C), AC current ($\dot{I}_b$, $\dot{I}_c$), DC voltage (U_{BE}, U_{CE}), and AC voltage ($\dot{U}_{be}$, $\dot{U}_{ce}$)?



[Q2] The direction of DC current (I_B, I_C), AC current ($\dot{I}_b$, $\dot{I}_c$), DC voltage (U_{BE}, U_{CE}), and AC voltage ($\dot{U}_{be}$, $\dot{U}_{ce}$)?

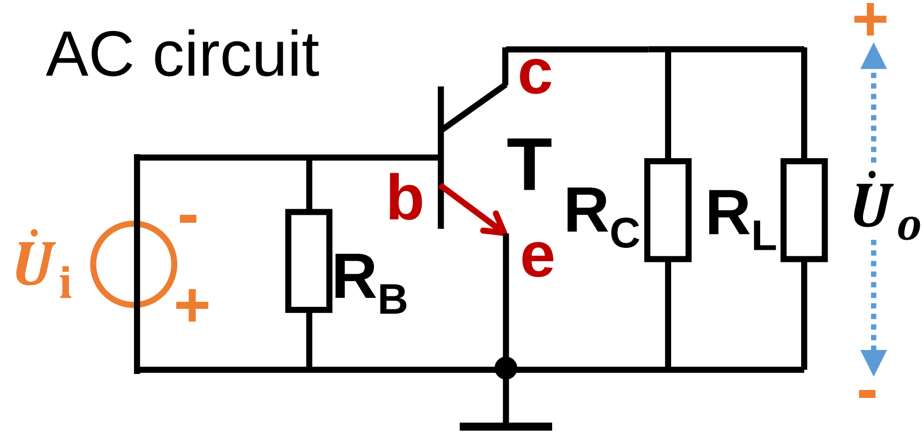
NPN





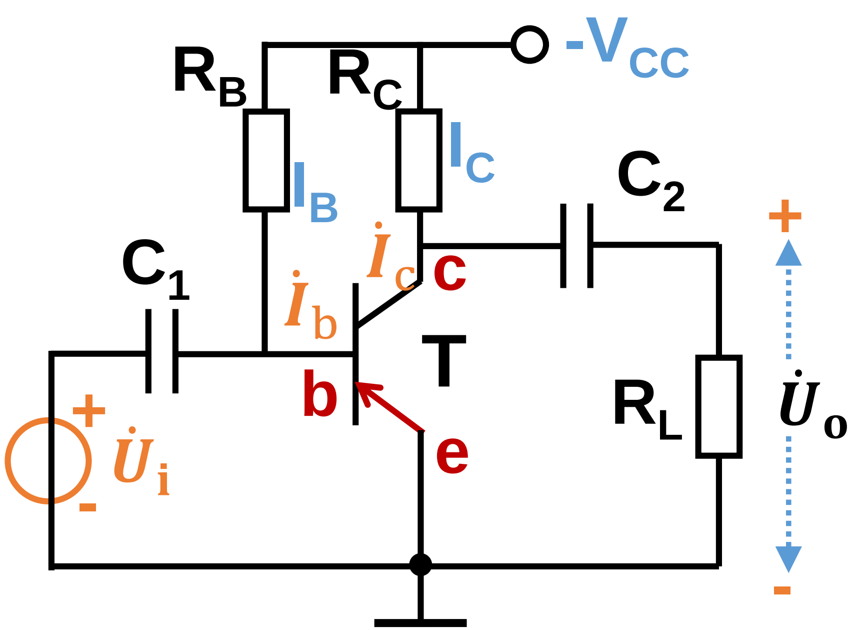
DC circuit

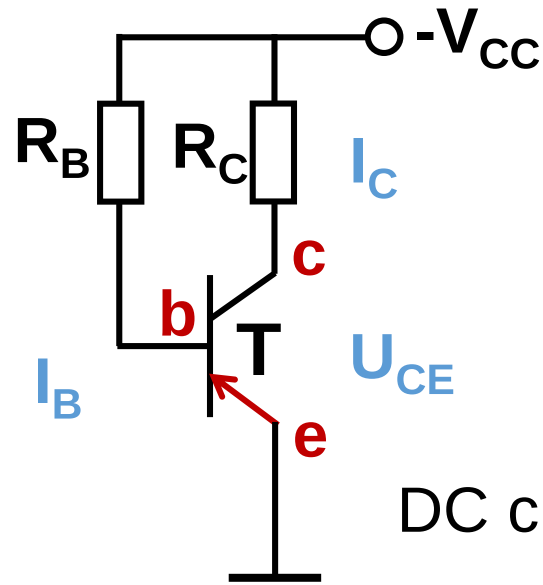
AC circuit



[Q2] The direction of DC current (I_B, I_C), AC current ($\dot{I}_b$, $\dot{I}_c$), DC voltage (U_{BE}, U_{CE}), and AC voltage ($\dot{U}_{be}$, $\dot{U}_{ce}$)?

PNP





DC circuit

AC circuit